Sharp Sample Complexity and Polynomial-Time Learning for Prophet Secretary

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Abstract

In the prophet-secretary problem, n independent nonnegative random variables arrive in uniformly random order. An online decision maker sees each variable's identity and value on arrival and must accept or reject it immediately. The goal is to maximize expected reward by selecting at most one value, with zero reward if none is selected. Here the decision maker receives independent samples from each unknown distribution before the arrivals. Let C_{PS} denote the optimal competitive ratio against the expected maximum when the distributions are known. For every sufficiently small $\varepsilon > 0$, we give a randomized learner that achieves ratio $C_{\mathrm{PS}} - \varepsilon$ in $\mathrm{poly}(n, 1/\varepsilon)$ time. Its sample budget is $O(1/\varepsilon)$ per distribution, independent of n, improving the $O(\varepsilon^{-5})$ sample bound of Cristi and Ziliotto (STOC 2024). A matching worst-case lower bound of $\Omega(1/\varepsilon)$ samples per distribution holds even with adaptive sample requests. The result applies to arbitrary distributions with positive, finite expected maximum. The reward guarantee averages over training and online randomness. The upper bound learns candidate stopping rules and a target expected stopping time from the same samples, then calibrates the rule on independent validation data. A quadratic bound on regret from threshold errors controls statistical loss, while a weighted Monte Carlo recursion approximates the required continuation values. Simultaneous guarantees on reward and the probability of reaching each variable before stopping protect against rare, large values missed by the samples. For the lower bound, we construct statistically close distributions that require different balances between reward and expected stopping time. We bound the information revealed by the samples and show that a learner with too few samples cannot adjust its stopping rule enough to reach the target ratio.

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Contents

1 Introduction 3

2 Preliminaries 6

3 Structural properties of the stopping problem 9

I The upper bound 13

4 The sampling guarantee and its reduction 14

5 Reduction to bounded values 14

6 Keeping scalar weights away from the endpoints 18

7 Empirical scalar optimization with a selected weight 20

8 Curvature under perturbations of the marginal distributions 25

9 A uniform first-moment bound near scalar criticality 28

10 A quadratic diameter bound for sublevel sets of the scalar gap 30

11 An explicit learner with reward and reach guarantees 34

12 Computing the scalar policies 41

II The lower bound 44

13 The sample lower bound 45

14 Finite instances for the lower bound 47

Discussion and formalization 52

15 Discussion 52

16 Formalization 52

A Reward and reach probabilities 53

B Bounded normalization 54

C Scalar curvature 55

D Stability of policy outcomes 57

E Limits of bounded instances 58

F Nondegeneracy at a critical limit 63

G Finite-bit implementation 66

1 Introduction

In the prophet-secretary problem, n independent nonnegative random variables arrive in uniformly random order. An online decision maker observes each variable's identity and value and must accept or reject it immediately, selecting at most one value. The reward is the selected value, or zero if nothing is selected. A competitive ratio is a factor of the expected maximum that a stopping policy guarantees in expected reward on every instance. Let C_{PS} denote the optimal competitive ratio when the distributions are known, uniformly over every finite n and every choice of distributions. Here the distributions are unknown; before the online arrivals begin, the decision maker receives samples from each distribution, independent of one another and of the online values.

For an accuracy $\varepsilon > 0$, we determine how many such samples are needed to approach C_{PS} within ε and show how to use them efficiently. The worst-case sample budget per distribution is linear in inverse accuracy, independently of the number of variables. A randomized algorithm achieves this bound in time polynomial in both n and inverse accuracy.[1]

The samples may miss precisely the values that matter most. For $0 < p < 1$, a variable equal to $1/p$ with probability p and zero otherwise has mean one, yet k samples are likely to be all zero when $kp \ll 1$. Such unseen values can make a substantial contribution to the expected maximum. Cristi and Ziliotto [12] show that $O(\varepsilon^{-5})$ samples per distribution suffice for a $(C_{\mathrm{PS}} - \varepsilon)$ guarantee, independently of the number of variables. We improve this bound to $O(1/\varepsilon)$ and prove a matching lower bound, even for adaptive sample requests.

A hard sample budget bounds the samples requested from each distribution on every training outcome, including when requests depend on earlier observations. The reward guarantee averages over training samples, online values, arrival order and internal randomness.

Theorem 1.1 (Sharp sample complexity)**.** *There are universal constants $0 < a \le A < \infty$, $d < \infty$ and $\varepsilon_0 > 0$ such that, for every accuracy $0 < \varepsilon < \varepsilon_0$, a single randomized learner achieves competitive ratio $C_{\mathrm{PS}} - \varepsilon$ on every finite number n of independent nonnegative variables with positive, finite expected maximum, with a hard budget of at most A/ε training samples per distribution. Its number of arithmetic operations, comparisons and internal random draws is at most $A(n + 1/\varepsilon)^d$ on every outcome. Every learner with this competitive guarantee requires a hard budget of at least a/ε samples per distribution in the worst case, even with adaptive training requests.*

The learner in Construction 11.1 uses the randomized routine of Theorem 12.1. Appendix G gives the corresponding bit-complexity bound for rationally represented data.

1.1 Technical highlights

The upper bound must protect reward from values absent from the samples. For this purpose, Cristi and Ziliotto [12] introduce simultaneous guarantees on reward and the probability of reaching each variable before stopping. We use their reduction and construct a learner with the sharp sample budget. Assign independent uniform arrival times in $[0, 1]$ to the variables; their order is still uniformly random. A policy's duration is its stopping time, with duration 1 if it selects nothing. We optimize a nonnegative weighted sum of reward and duration. An optimal policy accepts whenever stopping gives at least as much expected weighted payoff as continuing. These threshold policies reach every variable with probability at least their expected duration, connecting the two requirements. The optimal value is at least the weighted benchmark that assigns C_{PS} times the expected maximum to reward and C_{PS} to duration. We call the excess above this benchmark the *gap*. Small gaps are the difficult case in both bounds.

The first statistical step is to learn a policy for this weighted objective from k samples per distribution. Suppose values are bounded, the sum of the individual means is bounded, and the reward–duration weights

[1] We used AI tools in developing the analysis and algorithm. The Declaration of AI use gives details.

sum to one and are bounded away from zero. Optimization using the empirical distributions, which assign equal mass to the sampled values, then loses only $O(1/k)$ in expected weighted payoff relative to the optimum, even when the weight is chosen from the samples. A threshold error matters only when an observed value lies near the acceptance boundary, where a wrong decision has small loss. Random arrival times also limit how long this disagreement can last. Integrating over time makes the loss quadratic in the threshold error. Moment estimates for these threshold errors give a bound uniform over the chosen weight and the continuation problems after rejections. Bounded values alone do not bound the sum of the individual means. Comparing selection probabilities under value perturbations gives a uniform bound on this sum under a quantitative small-gap condition; outside that regime, the gap absorbs the larger empirical error.

A policy with small regret still needs to satisfy reward and reach together. Varying the weight gives an ordered family of empirical policies. The same sample block estimates a target from the empirical reward–duration tradeoff. We interpolate the policies' stopping times and extend the curve to cover every target in $[0, 1]$. An independent validation block selects a curve point whose empirical mean matches the estimated target. The point determines a mixture of neighboring policies or an endpoint policy. The signed difference between the true mean at the selected curve point and the target has expectation $O(1/k)$. The selected weight is correlated with this error. A quadratic bound on the distance between weights with small gaps controls the correlation in terms of the gap. The gap also controls upward error in estimating the target. For larger gaps, the excess expected payoff absorbs the remaining fluctuation. These estimates give a single learner with reward and reach deficits of order $1/k$.

Computing this policy presents a different difficulty. The continuation value, the best expected additional payoff from continuing, depends on the set of unseen identities, and there are exponentially many such sets. We expand the dynamic programming recursion over all arrivals and reweight it so that each recursive step shrinks errors by a fixed constant factor. A multilevel Monte Carlo estimator approximates each requested continuation value by estimating differences between successive recursive approximations. For a product of empirical distributions, its cost is exponential only in the sum of the probabilities of positive values. The reduction to bounded values controls the corresponding sum under the true distributions; a deterministic cutoff permits calls to this routine only when the empirical sum is $O(\log k)$. The total work is therefore polynomial in n and k. Ordering the estimated thresholds by duration coefficient keeps calibration valid, while coupling to the exact policies bounds the additional numerical loss.

The lower bound exploits the same reward–duration tradeoff. A universal reward guarantee also gives a duration guarantee after deterministic-zero variables are added, so the learner must satisfy positive weighted combinations of the two even when the weight is hidden. We pair two statistically close distributions with different evaluation weights. Their correlation creates a loss of order t when the distributions differ at scale t. Bounds on how the optimal value and policy change limit the expected benefit of adapting to k samples to $O(kt^2) + o(t)$, yielding $k = \Omega(1/t)$ and hence $\Omega(1/\varepsilon)$ when t is proportional to ε. The essential structural step is to find a distributional perturbation that changes the expected duration of a fixed optimal policy at an instance where the gap vanishes. The upper bound uses this same response to control how far apart weights with small scalar gaps can be.

1.2 Proof overview

The upper proof first reduces arbitrary nonnegative values to bounded instances with a controlled expected number of positive values. For such an instance, the learner uses two independent sample blocks. The first estimates both a duration target and a family of empirical policies ordered by duration; validation selects a policy or a mixture of two adjacent policies. The reward proof combines the empirical regret estimate, the target estimate and the bound on how far apart small-gap weights can lie. It treats bounded and large sums of individual means as two cases of the analysis of this same learner. The reach proof compares duration with the validation target and bounds the probability that the target duration exceeds the validation averages

available within the policy family. The continuation planner supplies the values used in these policies; its approximation and finite-bit errors are charged within the bounded reward and reach guarantees. Only then does the outer reduction recover reward from the original upper tails.

For the lower bound, a normalization places a sequence of nearly worst-case instances in a common bounded range. A compactness argument gives a limiting instance where the gap vanishes. We perturb its distribution in a direction that changes the duration of an optimal policy, then approximate the perturbed instances by finite ones. Convergence and stability give the derivative estimates required by the two-distribution experiment, while likelihood bounds control the information in adaptive training samples. The statistical criterion turns these estimates into the sample lower bound. Both proofs therefore use the shared structural results before proceeding to their separate statistical arguments.

1.3 Related work

Esfandiari, Hajiaghayi, Liaghat and Monemizadeh [14] introduced the prophet-secretary problem. Correa, Saona and Ziliotto [11] develop blind strategies, which use thresholds determined by the distribution of the maximum. Chen, Huang, Li and Tang [5] improve the competitive guarantee by also exploiting individual variables' identities, while Bubna and Chiplunkar [4] give upper bounds even when the algorithm knows the arrival order in advance. The exact value of C_{PS} remains open. Our result determines the sample complexity of approaching this constant.

Azar, Kleinberg and Weinberg [1] initiated the study of prophet inequalities with sample access in place of known distributions. For prophet secretary, Correa, Cristi, Epstein and Soto [6] obtain a competitive ratio of 0.63518 using one sample per distribution through their two-sided game of Googol. Cristi and Ziliotto [12] show that a number of samples independent of n suffices to approach the optimal full-information ratio in the prophet-secretary, free-order and i.i.d. models. Their prophet-secretary bound is $O(\varepsilon^{-5})$ samples per distribution, with polynomial running time in n for each fixed accuracy. We use their reward-and-reach reduction, replace the minimax policy selection by duration calibration, and achieve the sharp $\Theta(1/\varepsilon)$ sample bound with running time polynomial in both n and $1/\varepsilon$. The matching lower bound applies to every learner approaching the optimal competitive ratio, including learners that request samples adaptively.

The arrival model and the relation between distributions affect this cost. In the fixed-order prophet inequality, Rubinstein, Wang and Weinberg [24] attain the optimal ratio $1/2$ with one sample per distribution. In the i.i.d. model, the optimal full-information competitive ratio is known [21, 10]. Correa, Dütting, Fischer and Schewior [8] study selection from an unknown common distribution and show that, for each fixed accuracy, $O(n^2)$ total samples suffice to approach this ratio. Rubinstein, Wang and Weinberg [24] reduce this dependence on n to linear. Further work studies the guarantees achievable with a fixed linear sample budget [9] and obtains an $O(n/\varepsilon)$ total-sample bound in the large-horizon i.i.d. setting [7]. These samples come from one common distribution. Here the distributions may all differ, and the sample bound holds uniformly over every finite number of variables.

Learning an instance-optimal policy gives a different statistical target. For independent values in $[0, 1]$ arriving in a fixed order, Jin, Kesselheim, Ma and Singla [20] study the sample complexity of learning optimal posted prices. At fixed confidence, their welfare guarantee requires $\Theta(\varepsilon^{-2})$ samples per distribution for additive ε accuracy. Their benchmark is the optimal expected reward for the given instance. Our inverse-linear bound concerns approaching the optimal competitive ratio in random order.

Ezra [15] studies a different sampling model: the samples are pooled, and the algorithm is not told which distribution produced each one. In that model, even many samples may fail to support the fixed-order $1/2$ guarantee. Feng, Jin, Wu and Xiang [17] instead study policies given the exact expected maximum, without full distributional knowledge. Beyond single selection, Fu et al. [18] and Feldman, Svensson and Zenklusen [16] study sample complexity for selecting a feasible set under matroid constraints.

1.4 Organization

Both parts use the model and structural results in Sections 2 and 3. Part I gives a polynomial-time learner and proves the sample upper bound. Part II proves the matching sample lower bound by constructing finite instances that satisfy a statistical criterion. Figure 1 shows how the sections fit together. Appendices A to F prove the shared structural results, and Appendix G gives the finite-bit implementation. Sections 15 and 16 discuss extensions and describe the Lean formalization.

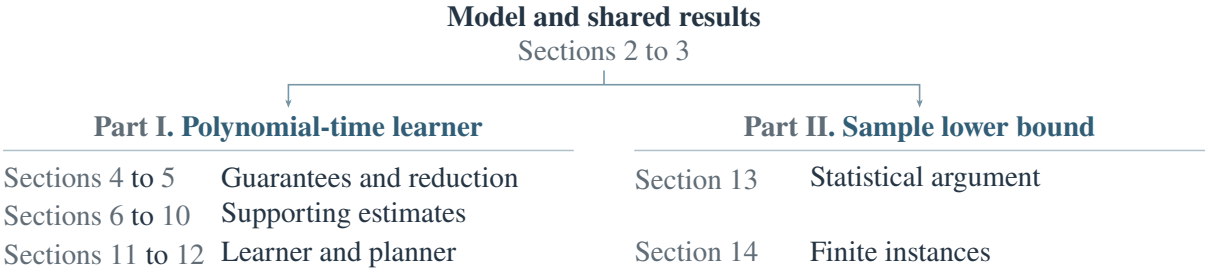


Figure 1: Organization of the paper.

2 Preliminaries

We first specify the random variables, observations and sample budget. Weighted combinations of reward and stopping time connect the reward and reach guarantees in the upper bound and define the statistical experiment in the lower bound. The final subsection describes the limiting instances and perturbations used by both arguments.

2.1 Random variables, samples, and the competitive benchmark

Write $[n] = \{1, \dots, n\}$. An instance consists of independent nonnegative random variables $V_1, \dots, V_n$, where P_i is the distribution of V_i and $P = \bigotimes_{i=1}^{n} P_i$ is their joint distribution. The arrival times $U_1, \dots, U_n$ are independent uniforms on $[0, 1]$, independent also of the values, training samples and policy randomness. Their order is a uniformly random permutation of $[n]$.

At each arrival, an online decision maker observes the variable's identity i, value V_i and arrival time U_i. A stopping policy π specifies the immediate accept or reject decision as a function of training information, revealed history and internal randomness. The history includes zero-valued arrivals. The decision maker may accept at most one value. Let X_t be the value available at time t: at an arrival $t = U_i < 1$, set $X_t = V_i$; at other times $t < 1$, set $X_t = 0$, and set $X_1 = 0$. The arrival times are distinct and lie in $(0, 1)$ almost surely. The stopping time $\tau_\pi \in [0, 1]$ under π is the time of acceptance, or 1 if no value is accepted. The selected value is X_{τ_π}. We write τ when the policy is fixed.

A *learner* uses independent samples from each distribution to choose a stopping policy before the online arrivals. The samples are independent of the online values and arrival times. A *full training round* contains one fresh sample per distribution. A hard budget k allows at most k samples per distribution on every outcome, even with adaptive requests. Given $k \geq 1$ samples per distribution, write $\widehat{P}_i$ for the empirical distribution assigning mass $1/k$ to each sample and $\widehat{P} = \bigotimes_i \widehat{P}_i$ for their product.

We write $\mathbb{E}_P$ and $\Pr_P$ for expectation and probability when the online values have distribution P, averaging also over training samples, arrival times and internal randomness unless these are explicitly held fixed. We omit the subscript when the underlying instance is clear. When Q is an empirical distribution, $\mathbb{E}_Q$ and $\Pr_Q$ hold its training data fixed and use fresh online values drawn from Q; an outer expectation averages over those data. The notation $\mathbb{E}_{\text{clocks}}$ averages only over arrival times. Write $X^* = \max_i V_i$. The competitive guarantee concerns $0 < \mathbb{E}_P[X^*] < \infty$; bounded auxiliary instances may have $\mathbb{E}_P[X^*] = 0$.

We measure running time in the sampling and real-arithmetic model, counting arithmetic operations, comparisons and internal random draws at unit cost. The latter include uniform draws on $[0, 1]$ and sampling from explicitly known distributions with finite support. Requests to the unknown distributions are counted separately by the hard sample budget. For rationally encoded observations and accuracy parameter, Appendix G gives a bit-complexity bound polynomial in n, inverse accuracy and the total input encoding length. Both bounds hold for every realization of the training data, online values and internal randomness.

The optimal full-information competitive ratio is

$$C_{\mathrm{PS}} = \inf_{0<\mathbb{E}_P[X^*]<\infty} \frac{\sup_\pi \mathbb{E}_P[X_{\tau_\pi}]}{\mathbb{E}_P[X^*]},$$

where the infimum ranges over every finite n and all product distributions P with $0 < \mathbb{E}_P[X^*] < \infty$; the policy in the supremum may depend on P.

To implement a policy using these times in the random-order model, sort n fresh uniform draws, independent of the values, training data, arrival permutation and original policy randomness, and assign them successively to arriving variables. The simulated policy receives only the times assigned to current and past variables. This reproduces the uniform-time model and preserves expected reward and every reach probability. Conversely, a policy observing arrival times can ignore their numerical values. All weighted objectives below use stopping time in this uniform-time representation.

The classical prophet inequality [25] gives $C_{\mathrm{PS}} \geq 1/2$, and Bubna and Chiplunkar [4, Theorem 3] prove $C_{\mathrm{PS}} \leq 0.7254$. Their theorem concerns the stronger order-aware model: given the arrival order, an algorithm can use the preceding simulation to implement any policy considered here. For the nondegeneracy argument in Appendix F, the numerical bound implies the strict separation $C_{\mathrm{PS}} < \sqrt{3} - 1$.

For a fixed policy, its reach probability at variable i is

$$q_i(P, \pi) = \Pr_P(\text{no value is accepted before variable } i \text{ arrives}).$$

We write q_i when the instance and policy are clear. For a learned policy, this probability also averages over training unless training data are explicitly held fixed.

For a threshold $b \geq 0$, the following quantities recur in the bounded-instance arguments:

$$S(P) = \sum_i \mathbb{E}_P[V_i], \qquad L(P) = \sum_i \Pr_P(V_i > 0), \qquad L_b(P) = \sum_i \Pr_P(V_i > b).$$

Thus S is the total first moment, L is the expected number of positive values, and L_b is the expected number exceeding b. We also call L_b an expected tail count. When an instance is fixed, we may omit P from these quantities and from expectation and probability subscripts. A uniform bound over a family is stated separately as a supremum of $L_b(P)$.

2.2 Reward, duration, and continuation values

To keep track of both reward and the time for which a policy remains active, assign coefficients $w, c \geq 0$ to reward and stopping time. The weighted objective is $wX_\tau + c\tau$, and its optimal expected value is

$$V(P; w, c) = \sup_\pi \mathbb{E}_P[wX_{\tau_\pi} + c\tau_\pi].$$

Normalized weights have the form $(\lambda, 1 - \lambda)$ for $\lambda \in [0, 1]$. Define the *scalar gap* above the competitive benchmark by

$$F_P(\lambda) = V(P; \lambda, 1 - \lambda) - C_{\mathrm{PS}}[\lambda \mathbb{E}_P[X^*] + 1 - \lambda].$$

For $z \geq 0$, we also use the objective $X_\tau + z\tau$, whose reward coefficient is 1; dividing by $1 + z$ gives normalized weights with $\lambda = 1/(1 + z)$. A policy's regret is the optimal value minus its expected weighted payoff. At

normalized weights, its excess over the competitive benchmark is $F_P(\lambda)$ minus its regret. Normalizations such as $\mathbb{E}_P[X^*] = 1$ are imposed only where stated.

In a state with unseen variable identities $A \subseteq [n]$ and remaining horizon $u \in (0, 1]$, use a relative clock on $[0, u]$. The remaining arrival times are independent uniforms on this interval, τ is the additional stopping time, and $\tau = u$ if no value is selected. The same payoff-process convention gives selected value X_τ, with $X_u = 0$. Rescaling to $[0, 1]$ gives the optimal *additional* value

$$v_A^P(u; w, c) = V(P_A; w, cu), \qquad P_A = \bigotimes_{i \in A} P_i.$$

For $A = \varnothing$ the value is cu. The same formula defines the reward-only value at $u = 0$. This value does not include duration payoff already earned. We may suppress the superscript P inside a proof when the distribution is fixed. The optimal Markov threshold policy accepts a candidate x from variable $i \in A$ when $wx \geq v^P_{A\setminus\{i\}}(u; w, c)$, including ties. We use this tie convention throughout and refer to this rule as the optimal threshold policy. It depends on unseen identities, the current value and remaining time, and rejects zero when $w, c > 0$ except for terminal decisions that do not affect reward or stopping time. Additional unseen options cannot lower continuation values.

In state calculations with $u > 0$, $T = \mathbb{E}[\tau]/u$ denotes the normalized expected duration of an optimal continuation policy, with τ measured from the state's start. The state's value is therefore $w\mathbb{E}[X_\tau] + cuT$. At an arrival, the *Bellman loss* of an action is the difference between the larger of the acceptance and continuation values and the value of the action taken.

2.3 Limits of bounded instances

Bounding the values does not bound the number of variables in a nearly worst-case sequence. Individual rare variables may disappear in the limit even though their combined contribution remains positive. The limit has three components. Some individual variables persist, while aggregate rare exceedances form a Poisson process. A terminal reward records a limiting value level that can be approached with probability tending to one in any fixed final time interval. The next definition specifies the observations and payoffs of this stopping problem. The convergence conditions below describe how finite instances approach it; Lemma 3.11 establishes that such limits exist.

Definition 2.1. *Fix $H < \infty$. A* limiting instance *with cap H consists of a terminal reward $a \in [0, H]$, countably many independent variables V_i with distributions Q_i on $[0, H]$ and independent uniform arrival times, and an independent marked Poisson process with intensity $dt\,\mu(dx)$, where μ is a measure on $(a, H]$, extended by zero to $(0, H]$. We call the variables with distributions Q_i the* individual variables*. Each Poisson point (t, x) reveals a value x at time t; x is its* mark*. For every $b > a$, the sum $\sum_i Q_i((b, H])$ and the measure $\mu((b, H])$ are finite. The payoff process X_t equals the arriving individual value or Poisson mark at each arrival and is zero at other times $t < 1$; its terminal value is $X_1 = a$. A policy is admissible if, almost surely, its prescribed accepting arrivals have an earliest time whenever there are any. Its stopping time τ is that earliest time, or 1 if there is no acceptance, and its reward is X_τ. Every individual arrival reveals its identity and value, including when the value is zero. In finite approximations, the observer may also see the identities revealed by all zero-valued arrivals.*

For a nonnegative integer N and $b > a$, keep only the first N individual variables and Poisson marks above b, together with the terminal reward a; we call this a *truncated instance*. Optimal values are defined by letting $N \to \infty$ and $b \downarrow a$. A policy for a truncated instance ignores the omitted observations when used on the full instance.

For nonnegative weights, the optimal value defined by truncation equals the supremum over all admissible history-dependent policies on the full limiting instance. The comparison in Appendix E proves this identification while allowing policies to use the identities revealed by zero-valued arrivals.

For a remaining horizon of length u, the Poisson intensity becomes $u\mu$ after time rescaling; the distributions of the remaining individual variables are unchanged, and the terminal reward remains available at the terminal time, so $X_u = a$ on the relative clock $[0, u]$. The realized maximum X^* includes the terminal reward and all realized values. The definitions of the optimal value and normalized gap use the same reward and duration conventions for limiting instances. An instance P, finite or limiting, with $\mathbb{E}_P[X^*] > 0$ is *critical* at weights (w, c) if

$$V(P; w, c) = C_{\mathrm{PS}}[w\mathbb{E}_P[X^*] + c].$$

A level is a continuity level of a measure if the measure has no atom there; when several measures are involved, we use a common continuity level. A continuity band has no mass at its boundary.

Convergence must preserve the individual distributions and the aggregate contribution of variables that become rare. The following conditions describe these two limits and identify the terminal reward.

Definition 2.2. *Finite instances P_n supported on a common interval $[0, H]$* converge to *a limiting instance $\mathfrak{B}$ with data $(a, (Q_i), \mu)$ when, after a choice of variable labels, the following conditions hold. Write V_{ni} for variable i of P_n and P_{ni} for its distribution.*

(i) Each distribution P_{ni} corresponding to an individual variable converges weakly to Q_i.

(ii) For every continuity level $b > a$, the sums of the distributions P_{ni} over $i > N$, restricted to $(b, H]$, converge, as $n \to \infty$ and then the number N of individual variables tends to infinity, to μ restricted to $(b, H]$, and $\lim_{N\to\infty} \limsup_{n\to\infty} \max_{i>N} \Pr(V_{ni} > b) = 0$.

(iii) Writing $L_n(x) = \sum_i \Pr(V_{ni} > x)$, one has $L_n(x) \to \infty$ for $0 \le x < a$ and $\sup_n L_n(x) < \infty$ for $x > a$.

Perturbations. To study how the optimal value responds to changes in distribution, both bounds perturb probabilities or values. We use the following conventions. For a probability distribution Q and an integrable function f, write $Qf = \int f\, dQ$. A *likelihood score* records the relative first-order change in probabilities: it is the derivative of the log likelihood at the base distribution. For $P_i h_i = 0$, the perturbation $P_i^\theta = (1 + \theta h_i)P_i$ has score h_i at $\theta = 0$; we call it a *likelihood tilt*. For a product perturbation, the likelihood score of the complete vector of values is $\sum_i h_i(V_i)$, including variables that arrive after the stopping time. An *intensity tilt* on a band B of finite Poisson intensity multiplies μ there by $1 + \theta$; its score is $N_B - \mu(B)$, where N_B counts the points in the band. All admissible parameter ranges are specified at use.

A *perturbation of values* replaces a value x by $x + h\phi(x)$. A symmetric perturbation replaces it by $x + h\varsigma\phi(x)$, where ς is an independent uniform sign in $\{-1, 1\}$, sampled separately for each affected value. The functions ϕ and their supports are specified at use.

Throughout the paper, KL is Kullback–Leibler divergence, used to bound the information supplied by training samples. We use natural logarithms unless a base is displayed. Constants $K, K', K'', \ldots$ and C_p may change from line to line, with dependence limited to the parameters in the enclosing statement. Constants with fixed roles have distinct symbols or subscripts.

3 Structural properties of the stopping problem

Expected duration measures how long a policy leaves future variables available. It connects the upper bound's reach requirement to the lower bound's reward guarantee. We first establish this connection, then control changes of the optimal value and of nearly optimal policies. The common destination is a perturbation that changes the expected duration of a fixed optimal policy at an instance where the competitive benchmark is tight. The proofs are in Appendices A to F.

3.1 Reward and reach

The first result extends the competitive benchmark from reward to any nonnegative weighted combination of reward and duration. Its proof represents duration payoff through rare, large values arriving at independent

uniform times.

Lemma 3.1. *For every finite bounded product distribution P and all $w, c \geq 0$,*

$$V(P; w, c) \geq C_{\mathrm{PS}}[w\mathbb{E}_P[X^*] + c].$$

Moreover,

$$\inf_{P,w,c\geq 0:\, w\mathbb{E}_P[X^*]+c>0} \frac{V(P; w, c)}{w\mathbb{E}_P[X^*] + c} = C_{\mathrm{PS}}.$$

Temporarily keep one variable among the unseen variables and run the policy on all other arrivals. The probability that its independent uniform clock rings before the auxiliary stopping time equals the expected duration of that run. When deleting unseen variables cannot raise the thresholds, the original stopping time is at most the auxiliary stopping time, giving the lower bound on reach. This connects duration calibration to reach in the upper bound; the strict term below rules out a tight reward-only endpoint in Section 6.

Lemma 3.2. *Under a finite product distribution with independent uniform arrival times, let q_i and p_i^{sel} be a policy's reach and selection probabilities at variable i, and put $T = \mathbb{E}[\tau]$. Every online policy satisfies $q_i \leq 2T$. If it is a Markov threshold policy whose acceptance threshold at a fixed time and candidate variable cannot decrease when unseen variables are added, then*

$$q_i - T \geq \frac{1}{2}(p_i^{\mathrm{sel}})^2. \tag{1}$$

For finite instances, an optimal threshold rule satisfies $q_i \geq T$ even when evaluated under another product distribution. For a limiting instance, both bounds hold for each individual variable under its own admissible optimal Markov threshold policy. On a remaining horizon $u > 0$, the same statements use $T = \mathbb{E}[\tau]/u$, with time measured from the state's start.

The limiting arguments need nearly worst-case instances in a common bounded range, with both weights positive. The next lemma supplies this analytic normalization by exchanging clipped tail reward for duration payoff. The learner's sample-based cutoff is constructed separately in Section 5.

Lemma 3.3. *Suppose a finite instance with $0 < \mathbb{E}_P[X^*] < \infty$ and weights $w > 0, c \geq 0$ has scalar ratio $r_0 = V(P; w, c)/(w\mathbb{E}_P[X^*] + c) < 1$. Set $h = V(P; w, c)/w$, let P^h be the joint distribution of the capped values $\min(V_i, h)$, and put*

$$c_h = c + w \sum_i \mathbb{E}[(V_i - h)_+].$$

Then $V(P^h; w, c_h) \leq V(P; w, c)$. Normalizing this capped instance gives, with notation P, w, c reused,

$$\mathbb{E}_P[X^*] = 1, \quad w + c = 1, \quad w, c \in [1 - r_0, r_0], \quad V_i \leq \frac{r_0}{1 - r_0},$$

and scalar ratio at most r_0.

A universal reward guarantee requires reaching each variable with substantial probability, since the samples may miss a rare, large value. The arrival times of zero variables then measure elapsed time before stopping. These two necessary conditions let the lower bound evaluate a reward guarantee through weighted reward and duration.

Lemma 3.4. *Suppose a learner with a finite hard training budget guarantees $\rho\mathbb{E}_P[X^*]$ on every independent nonnegative instance. At every fixed bounded base instance its probability q_i of reaching variable i is at least ρ.*

Lemma 3.5. *Append N deterministic-zero variables to a bounded base instance. If each is reached with probability at least ρ, then*

$$\mathbb{E}[\tau] \geq \rho - O(N^{-1/2}).$$

The bound is uniform over all online policies, including those that use the identities and arrival times of the zero variables. Moreover, any policy on the augmented instance can be represented by a randomized policy on the base instance with the same reward and no smaller duration.

These facts are proved in Appendices A and B.

3.2 Curvature and stability

Each fixed policy gives an affine function of the normalized weight, so the optimal value is convex. Policy changes could create corners in this upper envelope. Random arrival times smooth the changes enough to give the following curvature bound, uniformly in the number of variables and their distributions.

Theorem 3.6. *For every finite bounded product distribution P, the scalar value $V(\lambda) = V(P; \lambda, 1-\lambda)$ is continuously differentiable with locally Lipschitz derivative (locally $C^{1,1}$) on $(0,1)$ and satisfies*

$$0 \leq V''(\lambda) \leq \frac{1}{\lambda^2(1-\lambda)} \quad \text{for almost every } \lambda \in (0,1). \tag{2}$$

The same local $C^{1,1}$ bound holds for every finite pointwise limit of such scalar value functions.

A policy's slope in the normalized weight is expected reward minus expected duration. The next consequence bounds how much this slope can change at small regret. In Theorem 13.2, it limits the benefit of adjusting a policy to the training data.

Corollary 3.7. *Fix a compact interval $I \subset (0,1)$ and a cap $H < \infty$. For any finite product distribution on $[0,H]^n$ and any $\lambda \in I$, let π_* be scalar-optimal. For any randomized online policy π, let $r = V(P; \lambda, 1-\lambda) - \mathbb{E}[\lambda X_{\tau_\pi} + (1-\lambda)\tau_\pi]$. Then*

$$\left|(\mathbb{E}[X_{\tau_\pi}] - \mathbb{E}[\tau_\pi]) - (\mathbb{E}[X_{\tau_{\pi_*}}] - \mathbb{E}[\tau_{\pi_*}])\right| \leq K\sqrt{r}$$

for a finite constant K depending only on I and H.

Distributional derivatives require more than a comparison of expected reward–duration slopes: they involve correlations with likelihood scores. We therefore compare policies on the same values and arrival times. Outside a short terminal window and arrivals near an acceptance threshold, disagreement costs regret. The next theorem makes the resulting outcome stability uniform; the corollary converts it into stability of score-weighted expectations.

Theorem 3.8. *Consider a family of finite independent instances with values in $[0,H]$, weights $w, c \geq \delta > 0$, $w + c = 1$, and independent uniform arrival times. Assume that for every $b > 0$ there is a finite uniform bound*

$$\sup_P L_b(P) < \infty.$$

Let π^* be the optimal Markov threshold policy for $wX_\tau + c\tau$. Any competing policy π may use arbitrary independent auxiliary randomness and may observe every zero variable and its identity. Write

$$r = V(P; w, c) - \mathbb{E}_P[wX_{\tau_\pi} + c\tau_\pi].$$

When both policies are run on the same complete instance and arrival times, there is a modulus $\omega(r)$ tending to zero as $r \downarrow 0$, uniform over this family and over the competing policies, such that

$$\mathbb{E}\left[|X_{\tau_\pi} - X_{\tau_{\pi^*}}| + |\tau_\pi - \tau_{\pi^*}|\right] \leq \omega(r). \tag{3}$$

Corollary 3.9. *Let P, π, and π^* satisfy Theorem 3.8, with value bound H, scalar regret r, and outcome bound $\omega(r)$. For any $I \geq 0$ and any mean-zero likelihood score ℓ of the complete vector of values with $\mathbb{E}[\ell^2] \leq I$,*

$$\left|\mathbb{E}[\ell(X_{\tau_\pi} - X_{\tau_{\pi^*}})]\right| + |\mathbb{E}[\ell(\tau_\pi - \tau_{\pi^*})]| \leq \sqrt{2I(H+1)\omega(r)}.$$

The proofs are in Appendices C and D.

3.3 Critical limits and perturbations

A finite worst-case instance need not exist, so we first pass to a limit of increasingly hard instances. Rare independent variables aggregate into Poisson arrivals; values available arbitrarily late give the terminal reward. The first comparison below extends the usual Poisson approximation to the stopping problem. It accounts for information from zero-valued arrivals, which reveal that the corresponding identity cannot arrive again. This comparison gives the compactness statement that follows.

Lemma 3.10. *Start with a finite product distribution on $[0,H]^n$ for $H < \infty$, possibly with a fixed terminal reward, and leave any chosen set of variables unchanged. Write ν_i for the restriction of each remaining variable's distribution to $(0,H]$ and $p_i = \nu_i((0,H])$. Replace these variables by a Poisson process with intensity $dt\,\sum_i \nu_i(dx)$ restricted to marks above the common terminal reward, which is zero when no terminal reward is specified. Write V_{fin} and V_{poi} for the scalar values before and after this replacement. With the same terminal reward in the two models,*

$$|V_{\mathrm{fin}}(w,c) - V_{\mathrm{poi}}(w,c)| \leq \frac{\pi}{2}\, wH\Big(\sum_i p_i^2\Big)^{1/2}.$$

The upper comparison permits arbitrary finite-model policies using all observed zero identities.

Lemma 3.11. *Fix $H < \infty$. Every sequence of finite independent instances on $[0,H]$ has a subsequence and a limiting instance such that the expected maxima and all scalar values $V(w,c)$ converge. The convergence is uniform on compact weight sets.*

At a tight benchmark, the limit's terminal reward must vanish. At positive reward and duration weights, the remaining stopping problem has an optimal threshold policy whose outcomes obey the same stability principle as finite instances.

Lemma 3.12. *Let $\mathfrak{B}$ be a limit of finite instances in the sense of Definition 2.2. If $\mathfrak{B}$ is critical at weights $w > 0$, $c \geq 0$, then its terminal reward is zero.*

Lemma 3.13. *At positive weights w, c, a limiting instance with zero terminal reward has a measurable optimal Markov threshold policy. Ties between positive acceptance and continuation values have probability zero, and its selected outcome is unique up to null events. The outcome-stability conclusion of Theorem 3.8 extends to this instance, with a modulus depending on its positive-level tail counts.*

With a stable optimizer available, the envelope principle differentiates the optimal value by holding the base policy fixed. This turns changes of distribution into expectations that can be compared along finite approximations.

Lemma 3.14. *Let P be a limiting instance with zero terminal reward, let $w, c > 0$, and let π_0 be its optimal threshold policy. For a bounded individual-variable likelihood tilt or a finite positive-band intensity tilt P^θ, the scalar value is differentiable at $\theta = 0$ and*

$$\frac{d}{d\theta} V(P^\theta; w, c)\Big|_{\theta=0} = \mathbb{E}_P[Z(wX_{\tau_{\pi_0}} + c\tau_{\pi_0})],$$

where Z is the centered score of the perturbed individual variable, or $Z = N_B - \mu(B)$ for a band of Poisson marks B, with N_B its point count. A bounded continuous perturbation of values of one individual variable, or of marks in a finite positive band, supported away from zero has scalar derivative equal to w times its expected perturbation on the selected occurrence. Independent symmetric perturbations of such marks have scalar value change $o(h)$ when their amplitude is h.

At a critical instance, nonnegativity of the scalar gap forces its first derivative to vanish under the two-sided perturbations above. Expected duration cannot have zero derivative in all these directions. The next lemma supplies a direction in which it changes.

Lemma 3.15. *Let $\mathfrak{B}$ be a limit of finite instances in the sense of Definition 2.2, with zero terminal reward and $\mathbb{E}_{\mathfrak{B}}[X^*] > 0$. Suppose it is critical at weights $w, c > 0$. There is either a bounded likelihood-score tilt of one individual variable or an intensity tilt of one finite positive Poisson band for which the duration of the fixed optimal threshold policy has a nonzero derivative with respect to the perturbation parameter. In the Poisson case, the band can be chosen as $(b, H]$ for a positive continuity level b.*

This duration response has two uses. It forces weights with small gaps to be close in Theorem 10.1, and it supplies the statistically close distributions for the lower-bound experiment in Section 14. The limiting and perturbation results are proved in Appendices E and F.

Part I
The upper bound

The proof separates reduction, learning and computation. Section 4 states the general guarantee and the bounded reward-and-reach guarantee; Section 5 proves the reduction to bounded values. The empirical regret bound in Section 7 controls the loss from learning a policy for a weighted objective, while Sections 8 to 10 supply the structural estimates needed to choose its weight. Section 11 combines these estimates in a learner with two independent sample blocks: one estimates the policy family and target duration, and the other calibrates the choice of policy. Section 12 computes the required continuation approximations in polynomial time, and Appendix G completes the bit analysis. Figure 2 shows the main steps. All use the shared results in Section 3.

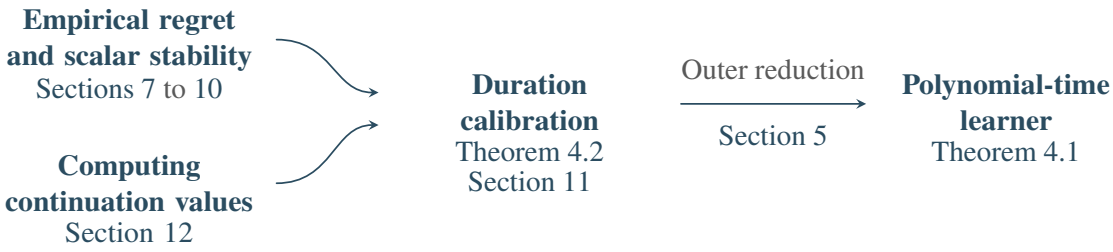


Figure 2: Main proof steps for the upper bound.

4 The sampling guarantee and its reduction

Theorem 4.1 (Polynomial-time sample upper bound)**.** *There is a universal constant $K_{\text{out}} > 0$ such that, for every integer $k \geq 1$, a randomized learner using at most k training samples per distribution satisfies*

$$\mathbb{E}[X_\tau] \geq \left(C_{\text{PS}} - \frac{K_{\text{out}}}{k}\right)\mathbb{E}[X^*]$$

for every finite number of independent nonnegative random variables with $0 < \mathbb{E}[X^*] < \infty$. Its number of arithmetic operations, comparisons and internal random draws is polynomial in n, k on every outcome. For rationally represented observations, its bit complexity is polynomial in n, k and their representation length. Expectations include training and online randomness.

Each distribution may produce a rare realization above the cap. Accepting every reached cap collects this excess whenever the policy reaches the corresponding variable, so the reach guarantee also lower-bounds the expected reward from the entire upper tail. A positive lower bound on the expected maximum then converts the bounded learner's additive error into competitive-ratio error. The reduction normalizes the values and bounds the positive tail count as required by the next theorem.

Here m is the number of samples per distribution in each of two independent blocks. The bounded learner uses $2m$ samples per distribution and has reward and reach errors $O(1/m)$. Taking m proportional to $1/\varepsilon$ gives the desired accuracy.

Theorem 4.2 (Bounded reward and reach)**.** *Fix $0 < \mu_{\min} < 1/80$. There is a constant K, depending only on $\mu_{\min}$, such that for every integer $m \geq 1$, every n, and every finite set of possible values $G \subset [0, 1]$ containing $0, 1$, a randomized learner uses at most $2m$ independent samples per distribution and has the following guarantee. For every product distribution P on G^n satisfying*

$$\mu_{\min} \leq \mathbb{E}_P\left[\max_i V_i\right] \leq 1/80, \qquad \sum_i \Pr_P(V_i > 0) \leq 2\log m,$$

its online policy always accepts a cap 1 when reached, always rejects 0, and satisfies

$$\mathbb{E}_P[X_\tau] \geq C_{\text{PS}}\mathbb{E}_P\left[\max_i V_i\right] - \frac{K}{m}, \qquad q_i \geq C_{\text{PS}} - \frac{K}{m} \quad (1 \leq i \leq n).$$

Here q_i is the probability of reaching variable i under the learned policy. The expectations and reach probabilities include training, arrival-order, and policy randomness. The learner runs in time polynomial in n, m in the computational model of Section 2.

5 Reduction to bounded values

The reduction supplies the bounded learner with a scale and a controlled expected number of positive values. One training block chooses a lower cutoff; a second estimates a high quantile of the maximum to set the cap. Capping and rounding give a finite set of bounded values. During online selection, the policy acts on these transformed values and collects the original selected value. Apply Theorem 4.2 with $\mu_{\min} = \mu_{\text{out}} := 10^{-7}$ and constant K: a product distribution supported on a finite set of values in $[0, 1]$ is eligible when

$$\mu_{\text{out}} \leq \mathbb{E}[X^*] \leq \frac{1}{80}, \qquad \sum_i \Pr(V_i > 0) \leq 2\log m,$$

and the set of possible values contains $0, 1$. On this class, the learner uses at most $2m$ full rounds and has

$$\mathbb{E}[X_\tau] \geq C_{\text{PS}}\mathbb{E}[X^*] - \frac{K}{m}, \qquad q_i \geq C_{\text{PS}} - \frac{K}{m} \quad \text{for every } i. \tag{4}$$

The policy rejects zero and accepts a reached cap on every training outcome, including when an endpoint has zero probability. On the original instance, the policy applies these acceptance rules to the transformed values.

Construction 5.1. *Fix an integer $m \geq 1$. Give every training and online observation an independent uniform tie-breaker $A_i \in [0,1]$, independently of the arrival times. In lexicographic order on (V_i, A_i), let T_{cut} be the lexicographically smallest maximum in m full training rounds. In each later round, set observations at or below the cutoff to zero:*

$$Y_i = V_i \mathbf{1}_{\{(V_i,A_i)>T_{\mathrm{cut}}\}}.$$

From $q_m = 6000\lceil \log_2(2m)\rceil$ fresh rounds with this cutoff applied, take the r_m-th largest numerical maximum B, where $r_m = q_m/500$. If $B = 0$, accept the first strictly positive value after applying the cutoff. Otherwise set $H = 1000B$, cap at H, round downward on mesh H/m leaving $0, H$ unchanged, and divide by H. Train the bounded learner on $2m$ fresh transformed rounds. Apply the same transformations during the online phase and collect the original selected value; observations whose transformed value is zero are rejected.

The construction uses at most

$$3m + 6000\lceil \log_2(2m)\rceil \tag{5}$$

samples per distribution.

Lemmas 5.2 and 5.3 justify the cutoff and scale; Lemma 5.4 converts the bounded learner's reward and reach guarantees into a reward guarantee for the original values.

Lemma 5.2. *Let P be the product distribution of independent nonnegative variables $V_1, \ldots, V_n$ with $\mathbb{E}[X^*] = \mathbb{E}[\max_i V_i] < \infty$, and let $m \geq 1$ be an integer. For the cutoff T_{cut} defined from m full training rounds in Construction 5.1, the variables Y_i after applying the cutoff are conditionally independent across variables given T_{cut} and satisfy*

$$\mathbb{E}\left[\max_i Y_i \,\middle|\, T_{\mathrm{cut}}\right] \leq \mathbb{E}_P[X^*], \qquad \mathbb{E}\left[\max_i Y_i\right] \geq \left(1 - \frac{1}{m+1}\right)\mathbb{E}_P[X^*].$$

Moreover, with probability at least $1 - 1/m$ over the cutoff,

$$\sum_i \Pr(Y_i > 0 \mid T_{\mathrm{cut}}) \leq 2\log m.$$

Proof. Fresh variables remain independent conditional on T_{cut}. The CDF of the lexicographically ordered maximum is

$$G(x,u) = \prod_i \{\Pr(V_i < x) + u\Pr(V_i = x)\}.$$

At each numerical atom, varying u continuously fills its CDF jump interval. Thus the maximum together with its tie-breaker has a uniform probability transform, and $Q := G(T_{\mathrm{cut}})$ is the minimum of m independent uniforms:

$$Q \sim \mathrm{Beta}(1,m), \quad \mathbb{E}[Q] = \frac{1}{m+1}, \quad \Pr(Q < m^{-2}) = 1 - (1-m^{-2})^m \leq \frac{1}{m}. \tag{6}$$

Conditional on T_{cut}, put $p_i = \Pr((V_i, A_i) > T_{\mathrm{cut}})$. Independence gives $\prod_i (1-p_i) = Q$. In particular,

$$\sum_i \Pr(Y_i > 0 \mid T_{\mathrm{cut}}) \leq \sum_i p_i \leq -\log Q.$$

This proves the tail-count bound on $Q \geq m^{-2}$.

Write $Z = \max_i V_i$ and Z^{tag} for its maximum together with the tie-breaker. Since $\max_i Y_i = Z\mathbf{1}_{\{Z^{\mathrm{tag}} > T_{\mathrm{cut}}\}}$ and numerical value increases with lexicographic order, the discarded lower Q fraction contributes at most

$Q\mathbb{E}[X^*]$ to the expected maximum. Hence

$$(1-Q)\mathbb{E}_P[X^*] \le \mathbb{E}[\max_i Y_i \mid T_{\text{cut}}] \le \mathbb{E}_P[X^*].$$

Taking expectations in this bound and using (6) proves the result. □

The scale estimate uses a quantile of the maximum, which can be learned even when the uncapped mean is hard to estimate. The next lemma makes the capped expected maximum a controlled fraction of the cap.

Lemma 5.3. *Fix the cutoff T_{cut} and the conditional product distribution of $Y = (Y_i)$ defined in Construction 5.1. For an integer $m \ge 1$, use the q_m fresh rounds and order statistic B of Construction 5.1. With conditional probability at least $1 - 1/m$,*

$$\Pr(\max_i Y_i > B) \le 0.003, \qquad \Pr(\max_i Y_i \ge B) \ge 0.001, \tag{7}$$

for a fresh round with the cutoff and realized cap-pilot data held fixed. All fresh-round expectations below use this conditioning. On this event:

1. *$B = 0$, and accepting the first strictly positive value after applying the cutoff has expected reward at least $0.997\mathbb{E}[\max_i Y_i]$; or*
2. *$B > 0$, and $H = 1000B$, $Z_i = \min(Y_i, H)$ satisfy*

$$10^{-6} \le \frac{\mathbb{E}[\max_i Z_i]}{H} \le \frac{1}{250}, \tag{8}$$
$$\Pr(\max_i Y_i \ge H) \le 0.003, \qquad H \le 10^6\mathbb{E}[\max_i Z_i] \le 10^6\mathbb{E}[\max_i Y_i].$$

Proof. Randomize ties among sample maxima with independent uniform tie-breakers, and let F be their numerical CDF. For a sampled maximum Z and its independent uniform tie-breaker A, the randomized rank $F(Z-) + A[F(Z) - F(Z-)]$ is uniform. The r_m-th smallest complementary rank W_{r_m} satisfies the binomial Chernoff bounds [3, Section 2.2]

$$\Pr(W_{r_m} < 0.001) = \Pr(\text{Bin}(q_m, 0.001) \ge 0.002q_m) \le e^{-q_m/3000},$$
$$\Pr(W_{r_m} > 0.003) = \Pr(\text{Bin}(q_m, 0.003) < 0.002q_m) \le e^{-q_m/6000}.$$

Their sum is at most $2e^{-q_m/6000} \le 1/m$. Otherwise the CDF rank lies in $[0.997, 0.999] \cap [F(B-), F(B)]$, proving (7). Tie-breaking leaves B unchanged. Hold the realized cap-pilot data fixed for the following fresh-round comparisons.

If $B = 0$, all other variables are zero with probability at least 0.997, so the policy that accepts the first positive value reaches each variable with at least that probability. Independence of reach and its value gives $\mathbb{E}[X_\tau] \ge 0.997 \sum_i \mathbb{E}[Y_i] \ge 0.997\mathbb{E}[\max_i Y_i]$; the sum is finite since $n < \infty$ and $\mathbb{E}[Y_i] \le \mathbb{E}[X^*]$.

If $B > 0$, put $Z = \max_i Y_i$. The bounds $B\mathbf{1}_{\{Z \ge B\}} \le \min(Z, H) \le B + H\mathbf{1}_{\{Z > B\}}$ yield

$$10^{-6} \le \frac{B}{H}\Pr(Z \ge B) \le \frac{\mathbb{E}[\max_i Z_i]}{H} \le \frac{B}{H} + \Pr(Z > B) \le \frac{1}{250}.$$

The lower bound gives the scale estimate; $H > B$ gives $\Pr(Z \ge H) \le 0.003$. □

Lemma 5.4. *Let $Y_1, \dots, Y_n$ be independent nonnegative variables with $\mathbb{E}[\max_i Y_i] < \infty$. Fix $H > 0$ and an integer $m \ge 1$. Let $Z_i = \min(Y_i, H)$, and round Z_i downward on the mesh H/m, preserving zero and H exactly. Call the result W_i. Then*

$$\mathbb{E}[\max_i Z_i] - H/m \le \mathbb{E}[\max_i W_i] \le \mathbb{E}[\max_i Z_i]. \tag{9}$$

Let a policy act on the rounded observations and accept every reached H, but receive the actual selected value. Let I be its selected identity, with $I = 0$ if it accepts nothing, and set $Y_0 = W_0 = 0$. Its actual and rounded selected values are Y_I and W_I. If q_i is its reach probability and $e_i = \mathbb{E}[(Y_i - H)_+]$, then

$$\mathbb{E}[Y_I] \geq \mathbb{E}[W_I] + \sum_i q_i e_i, \qquad \mathbb{E}[\max_i Y_i] \leq \mathbb{E}[\max_i W_i] + H/m + \sum_i e_i.$$

Thus, for $0 \leq \rho \leq 1$ and every real Δ, the two bounds $\mathbb{E}[W_I] \geq \rho\mathbb{E}[\max_i W_i] - \Delta$ and $q_i \geq \rho$ imply

$$\mathbb{E}[Y_I] \geq \rho\mathbb{E}[\max_i Y_i] - \Delta - \rho H/m. \tag{10}$$

These statements also hold after averaging over independent inner training and policy randomization.

Proof. Rounding changes the maximum by at most H/m. Reach is independent of Y_i, and every reached $Y_i \geq H$ is accepted. Receiving the actual selected value adds $\sum_i q_i e_i$ in expected tail reward and also restores any rounding loss, proving the reward inequality. The bound on the expected maximum follows from the pathwise inequalities

$$\max_i Y_i \leq \max_i Z_i + \sum_i (Y_i - H)_+ \leq \max_i W_i + H/m + \sum_i (Y_i - H)_+.$$

Multiplying by ρ and using the reward and reach bounds gives (10). The e_i are fixed conditional on the outer pilot, so independent inner training can be averaged out. Without rounding, the reward relation is an equality and the H/m terms vanish. □

Proof of Theorem 4.1. Use Construction 5.1 with m sufficiently large. Let G_0 be the event that the positive tail count after applying the cutoff is at most $2\log m$, and G_1 the cap-pilot event. By Lemmas 5.2 and 5.3, $\Pr(G_0^c) \leq 1/m$ and $\Pr(G_1^c \mid T_{\text{cut}}) \leq 1/m$.

Condition on the outer cutoff and cap-pilot data. Expectations of transformed values and policy outcomes in the following comparisons average over fresh online data and inner training. On $G_0 \cap G_1$ with $B > 0$, (8) and (9) give $\mathbb{E}[\max_i W_i]/H \in [10^{-6} - 1/m, 1/250] \subset [\mu_{\text{out}}, 1/80]$. Capping and rounding preserve the tail-count bound, and the normalized grid contains $0, 1$, so (4) applies. Let I be the selected identity, with $I = 0$ for no selection and $Y_0 = W_0 = 0$. With $\rho = C_{\text{PS}} - K/m \geq 0$, the bounded guarantee gives $\mathbb{E}[W_I] \geq \rho\mathbb{E}[\max_i W_i] - KH/m$ and $q_i \geq \rho$. By Lemma 5.4,

$$\mathbb{E}[Y_I] \geq \rho\mathbb{E}[\max_i Y_i] - (K + \rho)H/m \geq C_{\text{PS}}\mathbb{E}[\max_i Y_i] - D\mathbb{E}_P[X^*]/m$$

for a universal D, since $H \leq 10^6\mathbb{E}[\max_i Y_i] \leq 10^6\mathbb{E}_P[X^*]$. The $B = 0$ branch gives $0.997\mathbb{E}[\max_i Y_i] \geq C_{\text{PS}}\mathbb{E}[\max_i Y_i]$.

On failure, nonnegative reward and $\mathbb{E}[\max_i Y_i] \leq \mathbb{E}_P[X^*]$ bound the expected loss by $2C_{\text{PS}}\mathbb{E}_P[X^*]/m$. Averaging over the outer data and applying Lemma 5.2 gives

$$\mathbb{E}[X_\tau] \geq C_{\text{PS}}\mathbb{E}[\max_i Y_i] - (D + 2C_{\text{PS}})\mathbb{E}_P[X^*]/m \geq \left(C_{\text{PS}} - \frac{D + 3C_{\text{PS}}}{m}\right)\mathbb{E}_P[X^*].$$

The fresh blocks preserve the bounded learner's independent-sample model conditional on the outer stages. The budget is $3m + O(\log m)$ by (5); taking $m = \lfloor k/10 \rfloor$ for sufficiently large k gives budget at most k and deficit $O(1/k)$. Each outer stage takes polynomial time in n, m, as does the bounded learner. For the remaining finitely many k, reject every value and increase K_{out}. The rational-input implementation, including finite tie-breaking marks, is proved in Appendix G. □

Remark 5.5. *For almost surely finite random variables with $\mathbb{E}_P[X^*] = \infty$, mix the policy with probability $1/k$ with acceptance of a uniformly chosen variable. Some variable has infinite mean since $n < \infty$, so the mixture has infinite expected reward. For finite $\mathbb{E}_P[X^*]$, it costs only $O(\mathbb{E}_P[X^*]/k)$ and no samples.*

6 Keeping scalar weights away from the endpoints

The empirical analysis assigns a positive coefficient to stopping time. This makes continuation values vary with time and limits how long a threshold error can affect a decision. We therefore keep scalar weights in a fixed interior interval. To justify this restriction, we show that a bounded instance with positive expected maximum has uniform slack at the reward-only endpoint. We prove this gap estimate using the strict reach bound in Lemma 3.2.

Lemma 6.1. *Fix a cap $H < \infty$ and a positive lower bound on the expected maximum $\mu_{\min} > 0$. There is $d(H, \mu_{\min}) > 0$ such that every finite independent instance on $[0, H]$ with $\mathbb{E}_P[X^*] \geq \mu_{\min}$ satisfies*

$$V(P; 1, 0) - C_{\mathrm{PS}}\mathbb{E}_P[X^*] \geq d(H, \mu_{\min}). \tag{11}$$

Proof. Suppose equality holds in a bounded limit of finite instances with positive expected maximum, in the sense of Definition 2.2. Normalize it to expected maximum 1 and optimal expected reward C_{PS}. After clipping, increase the duration coefficient by the total upper-tail expectation. Equality in the resulting bounds then forces all excess reward into one variable, where strict reach gives a contradiction. By Lemma 3.12, also valid at duration coefficient zero, the terminal reward is zero.

Clip at C_{PS} and let e be the total expected excess, summing over individual variables and integrating against the Poisson intensity. Local finiteness above $C_{\mathrm{PS}}/2$ gives $e < \infty$, and $e \geq 1 - C_{\mathrm{PS}} > 0$. The capped maximum is $\min(X^*, C_{\mathrm{PS}})$. Write V_{new} for the optimal value at coefficients $(1, e)$. Then

$$V_{\mathrm{new}} \leq C_{\mathrm{PS}}.$$

Indeed, apply Lemma 3.3 to finite approximants of value $v_m \to C_{\mathrm{PS}}$, with cap v_m and tail total e_m. Above a continuity level below $C_{\mathrm{PS}}/2$, aggregate measures of values converge; the continuous kernel $(x - v_m)_+$ gives $e_m \to e$, including with atoms at C_{PS}. Capped values converge at this positive duration coefficient, so $v_{\mathrm{new},m} \leq v_m$ passes to the limit.

The scalar guarantee in Lemma 3.1 and the bound $\mathbb{E}[\min(X^*, C_{\mathrm{PS}})] + e \geq 1$ now give

$$C_{\mathrm{PS}} \;\leq\; C_{\mathrm{PS}}(\mathbb{E}[\min(X^*, C_{\mathrm{PS}})] + e) \;\leq\; V_{\mathrm{new}} \;\leq\; C_{\mathrm{PS}}.$$

Equality forces the sum of tail excesses to equal their maximum almost surely. Independent values above the cap from individual variables can occur together, and a positive upper tail of the Poisson process can produce two points. Hence only one individual variable j contributes to the excess e.

The capped optimal threshold policy exists since $e > 0$. At time s, its continuation excludes the accrued payoff es; rescaling the remaining arrival times reduces the duration coefficient and Poisson intensity by $1 - s$ and removes seen individual variables. It is therefore at most $V_{\mathrm{new}} = C_{\mathrm{PS}}$, so every reached cap is accepted.

In the capped instance, acceptable values before any fixed time below 1 lie in a locally finite positive band. Thus the auxiliary stopping time with j kept unseen is positive almost surely, giving $q_j > 0$ and $p_j^{\mathrm{sel}} \geq q_j P(V_j \geq C_{\mathrm{PS}}) > 0$. Apply the strict reach bound in this capped instance.

Lift the optimizer by applying its rule to the capped observations of the original instance. Under this coupling, the reach events and the stopping time τ agree with those in the capped instance. Write $T = \mathbb{E}[\tau]$. The expected reward on the original instance is

$$\mathbb{E}[X_\tau] = \mathbb{E}[\min(X_\tau, C_{\mathrm{PS}})] + eq_j = V_{\mathrm{new}} + e(q_j - T) > C_{\mathrm{PS}}$$

by (1), a contradiction.

If uniform slack failed, compact extraction would produce such a bounded limiting instance with positive expected maximum. Reward-only values converge by value convergence at positive duration coefficients and $|V(P;1,c) - V(P;1,0)| \le c$. □

The endpoint gaps are uniformly positive by (11) and $F_P(0) = 1 - C_{\mathrm{PS}}$. Since F_P is $2(H+1)$-Lipschitz, there are $a, d_0 > 0$, depending only on $H, \mu_{\min}, C_{\mathrm{PS}}$, with $F_P(\lambda) \ge d_0$ outside $[a, 1-a]$. For sufficiently small s, the sublevel set $\{\lambda : F_P(\lambda) \le s\}$ therefore lies in this fixed interior interval.

Lemma 6.2. *Fix $H < \infty$ and $\mu_{\min} > 0$. There are $a \in (0, 1/2)$ and $K < \infty$, depending only on $H, \mu_{\min}, C_{\mathrm{PS}}$, with the following property. For every finite instance on $[0, H]$ with $\mathbb{E}_P[X^*] \ge \mu_{\min}$, every target $\lambda \in [0,1]$, and every policy $\widehat{\pi}$ with true scalar regret r at $\rho = \max(a, \min(1-a, \lambda))$,*

$$\mathbb{E}_P[\lambda X_{\tau_{\widehat{\pi}}} + (1-\lambda)\tau_{\widehat{\pi}}] - C_{\mathrm{PS}}[\lambda \mathbb{E}_P[X^*] + 1 - \lambda] \ge \frac{1}{2} F_P(\lambda) - Kr. \tag{12}$$

Proof. Shrink the boundary-layer width a so that its uniform slack d_0 also satisfies $(H+1)a \le d_0/2$. Every policy's weighted payoff has slope $\mathbb{E}[X_\tau - \tau] \in [-1, H]$ in λ. The value is the supremum of these affine payoffs, so its secant slopes lie in the same interval. Thus the regret of a fixed policy changes by at most $(H+1)|\lambda - \rho|$ between the two weights, and

$$\mathbb{E}_P[\lambda X_{\tau_{\widehat{\pi}}} + (1-\lambda)\tau_{\widehat{\pi}}] - C_{\mathrm{PS}}[\lambda \mathbb{E}[X^*] + 1 - \lambda] \ge F_P(\lambda) - r - (H+1)|\lambda - \rho|.$$

For a boundary target, the last term is at most $d_0/2 \le F_P(\lambda)/2$. For an interior target it vanishes, and $F_P(\lambda) \ge 0$ by Lemma 3.1. This proves (12). □

The bound after projection holds for each training outcome, so Theorem 7.1 bounds the expected loss by $O(S(1+S)/k)$ using only a compact interval of positive coefficients of the stopping time.

Corollary 6.3. *Fix $\mu_{\min} > 0$. For instances on $[0,1]$ with $\mu_{\min} \le \mathbb{E}_P[X^*] \le 1/80$, there are $b \in (4/5, 1)$ and $K < \infty$, depending only on $\mu_{\min}, C_{\mathrm{PS}}$, such that for every target $\lambda \in [0,1]$, every policy π of true scalar regret r at $\rho = \max(4/5, \min(b, \lambda))$ satisfies*

$$\mathbb{E}_P[\lambda X_{\tau_\pi} + (1-\lambda)\tau_\pi] - C_{\mathrm{PS}}[\lambda \mathbb{E}_P[X^*] + 1 - \lambda] \ge F_P(\lambda)/40 - Kr.$$

The queried coefficients z lie in $[(1-b)/b, 1/4]$.

Proof. At the lower endpoint $a = 4/5$, write $T = \mathbb{E}[\tau_\pi]$. Waiting is feasible and $\mathbb{E}[X_{\tau_\pi}] \le \mathbb{E}[X^*]$, so $a\mathbb{E}[X_{\tau_\pi}] + (1-a)T \ge 1 - a - r$ implies $T \ge 1 - 4\mathbb{E}[X^*] - 5r \ge 19/20 - 5r$. For $\lambda \le 4/5$ and $C_{\mathrm{PS}} \le 3/4$,

$$\begin{aligned}\mathbb{E}_P[\lambda X_{\tau_\pi} + (1-\lambda)\tau_\pi] - C_{\mathrm{PS}}[\lambda \mathbb{E}[X^*] + 1 - \lambda] &\ge -\lambda C_{\mathrm{PS}} \mathbb{E}[X^*] + (1-\lambda)(19/20 - C_{\mathrm{PS}}) - 5r \\ &\ge 13/400 - 5r \ \ge F_P(\lambda)/40 - 5r.\end{aligned}$$

The last step uses $0 \le F_P(\lambda) \le 1$. The same boundary comparison as in Lemma 6.2, applied at the reward endpoint, gives an upper endpoint $b \in (4/5, 1)$. Interior targets use their own weight. The interval for z follows from $z = (1-\rho)/\rho$. □

The empirical policy can enforce cap acceptance at the same statistical rate.

Lemma 6.4. *For an instance on $[0,1]$ with $\mathbb{E}_P[X^*] \le 1/80$, let $Z \in [z_-, 1/4]$, where $z_- > 0$, be chosen from k independent samples per distribution and auxiliary information independent of those samples, with all this information independent of the online values and arrival times conditional on the instance distribution. The empirical optimizer for $X_\tau + Z\tau$ can be altered to accept every reached value 1 and reject zero for every*

training outcome, with an $O(1/k)$ loss in expected scalar payoff. The altered thresholds remain monotone under deletion of unseen variables.

Proof. For any square-integrable $f = f(V_1, \dots, V_n)$ and $A \subseteq \{1, \dots, n\}$, define

$$f_A = \sum_{B \subseteq A} (-1)^{|A \setminus B|} \mathbb{E}[f \mid (V_i)_{i \in B}].$$

Expanding the finite sum gives $f = \sum_A f_A$. Each nonconstant component is centered in every coordinate on which it depends, so distinct components are orthogonal. For $f = \max_i V_i$, its expectation under the product empirical distribution $\widehat{P}$ from k observations per variable is unbiased and satisfies

$$\mathrm{Var}(\mathbb{E}_{\widehat{P}}[X^*]) = \sum_{\varnothing \neq A} k^{-|A|} \sigma_A^2 \leq \frac{\mathrm{Var}(\max_i V_i)}{k} \leq \frac{\mathbb{E}[X^*]}{k},$$

where $\sigma_A^2 = \mathbb{E}[f_A^2]$ for $A \neq \varnothing$. Coordinatewise centering makes empirical averaging divide each variance by $k^{|A|}$, while preserving orthogonality.

On $\mathbb{E}_{\widehat{P}}[X^*] \leq 1/4$, every empirical continuation is at most $\mathbb{E}_{\widehat{P}}[X^*] + Z \leq 1/2$, so caps are accepted. Otherwise accept the first cap and wait to the end if none appears. This rule rejects zero and has thresholds monotone under variable deletion. Chebyshev gives $\Pr(\mathbb{E}_{\widehat{P}}[X^*] > 1/4) \leq \mathbb{E}[X^*]/[k(1/4 - \mathbb{E}[X^*])^2] = O(1/k)$; bounded payoffs give the same expected loss, even for a sample-selected scalar. The outer value rounding leaves the cap atom unchanged. □

7 Empirical scalar optimization with a selected weight

A threshold estimate incurs regret only when it changes an acceptance decision. Averaging over the random arrival time makes the resulting loss quadratic in the threshold error and gives the inverse-linear sample bound below. The bound is uniform in the scalar weight, so the weight may be chosen using the same observations as the empirical optimizer. We use the continuation values and parametrization $X_\tau + z\tau$ from Preliminaries.

Theorem 7.1. *Fix $0 < z_- < z_+ < \infty$ and a finite index set A with independent distributions P_i on $[0, 1]$. Draw $k \geq 1$ independent training samples per distribution, independently across variables and of the online values and arrival times. Let $\widehat{P}_i$ be their empirical distributions, let $\widehat{\pi}_z$ be the optimal empirical threshold policy of $X_\tau + z\tau$ for $z \in [z_-, z_+]$, and put $S = \sum_{i \in A} \mathbb{E}_P[V_i]$. There is $K < \infty$, depending only on z_-, z_+, for which the following holds. Let $Z \in [z_-, z_+]$ be measurable with respect to the entire training sample and any auxiliary randomness or further sample blocks independent of that sample conditional on P. All this information is independent of the online values and arrival times conditional on P. Then*

$$\mathbb{E}\left[v_A^P(1; 1, Z) - X_{\tau_{\widehat{\pi}_Z}} - Z\tau_{\widehat{\pi}_Z} \right] \leq \frac{KS(1+S)}{k}. \tag{13}$$

For each fixed $z \in [z_-, z_+]$,

$$0 \leq \mathbb{E}[v_A^{\widehat{P}}(1; 1, z)] - v_A^P(1; 1, z) \leq \frac{KS(1+S)}{k}. \tag{14}$$

When a distribution is known, use it in place of its empirical distribution; that variable then contributes zero estimation error.

Fix a candidate value x and the other arrival times. Where the true threshold decreases at speed at least $\kappa > 0$ and its estimation error is at most δ, the acceptance decisions can differ only when the true threshold is within δ of x. These times have total length $O(\delta/\kappa)$, and each disagreement costs at most δ. Averaging over the candidate's uniform arrival time gives a local loss $O(\delta^2/\kappa)$; see Figure 3. At a fixed positive slope, errors of scale $k^{-1/2}$ thus suggest regret of order k^{-1}.

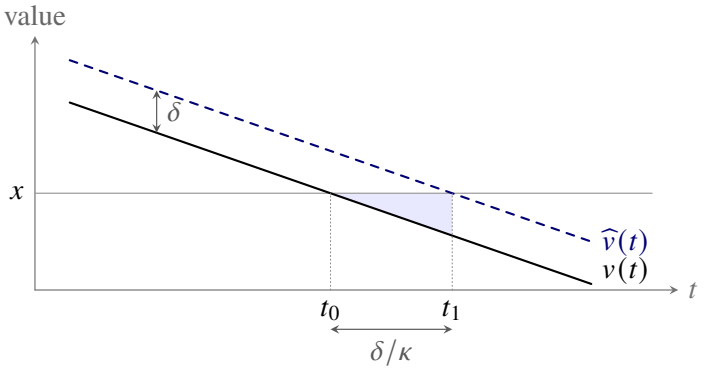


Figure 3: For a local linear threshold with slope magnitude κ, a shift δ creates a disagreement interval δ/κ and integrated Bellman loss $\delta^2/(2\kappa)$.

Proof. We reveal samples in reverse arrival order so that the local estimation errors are martingale differences. We bound second and sixth moments of the maximal partial sums in terms of the true continuation value, then make these bounds uniform over weights and later states. This uniformity permits the weight to depend on the samples and the estimates to be used after earlier arrivals. The second moments bound losses near a threshold, while the sixth moments bound the contribution of larger errors. Integrating over each candidate's arrival time proves (13). The final subsection applies the same estimates to the empirical regret of a fixed policy optimal for the true distribution and proves (14).

7.1 Continuation values and a martingale

For a policy in a state with unseen variables A and horizon $u > 0$, put $T = \mathbb{E}[\tau]/u$ and $s = \sum_{i\in A} \mathbb{E}_P[V_i]$. Rescaling Lemma 3.2 to this horizon gives $q_i \le 2T$. Reaching variable i is independent of its value, so

$$q_i \le 2T, \qquad \mathbb{E}[X_\tau] \le 2Ts.$$

Scaling arrival times to $[0,1]$ fixes the policy class, so $v_A^P(u;1,z)$ is convex and Lipschitz in u. Writing $v = v_A^P(u;1,z)$, the envelope identity gives, almost everywhere,

$$\partial_u v = zT, \qquad v \le T(2s+zu), \qquad \partial_u v \ge \frac{zv}{2s+zu}. \tag{15}$$

We also have $v \ge zu$; deleting variables or shortening the horizon cannot increase v.

Let $N = |A|$. Fix all arrival times and label the variables $1,\dots,N$ in arrival order. Let A_i be the variables remaining after i arrives and u_i the remaining horizon. Put

$$\widehat{v}_i(z) = v_{A_i}^{\widehat{P}}(u_i;1,z), \quad f_i(x;z) = (x-\widehat{v}_i(z))_+, \quad \xi_i(z) = (\widehat{P}_i - P_i) f_i(\cdot;z).$$

For each fixed z, reveal the sample blocks for the variables in reverse arrival order. The threshold for variable i depends only on later variables, so it is fixed before variable i's samples are revealed. Their centered contribution ξ_i is therefore a martingale difference. Revealing variable i's observations $Y_{i1},\dots,Y_{ik}$ successively refines them to

$$d_{ij}(z) = \frac{f_i(Y_{ij};z) - P_i f_i(\cdot;z)}{k}, \qquad |d_{ij}(z)| \le 1/k.$$

At the end of each sample block, the refined martingale equals a suffix sum over variables.

For fixed training data and any policy π evaluated under the true distribution, the Bellman residual identity at state (A, u) and scalar z, is

$$v_A^{\widehat{P}}(u;1,z) - \mathbb{E}_P[X_{\tau_\pi} + z\tau_\pi] = \mathbb{E}_{\text{clocks}}\left[\sum_i r_i^\pi \xi_i\right] + \mathbb{E}_P\left[\sum_{\text{reached } i} \ell_i^{\widehat{P}}(\pi)\right]. \tag{16}$$

Here r_i^π is the reach probability conditional on arrival times and training, and $\ell_i^{\widehat{P}}(\pi) \geq 0$ is the action loss relative to $\max(x, \widehat{v}_i)$, zero for $\widehat{\pi}_z$. Apply the arrival-process generator up to stopping to the empirical Bellman equation

$$u\partial_u v_A^{\widehat{P}}(u;1,z) = zu + \sum_{i\in A}\{\widehat{P}_i \max(x, v_{A\setminus\{i\}}^{\widehat{P}}(u;1,z)) - v_A^{\widehat{P}}(u;1,z)\}.$$

Replacing empirical transitions by true ones produces $(\widehat{P}_i - P_i)(x - \widehat{v}_i)_+$; the arrival-process compensator converts its hazard integral to the arrival sum. This proves (16) for history-dependent randomized policies as well.

For fixed arrival times, $1 \geq r_1^\pi \geq \cdots \geq r_N^\pi \geq 0$. Set $\Xi_j(z) = \sum_{i\geq j} \xi_i(z)$ and $\Xi_*(z) = \max_j |\Xi_j(z)|$. Summation by parts gives $\left|\sum_i r_i^\pi \xi_i\right| \leq 2\Xi_*(z)$. Applying (16) to the true and empirical optimizers, with their nonnegative action losses, gives the following comparison for every training outcome

$$|v_A^{\widehat{P}}(u;1,z) - v_A^{P}(u;1,z)| \leq 2\mathbb{E}_{\text{clocks}}[\Xi_*(z)]. \tag{17}$$

7.2 Localized second and sixth moments

Set $\widehat{U} = \sum_i \widehat{P}_i f_i$ and $U = \sum_i P_i f_i$, so $\Xi_1 = \widehat{U} - U$. Following all arrival times without stopping, the Bellman generator telescopes to

$$\mathbb{E}_{\text{clocks}}[\widehat{U}] = \widehat{v} - zu, \qquad \widehat{v} = v_A^{\widehat{P}}(u;1,z). \tag{18}$$

Each summand $a_i = \widehat{P}_i f_i$ is the empirical expected excess above the continuation threshold. By (18), the conditional expectation of their remaining sum is at most the current value $\widehat{v}$. Also $a_i \leq \widehat{P}_i x \leq \widehat{v}$, since waiting for variable i is feasible. For the subsequent sums $B_i = \sum_{j>i} a_j$, expand $\widehat{U}^2 = \sum_i (a_i^2 + 2a_i B_i)$ and $\widehat{U}^3 = \sum_i (a_i^3 + 3a_i^2 B_i + 3a_i B_i^2)$ to obtain

$$\mathbb{E}_{\text{clocks}}[\widehat{U}^2] \leq 3\widehat{v}^2, \qquad \mathbb{E}_{\text{clocks}}[\widehat{U}^3] \leq 13\widehat{v}^3. \tag{19}$$

The cubic terms contribute $1, 3, 9$, using the conditional future moment bounds $\widehat{v}$ and $3\widehat{v}^2$.

For martingale differences d_j adapted to a filtration $(\mathcal{F}_j)$, let $Q = \sum_j \mathbb{E}[d_j^2 \mid \mathcal{F}_{j-1}]$ be their predictable quadratic variation. Doob's maximal inequality [13, Section 4.4] and the sixth-moment argument below give

$$\mathbb{E}\left[\max_m \left|\sum_{j\leq m} d_j\right|^2\right] \leq 4\mathbb{E}[Q], \quad \mathbb{E}\left[\max_m \left|\sum_{j\leq m} d_j\right|^6\right] \leq K\left(\mathbb{E}[Q^3] + \sum_j \mathbb{E}[|d_j|^6]\right).$$

For the sixth moment, put $S_m = \sum_{j\leq m} d_j$ and $S_* = \max_m |S_m|$. Conditional sixth-power expansion cancels the linear term; bounding each remaining monomial by $K(S_{j-1}^4 d_j^2 + |d_j|^6)$ gives $\mathbb{E}[|\sum_j d_j|^6] \leq K\mathbb{E}[S_*^4 Q] + K\sum_j \mathbb{E}[|d_j|^6]$. Doob, Hölder, and Young then absorb the factor $(\mathbb{E}[S_*^6])^{2/3}$. All moments are finite here.

For the sample-by-sample reverse martingale, $Q \leq U/k$, and $\sum_j \mathbb{E}[|d_j|^6 \mid \mathcal{F}_{j-1}] \leq U/k^5$. Hence

$$\mathbb{E}[\Xi_*^2] \leq K\,\mathbb{E}[U]/k, \qquad \mathbb{E}[\Xi_*^6] \leq K(k^{-3}\mathbb{E}[U^3] + k^{-5}\mathbb{E}[U]). \tag{20}$$

All expectations now include arrival times and training. With $v = v_A^P(u;1,z)$, (17) and (18) give a bound in terms of the true value:

$$\mathbb{E}[\Xi_*^2] \le (K/k)\{v + 2(\mathbb{E}[\Xi_*^2])^{1/2}\} \le K(v/k + k^{-2}).$$

For $H_6 = \mathbb{E}[\Xi_*^6]$, use $U = \widehat{U} - \Xi_1$, (19), and Jensen in (17) to obtain

$$\mathbb{E}[U^3] \le K(\mathbb{E}[\widehat{v}^3] + \sqrt{H_6}), \qquad \mathbb{E}[\widehat{v}^3] \le K(v^3 + \sqrt{H_6}).$$

Substitution in (20), absorption of $Kk^{-3}\sqrt{H_6}$, and $v/k^5 \le K(v^3/k^3 + k^{-6})$, give

$$\mathbb{E}[\Xi_*^p] \le C_p\{(v/k)^{p/2} + k^{-p}\}, \qquad p \in \{2,6\}. \tag{21}$$

7.3 Uniform weights and future suffixes

The coefficient may be chosen after seeing the samples, and earlier decisions lead to later continuation states. We therefore make the bounds uniform over coefficients and over the later states along an arrival path.

For $z < z'$, put $\delta = z' - z$ and

$$b_i(x) = \frac{(x - \widehat{v}_i(z))_+ - (x - \widehat{v}_i(z'))_+}{\delta}.$$

The value is u_i-Lipschitz in z, so $0 \le b_i \le u_i \le u$. At any remaining state (A,u), subtracting (18) at the two scalars shows that the conditional mean of its empirical future sum is

$$u - \frac{v_A^{\widehat{P}}(u;1,z') - v_A^{\widehat{P}}(u;1,z)}{\delta} \in [0,u].$$

The preceding moment expansion therefore gives third moment at most $13u^3$. Write $U_b = \sum_i P_i b_i$. The centered reverse sum has jumps at most u/k and quadratic variation at most uU_b/k. The same absorption argument, now with a deterministic bound u for its expected empirical sum, gives

$$\left\| \max_j |\Xi_j(z') - \Xi_j(z)| \right\|_p \le C_p \delta u/\sqrt{k}, \qquad p = 2,6. \tag{22}$$

We pass from the increment estimate to all coefficients by successive dyadic refinements. Dyadic chaining has 2^j increments of length $(z_+ - z_-)2^{-j}$ at level j. Their L^p maximum is bounded by the ℓ^p sum of their norms, giving the convergent series $\sum_j 2^{-j(1-1/p)}$. Continuity extends the bound to all weights.

For any $a > 0$, if $v_A^P(u;1,z_+) \le a$, then $u \le \min(1, a/z_+)$. Thus $u^p \le C_p a^{p/2}$, and (21) together with (22) bounds the residual envelope over the complete path and all weights by $C_p\{(a/k)^{p/2} + k^{-p}\}$ in p-th moment.

At any later state, the reverse partial sums for its future suffix form a subset of those for the complete path. With training held fixed, (17) therefore bounds its value error by twice the conditional expectation of the complete-path envelope. Doob's inequality for this nonnegative martingale in the arrival-time filtration gives a random variable E_a such that

$$\sup_{\text{later states}} \sup_{z \in [z_-, z_+]} |v_A^{\widehat{P}}(u;1,z) - v_A^P(u;1,z)| \le E_a, \qquad \mathbb{E}[E_a^p \mid \text{initial true state}] \le C_p\{(a/k)^{p/2} + k^{-p}\}. \tag{23}$$

We apply this bound at entrances determined by true distributions and arrival times. Conditional on the entrance state, the remaining arrival times are uniform and the remaining sample blocks are still independent, with their original distributions. We can therefore apply the localized moment bounds conditional on that state.

7.4 Integrating the losses

Fix a candidate variable and temporarily exclude it while following the other arrival times without stopping. Its independent uniform arrival time is the variable of integration; the other variables determine the continuation threshold it faces. Let A_t be their unseen variable set, $u_t = 1 - t$, and $s_t = \sum_{j \in A_t} \mathbb{E}_P[V_j]$. Let σ_a be the first time with $v^P_{A_t}(u_t; 1, z_+) \le a$, and put $B_a = s_{\sigma_a} + u_{\sigma_a}$. The process s_t/u_t is a nonnegative martingale before time 1. Apply optional stopping [13, Section 4.8] on time grids bounded away from 1, then use right continuity and Fatou's lemma. Together with $u_{\sigma_a} \le a/z_+$, this gives

$$\mathbb{E}[B_a] \le a(1 + S)/z_+. \tag{24}$$

The sum of the remaining means is zero at time 1. The entrance uses true distributions and arrival times, so B_a is measurable there and can multiply the conditional bounds in (23). Both the remaining first moment and the remaining time are therefore small on average when the continuation value first becomes small.

Fix a candidate value x and its path of the other arrivals. For z chosen from the training data, put $v_z(t) = v^P_{A_t}(1 - t; 1, z)$ and $\widehat{v}_z(t) = v^{\widehat{P}}_{A_t}(1 - t; 1, z)$. The true value curve decreases, with downward jumps when other variables arrive. On an action disagreement, the empirical rule incurs true Bellman loss $|x - v_z(t)|$, while the true rule incurs empirical Bellman loss $|x - \widehat{v}_z(t)|$; both are at most $|v_z(t) - \widehat{v}_z(t)|$. After entering the region $v_{z_+} \le a$, either loss is therefore bounded by

$$E_a \mathbf{1}_{\{|x - v_z(t)| \le E_a\}}.$$

The scalar stays fixed during integration over arrival times, with (23) uniform over its choice. Reach probabilities are at most one. Thus the corresponding local losses bound both the true regret of the empirical rule and the empirical regret of the true rule. Near the acceptance threshold, a lower bound on the magnitude of the time derivative bounds the length of the disagreement interval. Farther away, disagreement requires a large error, whose contribution is bounded by the sixth moment.

If $x \le 1/k$, both rules reject whenever $u > x/z_-$. On the remaining interval the two local losses are at most $x + 2s_t + z_+u$ and $x + 2\widehat{s}_t + z_+u$, respectively, where $\widehat{s}_t = \sum_{j \in A_t} \widehat{P}_j x$. Both sums of remaining means have expectation at most uS, so either expected integral over arrival times is at most

$$K(1 + S)x^2 \le K(1 + S)x/k.$$

Suppose now $x \ge 1/k$, set $a = 4z_+x/z_-$, and enter the region $v_{z_+} \le a$. While $v_z \ge x/2$, (15), with the remaining mean and horizon bounded by their entrance values, bounds the reciprocal of the slope magnitude by KB_a/x. Downward jumps skip value intervals and cannot enlarge an integral. Consequently

$$\int E_a \mathbf{1}_{\{|x - v_z(t)| \le E_a,\ v_z(t) \ge x/2\}}\, dt \le KB_a E_a^2/x.$$

Conditional second moments and (24) bound its expectation by $K(1 + S)x/k$. When $v_z < x/2$, the available arrival time interval has length at most $x/(2z_-)$, and disagreement requires $E_a \ge x/2$. Sixth moments give

$$Kx\, \mathbb{E}[E_a \mathbf{1}_{\{E_a \ge x/2\}}] \le K\{(xk^3)^{-1} + (x^4k^6)^{-1}\} \le Kx/k.$$

Consider the earlier dyadic bands $v_{z_+} \in [a', 2a']$, where $a' \ge 4z_+x/z_-$. Since $v_{z_+} \le (z_+/z_-)v_z$, the true threshold exceeds x by at least a constant times a', and the inverse slope is at most $KB_{2a'}/a'$. For a constant $c_0 > 0$ depending only on z_-, z_+, the integral over arrival times is at most

$$\frac{KB_{2a'}}{a'} E^2_{2a'} \mathbf{1}_{\{E_{2a'} \ge c_0 a'\}}.$$

Conditional sixth moments followed by (24) bound its expectation by

$$K(1+S)\{(a'k^3)^{-1} + ((a')^4k^6)^{-1}\}.$$

Summing over $a' = 2^j a$, $j \geq 0$, gives at most $K(1+S)x/k$. Integrating the bounds for all regions against P_i and summing over candidates proves (13). The error envelopes depend only on the other variables' training data. Their uniformity in the scalar therefore permits the candidate's data to influence Z without changing the bound used to integrate its value.

Fix z independently of the training sample and let π_P be a true optimal policy. This policy is then fixed when the empirical distributions are drawn, so evaluating it under their product is unbiased: $\mathbb{E}[\mathbb{E}_{\widehat{P}}[X_{\tau_{\pi_P}} + z\tau_{\pi_P}]] = \mathbb{E}_P[X_{\tau_{\pi_P}} + z\tau_{\pi_P}] = V(P; 1, z)$. This identifies expected upward bias with the expected empirical regret of π_P, which is nonnegative.

Bound empirical reach probabilities by one in that regret identity, and apply the preceding bounds from integration over arrival times to the empirical action loss. Each bound depends only on the candidate value and the other variables' training data. Averaging over the candidate's samples therefore replaces its empirical distribution by P_i; integrating and summing proves (14). □

8 Curvature under perturbations of the marginal distributions

Here the weights of reward and stopping time stay fixed while the marginal distributions change. This affects both a policy's payoff and which policy is optimal. For a fixed policy, differentiating its payoff twice along a product likelihood tilt gives a score identity that controls the second derivative. The additional curvature comes from moving acceptance thresholds. We bound this contribution by the total first moment, uniformly in the number of variables and the atom probabilities.

Theorem 8.1. *Fix $H < \infty$ and compact positive ranges for w, c. Let $P = \bigotimes_{i=1}^n P_i$, where every P_i is supported on $[0, H]$. Let the likelihood scores h_i for the marginal distributions satisfy $P_i h_i = 0$ and $|h_i| \leq 1$. Define $P^t = \bigotimes_i P_i^t$, where*

$$P_i^t(dx) = (1 + t h_i(x)) P_i(dx), \quad |t| \leq \tfrac{1}{4}, \qquad S = \sum_i \mathbb{E}_P[V_i], \qquad I_h = \sum_i P_i h_i^2.$$

There is $K < \infty$, depending only on H and the coefficient ranges, such that $t \mapsto V(P^t; w, c)$ is $C^{1,1}$ and satisfies the integrated almost-everywhere bound

$$\left| \frac{d^2}{dt^2} V(P^t; w, c) \right| \leq K(1+S) I_h, \qquad |t| \leq \tfrac{1}{4}. \tag{25}$$

For any fixed policy π, the second derivatives of $\mathbb{E}_{P^t}[wX_{\tau_\pi} + c\tau_\pi]$ and $\mathbb{E}_{P^t}[X^*]$ are bounded in absolute value by KI_h, uniformly in the number of variables, atoms and their probabilities.

Proof. We first work with finite support. A threshold crossing contributes a positive semidefinite matrix whose trace can be charged to the candidate's value. Summing these charges bounds the curvature from policy changes; the product-score identity bounds the fixed-policy term, and rounding extends the result to general distributions.

8.1 Envelope gradients and their squared norm

For finite support, use orthonormal centered $L^2(P_i)$ coordinates in each marginal distribution, omitting zero-probability atoms. The Fisher norm is the $L^2(P)$ norm of the product likelihood score; its square is the sum of the marginal score variances. In these coordinates it coincides with the Euclidean norm.

In a continuation state with unseen variables A and horizon $u > 0$, let τ be its optimizer's stopping time, measured from the state's start, and put $T = \mathbb{E}[\tau]/u$. Let q_i be its reach probabilities. If variable i arrives and is rejected, write $v_{\text{child}(i)}$ for the optimal additional value of the resulting continuation state. Up to a constant annihilated by centered scores, the component of the value gradient corresponding to changes in variable i's distribution is

$$g_i(x) = \mathbb{E}\left[\mathbf{1}_{\{\text{reach } i\}}(wx - v_{\text{child}(i)})_+\right]. \tag{26}$$

Reach is independent of the candidate value. Subtracting the x-independent continuation value from $\max(wx, v_{\text{child}(i)})$ proves (26); the envelope theorem applies because threshold ties are null, as justified below.

By Lemma 3.2, $0 \le g_i \le wHq_i \le 2wHT$. The positive part $(wx - v_{\text{child}(i)})_+$ of the acceptance gap is nonzero at a reached variable only if that variable is selected. At most one variable is selected, and its gap is at most w times its value. Thus

$$\sum_i P_i g_i \le w\mathbb{E}[X_\tau], \qquad \sum_i \mathrm{Var}_{P_i}(g_i) \le \sum_i P_i g_i^2 \le 2w^2 H T \mathbb{E}[X_\tau].$$

Identifying the variance sum with the squared gradient norm gives

$$\|\nabla v_A^P(u; w, c)\|^2_{\text{Fisher}} \le 2w^2 H T \mathbb{E}[X_\tau]. \tag{27}$$

8.2 The adaptation Hessian and its trace

At points of twice differentiability, write H_V for the Hessian of the optimal value and subtract the Hessian obtained by holding the base-optimal policy fixed. We call the remainder H_{adapt} the *adaptation Hessian*; it measures the effect of changing the optimal policy.

In a compact neighborhood of a finite instance's positive atom probabilities, total variation is Lipschitz in the coordinates. Bounded payoffs make every continuation locally Lipschitz as well. Zero-rejecting policies have normalized duration at least $1/[4(n+1)]$, giving a derivative of the continuation value with respect to the remaining horizon of at least $c/[4(n+1)]$.

After a coordinate displacement of norm h, optimal actions can differ only within $K|h|$ of the old acceptance boundary. Conditional on other arrival times, a candidate's continuation decreases at the preceding rate and jumps only downward. The disagreement interval therefore has length $O(|h|)$, with new action loss $O(|h|)$. Bounding reach probabilities by one and summing candidates gives adaptation regret $O(h^2)$.

Fixed-policy second derivatives are bounded in this neighborhood. Applying the regret estimate in both directions gives local $C^{1,1}$ regularity in every continuation state, with finite-instance constants.

Hold variable i pending with candidate value x, and abbreviate its continuation value as v_{child}. Let τ_{child} be the child-state optimizer's stopping time, measured from that state's start, and put $T_{\text{child}} = \mathbb{E}[\tau_{\text{child}}]/u$. At a continuous interior crossing $v_{\text{child}} = wx$, the acceptance gap has time derivative cT_{child} and gradient with respect to the perturbation coordinates $-\nabla v_{\text{child}}$. Its adaptation Hessian contribution is

$$\mathbf{1}_{\{\text{continuous crossing}\}}\, B_i\, \frac{\nabla v_{\text{child}} \nabla v_{\text{child}}^{\mathsf{T}}}{cT_{\text{child}}}, \tag{28}$$

averaged over other arrival times and values and integrated against $P_i(dx)$; B_i indicates survival with variable i pending.

Under a perturbation ha in a fixed coordinate direction a, the fixed base-optimal policy's Bellman regret has gap displacement $-h\, a \cdot \nabla v_{\text{child}} + o(h)$. Integrating over the uniform candidate arrival time gives the triangular loss

$$\frac{h^2}{2} B_i \frac{(a \cdot \nabla v_{\text{child}})^2}{cT_{\text{child}}} + o(h^2).$$

Between other arrivals the old reach indicator is constant. Replacing perturbed densities by original ones costs $O(h^3)$; the finite-instance lower bound on the slope magnitude bounds the quotients and permits dominated convergence.

Crossings that jump strictly past zero contribute nothing. Equality at another continuously distributed arrival has probability zero, by strict monotonicity in arrival time. A root at initial time can occur on a set of distribution parameters of positive measure only with zero derivative of its gap there; the squared directional numerator then vanishes almost everywhere on that set. Local $C^{1,1}$ regularity excludes any additional singular Hessian.

The full Hessian is the sum of the fixed-policy Hessian and the positive semidefinite adaptation Hessian. A fixed policy's payoff is affine in each marginal distribution, so its Hessian has zero diagonal blocks. The full and adaptation traces therefore coincide. We use this trace to bound the adaptation term in every direction and bound the fixed-policy term separately below.

The duration factor in the gradient estimate cancels the duration factor in the derivative with respect to the remaining horizon. At a crossing $v_{\text{child}} = wx$, we have $w\mathbb{E}[X_{\tau_{\text{child}}}] + cuT_{\text{child}} = wx$, hence $\mathbb{E}[X_{\tau_{\text{child}}}] \le x$. (27) and (28) therefore bound the trace contribution by

$$B_i \frac{2w^2 H T_{\text{child}} \mathbb{E}[X_{\tau_{\text{child}}}]}{cT_{\text{child}}} \le \frac{2w^2 H x}{c}.$$

For fixed other arrival times and fixed x, the decreasing continuation curve has at most one continuous crossing. Bounding reach probabilities by one and integrating proves

$$0 \le \operatorname{tr} H_{\text{adapt}} = \operatorname{tr} H_V \le \frac{2w^2 H}{c} \sum_i P_i x = \frac{2w^2 HS}{c}. \tag{29}$$

Positive semidefiniteness of H_{adapt} gives $\|H_{\text{adapt}}\|_{\text{op}} \le 2w^2 HS/c$.

8.3 Completing the curvature bound

For any fixed bounded payoff U and $\ell = \sum_i h_i(V_i)$, the product tilt gives

$$\frac{d^2}{dt^2} \mathbb{E}_{P^t}[U]\Big|_{t=0} = \mathbb{E}_P \left[U \left(\ell^2 - \sum_i h_i(V_i)^2 \right) \right].$$

Independence and centering give $\mathbb{E}[\ell^2] = I_h$, so its absolute value is at most $2\|U\|_\infty I_h$. At an ambient point of twice differentiability where (29) holds, the fixed-function bound and the trace estimate give the directional bound

$$|V''(0)| \le \left\{ 2(wH + c) + \frac{2w^2 HS}{c} \right\} I_h.$$

At any $|t| \le 1/4$, the score $h_i/(1 + th_i)$ is centered under P_i^t, its total squared norm is at most $4I_h/3$, and the total first moment $S_t = S(P^t)$ satisfies $S_t \le 5S/4$. Thus the right side of the directional bound, evaluated along the line, is at most $K(1 + S)I_h$ throughout the interval.

To obtain the bound on the prescribed line, convolve the locally $C^{1,1}$ value in the positive atom coordinates. The smoothed second derivative along the line averages the ambient directional Hessians in the same fixed direction. The Fisher norm of this direction and the total first moment vary continuously with the atom probabilities. As the convolution radius shrinks, the nearby bounds converge uniformly to the bounds on the compact line segment. Passing the integrated second-difference bounds to the limit gives (25) almost everywhere along the line. Taking $U = wX_{\tau_\pi} + c\tau_\pi$ or $U = \max_i V_i$ gives the other bounds.

Round each value downward on a mesh of width $\delta > 0$, and replace each h_i by its conditional mean in its rounding cell. The resulting finite-support tilted family is exactly the distribution of the rounded values

under P^t; the scores are still centered, $|h_i| \le 1$, and the Fisher norm and total mean do not increase. Write $V_\delta(P^t; w, c)$ for the optimal value of the rounded instance. Coupling rounded and original observations gives uniformly in $|t| \le 1/4$

$$|\mathbb{E}_{P^t}[\delta\lfloor X^*/\delta\rfloor] - \mathbb{E}_{P^t}[X^*]| \le \delta, \qquad |V_\delta(P^t; w, c) - V(P^t; w, c)| \le w\delta.$$

To run an original policy using only rounded observations, simulate an original value from its conditional distribution within the observed cell, and feed these simulated values to the policy. The selected reward changes by at most δ and the stopping time is unchanged.

The uniform curvature constants let symmetric second-difference inequalities pass to the limit, proving $C^{1,1}$ regularity and (25). The product-score identity gives the fixed-policy and maximum bounds. □

9 A uniform first-moment bound near scalar criticality

The empirical regret bound in Theorem 7.1 grows with the total first moment S. Bounded values and a bounded expected maximum do not control S: if $V_i \equiv 1$ for all i, then $\mathbb{E}[X^*] = 1$ but $S = n$. The next result shows that S is uniformly bounded under a quantitative small-gap condition.

Theorem 9.1. *Fix $0 < \delta \le 1/2$. There are $\eta > 0$ and $B < \infty$, depending only on C_{PS}, δ, such that every finite independent instance on $[0, 1]$ with $w + c = 1$, $w, c \ge \delta$ and $\ell = 1 + L(P)$ satisfies $S(P) \le B$ whenever*

$$F_P(w) \le \eta \ell^{-4}. \tag{30}$$

No positive lower bound on $\mathbb{E}_P[X^*]$ is required.

Proof. Write $F = F_P(w)$ and let π be the optimal threshold policy. Perturbing the values and using nonnegativity of the scalar gap lets us compare the optimizer's selection probabilities with C_{PS} times the prophet's maximum probabilities. Near criticality these probabilities must be close. We lower-bound the probability that each positive value arrives late enough to be selected. A value below many competing values is unlikely to be the maximum; combining these estimates bounds the aggregate tail.

Perturbations and selection probabilities. Let $0 < h \le 1/4$, let $|\phi_i| \le 1$, and assume $\phi_i(x) = 0$ for $x < 2h$. Let P' be the product distribution of $V_i' = V_i + h\phi_i(V_i)$, and define

$$A(\phi) = \mathbb{E}\left[\sum_i \mathbf{1}_{\{\pi \text{ selects } i\}} \phi_i(V_i)\right].$$

With $A_0 = 32/\delta$, this value perturbation satisfies

$$0 \le V(P'; w, c) - V(P; w, c) - whA(\phi) \le A_0 \ell^2 h^2. \tag{31}$$

Couple each variable with its perturbed value, so $|V_i - V_i'| \le h$. Using the conditional distributions in this coupling, either policy can be simulated under the other distribution without seeing future observations. This leaves duration unchanged and changes reward by at most h, even when the perturbation map is not injective. Thus continuation values differ by at most wh, and simulating π under P' proves the lower bound.

For the upper bound, put $p_j = \Pr(V_j > 0)$. A zero-rejecting policy waits at least until the first positive arrival, so its normalized duration is at least $1/(4\ell)$: indeed, $\prod_j (1 - sp_j) \ge 1 - s\sum_j p_j \ge 1/2$ for $s \le 1/(2\ell)$. For a remaining horizon $u > 0$, the optimal continuation value is a supremum of affine functions $w\mathbb{E}[X_\tau] + cuT$, where $T = \mathbb{E}[\tau]/u \ge 1/(4\ell)$ and time is measured from the state's start. It therefore increases with horizon at slope at least $c/(4\ell)$.

Fix a candidate value x and all other arrival times, with A_t the other unseen variables. Its continuation $v_{A_t}(1 - t; w, c)$ decreases at this rate and jumps only downward. Actions can differ only where $|wx - v_{A_t}(1 -$

$t; w, c)| \le 2wh$, an interval of total length at most $16wh\ell/c$ with Bellman loss at most $2wh$. Both policies reject zero. Bounding reach probabilities by one and summing positive candidates bounds the Bellman regret by $32w^2L\ell h^2/c \le A_0\ell^2h^2$, proving (31).

Fix a measurable tie-break for the original maximum, with winning variable W, and set $D(\phi) = \mathbb{E}[\phi_W(V_W)]$, taking the integrand zero at a zero maximum. Pointwise the new maximum is at least the transported old winner, whence $\mathbb{E}_{P'}[X^*] \ge \mathbb{E}_P[X^*] + hD(\phi)$. Apply Lemma 3.1 to P' and use the upper bound in (31); repeat with $-\phi$. It follows that

$$|A(\phi) - C_{\mathrm{PS}}D(\phi)| \le d_h, \qquad d_h = \frac{F}{wh} + \frac{A_0\ell^2 h}{w}. \tag{32}$$

Set $a_i(x) = \Pr(\pi \text{ selects } i \mid V_i = x)$ and $b_i(x) = \Pr(W = i \mid V_i = x)$. Testing (32) with the signs of $a_i - C_{\mathrm{PS}}b_i$ makes this comparison simultaneous across all variables:

$$\sum_i \int_{[2h,1]} |a_i(x) - C_{\mathrm{PS}}b_i(x)|\, dP_i(x) \le d_h. \tag{33}$$

Survival and positive selection probabilities. Apply (31) with $\phi_i(x) = \sigma_i \mathbf{1}_{\{x \ge 2h\}}$, where the variable signs σ_i are independent fair signs, and average over these signs. The linear scalar term has mean zero. On an outcome with at least two maximizers of value at least $2h$, the maximum of their signs has mean at least $1/2$. On any other outcome the expected prophet change is nonnegative, by evaluating the perturbed value of the original winner. Therefore

$$\Pr(\max_i V_i \ge 2h \text{ is tied}) \le \frac{2d_h}{C_{\mathrm{PS}}}.$$

An accepted value below $2h$ must arrive in the final window of length $2wh/c$, since waiting alone gives continuation cu. A union bound and (32) with the common indicator $[2h, 1]$ show

$$\Pr(\pi \text{ accepts}) \le C_{\mathrm{PS}} + d_h + \frac{2w}{c}Lh.$$

Put $E_0 = \max\{1, 2/C_{\mathrm{PS}}, 2/\delta\}$, $e_h = E_0(d_h + \ell h)$ and $\rho = (1 - C_{\mathrm{PS}})/2$. Then e_h bounds d_h, the tie probability and the acceptance excess over C_{PS}. If $e_h \le \rho$, the policy accepts nothing with probability at least ρ. Its survival probability at every time is therefore at least this large.

Set $\gamma = \min\{\rho/4, \delta/2\}$ and $\kappa = \gamma\rho/2$. When $e_h \le \rho$, every $0 < x \le 1$ satisfies

$$a_i(x) \ge \frac{\kappa x}{1 + S}. \tag{34}$$

Condition on i arriving at $t = 1 - u$. Keeping it pending until t only raises other acceptance thresholds, so its reach probability is at least ρ, independently of $V_i = x$. If s_t sums the other unseen means, continuation is at most $ws_t + cu$ and $\mathbb{E}[s_t] \le uS$. For $u \le \gamma x/(1 + S)$, we have $cu \le wx/2$ and

$$\Pr(s_t > x/2) \le \frac{2uS}{x} \le 2\gamma \le \rho/2.$$

The union bound gives reach and acceptance probability at least $\rho/2$. Integrating over this arrival time interval proves (34).

Tail counts and the first moment. Let $\Lambda = \sum_i P_i|_{(0,1]}$ and $m(x) = \Lambda(\{x\})$. For an atom $x \ge 2h$, (33) and (34) imply

$$\frac{\kappa x m(x)}{1 + S} \le C_{\mathrm{PS}} \Pr(\max_i V_i = x) + d_h. \tag{35}$$

If $m(x) \geq 2$, condition on all $V_i \leq x$. When that event has positive probability, the number N equal to x is a sum of independent Bernoulli variables with mean $r \geq m(x) \geq 2$. For $Y = (N-1)_+$,

$$\mathbb{E}[Y] \geq r - 1 \geq r/2, \qquad \mathbb{E}[Y^2] \leq \mathbb{E}[N^2] \leq r^2 + r \leq 3r^2/2.$$

Cauchy–Schwarz gives $\Pr(N \geq 2) \geq 1/6$. Hence

$$\Pr(\max_i V_i = x) \leq 6 \Pr(\text{maximum tied at } x) \leq 6e_h.$$

If the conditioning event has probability zero, the displayed probability bound holds automatically. (35) now gives

$$x \leq \frac{6C_{\mathrm{PS}} + 1}{2\kappa}(1 + S)e_h.$$

Choose $K = \max\{2, (6C_{\mathrm{PS}} + 1)/\kappa, 2/\kappa\}$ and $a_* = K(1 + S)e_h$. Then $a_* \geq 2h$, and every aggregate atom at or above a_* has mass less than two.

For $a \in [a_*, 1]$, put $K_0 = \max\{\mathrm{e}, 2C_{\mathrm{PS}}\mathrm{e}/\kappa\}$ and $B_a = \log[K_0(1 + S)/a]$. Independence gives

$$b_i(x) \leq \prod_{j \neq i} \Pr(V_j \leq x) \leq \mathrm{e} \exp[-L_x(P)].$$

At $x \geq a$ with $L_x(P) \geq B_a$, (34) therefore gives $a_i(x) - C_{\mathrm{PS}} b_i(x) \geq \kappa a/[2(1 + S)]$. By (33), the total Λ-mass of such values is at most

$$\frac{2(1 + S)d_h}{\kappa a} \leq \frac{2}{\kappa K} \leq 1.$$

The complementary set $A = \{x \geq a : L_x(P) < B_a\}$ is an upper interval, with $\Lambda([x, 1]) = L_x(P) + m(x) < B_a + 2$ for $x \in A$. Letting x decrease to its lower endpoint (using its minimum when present) gives $\Lambda(A) \leq B_a + 2$. Combining the two bounds gives

$$\sum_i \Pr(V_i \geq a) \leq \log \frac{K_0(1 + S)}{a} + 3.$$

Layer-cake integration yields

$$S \leq KL(1 + S)e_h + \log[K_0(1 + S)] + 4. \tag{36}$$

Indeed the integral below a_* is at most La_*, and $\int_0^1 \log(1/a)\, da = 1$. If $a_* > 1$, the first term alone is at least $L \geq S$, so (36) still holds.

Take $h = \sqrt{\eta}\, \ell^{-3}$. By (30) and (32), $e_h \leq K_1 \sqrt{\eta}/\ell$ for $K_1 = E_0((1 + A_0)/\delta + 1)$. Choose $\sqrt{\eta} \leq \min\{1/4, \rho/K_1, 1/(4KK_1)\}$. Then $h \leq 1/4$, $e_h \leq \rho$ and $KLe_h \leq 1/4$, so (36) gives

$$S \leq \frac{1 + S}{4} + \log[K_0(1 + S)] + 4.$$

Using $\log(1 + S) \leq S/4 + \log 4$ gives $S \leq B := 17/2 + 2\log(4K_0)$. The same positive gap threshold works when $F = 0$, and all constants depend only on C_{PS}, δ. □

10 A quadratic diameter bound for sublevel sets of the scalar gap

Convexity alone permits a nonnegative function to vanish on an interval. For the scalar gap, the next theorem shows that two weights with gaps at most $s \geq 0$ are within $O(\sqrt{s})$ of each other, uniformly over the stated family of instances. In Section 11, this proximity controls the correlation between a selected weight and its calibration error. We compare two weights by perturbing the distribution so that the expected duration of a fixed base-optimal policy changes, then apply the curvature bound from Theorem 8.1.

Theorem 10.1. *Fix $H, S_0, \mu_{\min} > 0$. For some $K = K(H, S_0, \mu_{\min}, C_{\mathrm{PS}}) < \infty$, every finite independent instance on $[0, H]$ with $S(P) \le S_0$ and $\mathbb{E}_P[X^*] \ge \mu_{\min}$ satisfies*

$$|\lambda - \nu|^2 \le K\{F_P(\lambda) + F_P(\nu)\}, \qquad \lambda, \nu \in [0, 1]. \tag{37}$$

At a critical limiting instance, Lemma 3.15 gives a perturbation with a nonzero duration derivative. To use it for the diameter bound, we must approximate this derivative using optimizers along the original finite sequence. The next lemma proves this convergence.

10.1 Scores on the original finite sequence

Lemma 10.2. *Let a sequence P_m on $[0, H]$, with total first moments at most S_0, converge as in Lemma 3.11 to a limiting instance with zero terminal reward. Write $P_m = \bigotimes_i P_{mi}$ and V_{mi} for its variable values. Suppose $\lambda_m \to \lambda_* \in (0, 1)$, and let π_m, π_* be the optimal threshold policies at normalized weights $(\lambda_m, 1 - \lambda_m)$ and $(\lambda_*, 1 - \lambda_*)$, respectively. If Z is either a centered continuous bounded score of one limiting individual variable, or the centered count in an upper positive continuity band of the limiting Poisson process, then there are likelihood scores h_{mi} with*

$$P_{mi} h_{mi} = 0, \qquad |h_{mi}| \le 1, \qquad \sup_m \sum_i P_{mi} h_{mi}^2 < \infty,$$

after a fixed scaling of Z if necessary, for which $Z_m = \sum_i h_{mi}(V_{mi})$ satisfies

$$\mathbb{E}[Z_m X_{\tau_{\pi_m}}] \longrightarrow \mathbb{E}[Z X_{\tau_{\pi_*}}], \tag{38}$$

$$\mathbb{E}[Z_m \tau_{\pi_m}] \longrightarrow \mathbb{E}[Z \tau_{\pi_*}]. \tag{39}$$

The distributions $P_{mi}^t = (1 + t h_{mi}) P_{mi}$ exist on a common fixed interval $|t| \le 1/4$, have uniformly bounded Fisher information $\sum_i P_{mi} h_{mi}^2$, and satisfy the uniform curvature bounds under distributional perturbations of Theorem 8.1.

Proof. The bound $L_b(P_m) \le S_0/b$ gives zero terminal reward and the tail counts required by outcome stability. Write Q_i for the distributions of the limiting individual variables and μ for the Poisson intensity measure.

Constructing the finite scores. For an individual variable, center its continuous score under P_{mi} and scale it by a fixed factor to absolute value at most one. Weak convergence makes the centering constants converge.

For Lemma 3.15, use $(x - \mathbb{E}_{P_{mi}}[V_{mi}])/H$: the nonincreasing function $\mathbb{E}[\tau_{\pi_*} \mid V_i = x]$ has negative covariance with x unless constant on the support, in which case every centered score has zero covariance.

For a Poisson process score, write its upper continuity band as $B = (b, H]$, with $b > 0$ a continuity level. Choose $N_m \uparrow \infty$ slowly so the residual variables $i > N_m$ converge to the Poisson process on every positive continuity band. Aggregate-measure convergence for each fixed initial set of individual variables and each cutoff, and summability of exceedance probabilities of the limiting individual variables, permit a diagonal choice over countably many cutoffs and continuous test functions. The extraction's variable ordering also gives

$$\max_{i > N_m} P_{mi}(V_{mi} > d) \longrightarrow 0 \quad \text{for every } d > 0.$$

Let $p_{mi} = P_{mi}(B)$ for $i > N_m$ and set

$$h_{mi}(x) = \mathbf{1}_B(x) - \frac{p_{mi}}{1 - p_{mi}} \mathbf{1}_{B^c}(x) = \frac{\mathbf{1}_B(x) - p_{mi}}{1 - p_{mi}}, \qquad i > N_m.$$

Use score zero for the individual variables outside this residual set. For sufficiently large m, all $p_{mi} \le 1/2$, so these scores are centered, bounded by one, and

$$\sum_i P_{mi} h_{mi}^2 = \sum_{i>N_m} \frac{p_{mi}}{1-p_{mi}} \le 2 \sum_{i>N_m} p_{mi} \le 2S_0/b.$$

Compensating on B^c makes the construction valid even when the distribution has no zero atom.

Tilting multiplies the mass in B by $1+t$. The compensating change outside B, above a fixed $d>0$, has total variation at most

$$|t| \max_{i>N_m} \frac{p_{mi}}{1-p_{mi}} \sum_{i>N_m} P_{mi}(V_{mi} > d) \longrightarrow 0.$$

Hence the tilted Poisson process limit changes only the B-intensity, by the factor $1+t$.

At $t=0$, if N_{mB} is the residual B-count, then

$$Z_m = N_{mB} - \sum_{i>N_m} p_{mi} + o_{L^2}(1) \implies N_B - \mu(B). \tag{40}$$

The error is a centered sum with variance $O(\sum_i p_{mi}^2) = o(1)$. Bernoulli factorial-moment bounds give uniform moments of every fixed order for positive-band counts.

Comparing policies through finite configurations. Fix an initial set of individual variables $1, \dots, K$ and continuity cutoff $d>0$. Run the optimal threshold policy for these variables and the Poisson process above d on P_m, using only the first K variables and values above d from variables $i > N_m$. Ignore intermediate variables. This uses only observed identities and values.

This observed configuration converges to the truncated instance. Conditional on other arrivals and a positive candidate value, its continuous arrival time hits the strictly monotone continuation threshold with probability zero. Distinct arrival times and null threshold ties give almost-sure continuity of reward and duration.

Truncate the positive count, couple configurations and apply bounded convergence. The uniform count moments and (40) let the count cutoff tend to infinity, giving convergence of the joint expectations of the score with reward and stopping time for each policy for the truncated instance. Continuous scores of individual variables satisfy the same argument.

Write $w = \lambda_*$ for the limiting reward weight. As $K \to \infty$ and $d \downarrow 0$, the limiting scalar regret of this policy tends to zero. The omitted low values cost at most wd, while the omitted individual variables contribute at most

$$w\,\mathbb{E}\left[\max_{i>K} V_i\right] \le w\left\{\eta + H \sum_{i>K} Q_i((\eta, H])\right\} \longrightarrow w\eta$$

for every fixed $\eta > 0$; then let $\eta \downarrow 0$.

Scalar value convergence and $\lambda_m \to \lambda_*$ give the same limiting regret bound on P_m. Corollary 3.9 compares this policy's score expectations to those of π_m. For π_*, Lemma 3.13 and bounded outcomes give the same comparison by Cauchy–Schwarz. Let $m \to \infty$ and then refine the policy for the truncated instance to obtain (38)–(39) along the original sequence.

Finally $|h_{mi}| \le 1$ guarantees that all these tilts are defined on one common two-sided interval, and $S(P_m^t) \le 5S_0/4$. Theorem 8.1 gives the uniform curvature bounds. □

Proof of Theorem 10.1. We evaluate a midpoint optimizer at both endpoints. Its regret is small at each endpoint, so curvature under distributional perturbations forces both score derivatives to be small. Eliminating their reward terms leaves the product of the weight separation and a duration response. A response bounded away from zero then bounds the separation.

10.2 A smooth-envelope inequality

Let $\lambda_1 < \lambda_2$ lie in a fixed compact interval $I \subset (0,1)$, put

$$s = F_P(\lambda_1) + F_P(\lambda_2), \qquad \bar{\lambda} = (\lambda_1 + \lambda_2)/2,$$

and fix an optimizer π at $\bar{\lambda}$. Its endpoint regrets are $r_j = V(P;\lambda_j, 1-\lambda_j) - \mathbb{E}_P[\lambda_j X_{\tau_\pi} + (1-\lambda_j)\tau_\pi]$. The fixed-policy payoff and benchmark are affine in λ, so midpoint optimality gives

$$0 \le r_j \le r_1 + r_2 = s - 2F_P(\bar{\lambda}) \le s.$$

Take a product likelihood tilt $P^t = \bigotimes_i (1 + t h_i) P_i$ with centered scores $|h_i| \le 1$ and uniformly bounded Fisher information. For $j = 1, 2$, define

$$F_j(t) = V(P^t;\lambda_j, 1-\lambda_j) - C_{\mathrm{PS}}\{\lambda_j \mathbb{E}_{P^t}[X^*] + 1 - \lambda_j\},$$
$$G_j(t) = V(P^t;\lambda_j, 1-\lambda_j) - \mathbb{E}_{P^t}[\lambda_j X_{\tau_\pi} + (1-\lambda_j)\tau_\pi].$$

Both functions are nonnegative on the perturbation interval. Their values at zero are at most s, and Theorem 8.1 gives a common upper second-derivative bound Q, uniformly when $S \le S_0$ and the scores' information is uniformly bounded.

For a nonnegative $C^{1,1}$ function f with $f'' \le Q$ almost everywhere on $[-h, h]$, where $h > 0$, the two Taylor inequalities give $|f'(0)| \le f(0)/h + Qh/2$. Apply this to F_j, G_j, subtract, and take h proportional to $\sqrt{s}$ within the common perturbation interval:

$$|d_j| \le K\sqrt{s}, \qquad d_j = \frac{d}{dt}\left[\mathbb{E}_{P^t}[\lambda_j X_{\tau_\pi} + (1-\lambda_j)\tau_\pi] - C_{\mathrm{PS}}\{\lambda_j \mathbb{E}_{P^t}[X^*] + 1 - \lambda_j\}\right]\Big|_{t=0}.$$

For $s = 0$, let $h \downarrow 0$ to obtain the same bound. With $\ell = \sum_i h_i(V_i)$, the product-score identity gives

$$A = \mathbb{E}[\ell X_{\tau_\pi}] - C_{\mathrm{PS}}\mathbb{E}\left[\ell \max_i V_i\right], \qquad B = \mathbb{E}[\ell \tau_\pi], \qquad d_j = \lambda_j A + (1-\lambda_j)B.$$

Eliminating A gives the exact identity

$$(\lambda_2 - \lambda_1)B = \lambda_2 d_1 - \lambda_1 d_2.$$

Thus $|B| \ge b_0 > 0$ implies $|\lambda_2 - \lambda_1|^2 \le K_{b_0} s$.

10.3 Compactness makes the constant uniform

Suppose the theorem failed on a compact interior interval I. There would be a sequence $P_m, \lambda_{m1}, \lambda_{m2}$ with

$$|\lambda_{m2} - \lambda_{m1}|^2 > m\{F_{P_m}(\lambda_{m1}) + F_{P_m}(\lambda_{m2})\}.$$

The sum $s_m = F_{P_m}(\lambda_{m1}) + F_{P_m}(\lambda_{m2})$ tends to zero. By Lemma 3.11, extract a limiting instance and weights. Its expected maximum is at least $\mu_{\min}$, its terminal reward is zero since $L_b(P_m) \le S_0/b$, and its midpoint weight is interior. Convexity gives $F_{P_m}(\bar{\lambda}_m) \le s_m/2$, so value convergence makes the limit critical there.

By Lemmas 3.15 and 10.2, choose a linear individual-variable or upper-band score whose duration covariance along the original midpoint optimizers is eventually bounded below in absolute value by $b_0 > 0$. The information and curvature bounds are uniform, and the perturbations share a common interval. The preceding calculation gives $|\lambda_{m2} - \lambda_{m1}|^2 \le K s_m$, contradicting the assumed inequalities, also when $s_m = 0$. This proves (37) on I.

Finally, (11), $F_P(0) = 1 - C_{\mathrm{PS}}$ and uniform Lipschitz continuity give an interior I and $d_0 > 0$ with $F_P \ge d_0$ outside I. If either weight lies outside I, use $|\lambda - \nu|^2 \le 1 \le [F_P(\lambda) + F_P(\nu)]/d_0$; otherwise use the interior bound. □

11 An explicit learner with reward and reach guarantees

Two independent sample blocks suffice. The first estimates a target duration and supplies an ordered family of stopping rules; validation chooses a point in that family. Calibration has expected signed error of order $1/m$ and mean-square error $O(1/m)$. The scalar-gap diameter bound controls the interaction of this error with the selected weight, giving the simultaneous reward and reach bounds in Theorem 4.2.

Fix the bounded class

$$\mu_{\min} \le \mathbb{E}_P[X^*] \le 1/80, \qquad L(P) \le 2\log m.$$

We consider $m \ge 2$, since the class is empty at $m = 1$. Set $a = 4/5$ and choose a fixed rational $b \in (a, 1)$ sufficiently close to 1 that Corollary 6.3 holds and F_P is uniformly positive outside $I = [a, b]$. For $0 \le \nu \le a$, waiting gives

$$F_P(\nu) \ge (1 - C_{\mathrm{PS}})(1 - \nu) - C_{\mathrm{PS}}\nu\mathbb{E}_P[X^*] \ge 17/400,$$

using $C_{\mathrm{PS}} \le 3/4$ and $\mathbb{E}_P[X^*] \le 1/80$. Near 1, Lemma 6.1 and uniform Lipschitz continuity give the required slack. Put

$$z_- = (1 - b)/b, \qquad z_+ = 1/4.$$

All constants in this section may depend on $\mu_{\min}$ and these fixed endpoints.

11.1 The two-block rule

The target estimates a full-information ratio. For $\mathbb{E}_P[X^*] > 0$, define

$$\Gamma(P) = \min_{0\le\lambda\le1} \frac{V(P;\lambda,1-\lambda)}{\lambda\mathbb{E}_P[X^*] + 1 - \lambda} = \inf_{z\ge0} \frac{V(P;1,z)}{\mathbb{E}_P[X^*] + z}. \tag{41}$$

The normalized minimum exists, and Lemma 3.1 gives $C_{\mathrm{PS}} \le \Gamma(P) \le 1$. The ratio is a useful duration target: for $X_\tau + z\tau$, an optimal policy with expected duration $\Gamma(P)$ earns at least that fraction of the expected maximum.

The rule uses the randomized continuation routine of Theorem 12.1, proved in Section 12. For an empirical product distribution Q on $[0, 1]$ represented by at most m observations per distribution, a prescribed state (A, u, z) with $u \in [0, 1]$ and $z \in [0, 4]$, error tolerance $0 < h < 1$ and failure probability $0 < p < 1/2$, it estimates $v_A^Q(u; 1, z)$ within h except with probability p. Each call uses fresh randomness and has work

$$e^{O(L(Q))}\,\mathrm{poly}(n, m, h^{-1}, \log(1/p)).$$

A cutoff on $L(Q)$ makes this work polynomial on every sample outcome. The routine evaluates empirical expectations from stored observations.

Construction 11.1. Take two independent blocks of m samples per distribution. Let Q be the product empirical distribution from the first block; the second consists of m complete vectors of sampled values, used for validation. Set the empirical cutoff

$$\bar{L} = 4\lceil\log_2 m\rceil + 2.$$

If $L(Q) > \bar{L}$ or $\mathbb{E}_Q[X^*] < \mu_{\min}/2$, set $\widehat{\rho} = 4/5$. Otherwise use the grid $\mathcal{Z}_0 = \{j/m : 0 \le j \le 4m\}$ and query the continuation routine at $([n], 1, z)$ for each $z \in \mathcal{Z}_0$, with error tolerance $1/m$ and failure probability $1/[m^2(4m + 1)]$. If the returned values are $\widehat{v}_{[n]}(1; z)$, set

$$\widehat{\rho} = \mathrm{clip}_{[1/2,4/5]}\left(\min_{z\in\mathcal{Z}_0} \frac{\widehat{v}_{[n]}(1;z) + 1/m}{\mathbb{E}_Q[X^*] + z}\right). \tag{42}$$

Here $\mathrm{clip}_{[1/2,4/5]}$ projects its argument onto $[1/2, 4/5]$. Compute $\mathbb{E}_Q[X^*]$ exactly by sorting the observed values and integrating the product of their empirical CDFs.

Choose $J = \lceil m(z_+ - z_-)\rceil$ equally spaced coefficients $z_- = z_0 < \cdots < z_J = z_+$, so the mesh is at most $1/m$. If $L(Q) > \bar{L}$ or $\mathbb{E}_Q[X^*] > 1/4$, use the cap-only policy for every coefficient: accept the first reached 1, and otherwise wait to the end. In all other cases, at an arrival with remaining horizon u, let A be the unseen identities whose empirical distributions have positive mass above zero, excluding the candidate. Query all grid coefficients with fresh independent randomness, with error tolerance

$$h = \frac{z_-}{m(1+\bar{L})^2}$$

and failure probability $1/[m^2 n(J+1)]$ per call. If the estimates of $v_A^Q(u; 1, z_j)$ are $\widehat{v}_A(u; z_j)$, define

$$\widetilde{v}_j(A, u) = \max\left\{z_j u,\ \max_{0\le \ell \le j} \widehat{v}_A(u; z_\ell)\right\}. \tag{43}$$

The grid policy $\widetilde{\pi}_j$ rejects zero, accepts every reached cap 1, and accepts any other value x exactly when $x \ge \widetilde{v}_j(A, u)$. Identities absent from the empirical continuation set are still processed online; their positive values are tested by the same rule.

Give each validation profile independent uniform clocks and an array of independent random seeds for continuation queries, indexed by arrival position and coefficient. These arrays are also independent of the target queries. Simulate every grid policy on each profile, sharing the threshold vector at each arrival. If the resulting durations are $\tau_0, \dots, \tau_J$, concatenate the segments

$$\alpha\tau_0, \qquad (1-\alpha)\tau_j + \alpha\tau_{j+1}\ (0 \le j < J), \qquad (1-\alpha)\tau_J + \alpha, \qquad 0 \le \alpha \le 1. \tag{44}$$

Parametrize this concatenation on $[0, 1]$ and choose the smallest point whose validation average is $\widehat{\rho}$. On the left extension output $\widetilde{\pi}_0$; in a middle segment choose its endpoint policies with probabilities $1-\alpha, \alpha$, once before the online run; on the right extension output $\widetilde{\pi}_J$. Online clocks and continuation seeds are independent of training and validation and have the same distributions as those used in validation.

For every realization of the numerical estimates, the thresholds in (43) are nondecreasing in j. Before two policies first differ they see the same state, so a larger coefficient cannot stop first. Their durations are therefore ordered on each validation trajectory. The curves in (44) are continuous and nondecreasing from zero to one; the required root exists and is found by linear interpolation. The endpoint extensions allow every target in $[0, 1]$, while their output policies remain the corresponding endpoints.

Assign to the selected point the duration coefficient

$$Z = \begin{cases} 0, & \text{on the left extension,} \\ (1-\alpha)z_j + \alpha z_{j+1}, & \text{in a middle segment,} \\ z_+, & \text{on the right extension,} \end{cases} \qquad \lambda = (1+Z)^{-1}.$$

Thus $\lambda \in [a, 1]$. Given the first block and target-query randomness, Z determines the output policy, including its mixture probabilities. Let $\widetilde{T}$ be the selected curve's mean on a fresh trajectory, conditional on this information and the selected point, and put $g = \widetilde{T} - \widehat{\rho}$. This mean agrees with the output policy's duration on middle segments. On the left it is no larger, and on the right it is no smaller.

11.2 Calibration and accuracy of the target and policies

A leave-one-out identity controls the calibration bias.

Lemma 11.2. *Let $f_\theta(Y) \in [0, 1]$, $\theta \in [0, 1]$, be measurable and, for every Y, continuous and nondecreasing in θ, with $f_0 = 0, f_1 = 1$. Fix $\rho \in (0, 1)$ and an integer $m \ge 1$, draw $Y_1, \dots, Y_m$ independently with the*

distribution of Y, and choose the smallest root $\widehat{\theta}$ of $m^{-1}\sum_{j=1}^{m} f_\theta(Y_j) = \rho$. For $g(\theta) = \mathbb{E}[f_\theta(Y)] - \rho$,

$$|\mathbb{E}[g(\widehat{\theta})]| \le \frac{1}{m+1}, \qquad \mathbb{E}[g(\widehat{\theta})^2] \le \frac{K}{m}.$$

The same bounds hold conditionally on independent information specifying the family and target.

Proof. Take $m+1$ observations and let θ_{-i} be the root omitting observation i. If i_* gives the largest root, monotonicity and its root equation imply

$$\begin{aligned}\sum_i \big(f_{\theta_{-i}}(Y_i) - \rho\big) &\le \sum_i \big(f_{\theta_{-i_*}}(Y_i) - \rho\big) \\ &= f_{\theta_{-i_*}}(Y_{i_*}) - \rho \le 1.\end{aligned}$$

The smallest root gives a lower bound of -1. Divide by $m+1$ and use exchangeability: an omitted observation is independent of its root, so the expectation of each term is $\mathbb{E}[g(\widehat{\theta})]$.

For the second estimate add independent uniforms U_j. The indicators $\mathbf{1}_{\{U_j \le f_\theta(Y_j)\}}$ form a nested family and hence an empirical CDF of threshold variables. Its population CDF is $\mathbb{E}[f_\theta(Y)]$. Conditional expectation over the uniforms gives the empirical mean of f_θ. Jensen and the Dvoretzky–Kiefer–Wolfowitz inequality [23] give the bound

$$\mathbb{E}\left[\sup_\theta \left|\frac{1}{m}\sum_{j=1}^{m} f_\theta(Y_j) - \mathbb{E}[f_\theta(Y)]\right|^2\right] \le K/m.$$

At the empirical root, this deviation equals $-g(\widehat{\theta})$. □

Let $\mathcal{T}$ be the information generated by Q and the target-query randomness. Conditional on $\mathcal{T}$, both the target and the policy family are fixed, and the augmented validation observations are independent with the same distribution as a fresh online trajectory. Their duration curves are ordered for every realization of the validation seeds. Thus

$$|\mathbb{E}[g \mid \mathcal{T}]| \le \frac{1}{m+1}, \qquad \mathbb{E}[g^2 \mid \mathcal{T}] \le K/m. \tag{45}$$

We next bound the target's upward error. A coarse estimate applies to every bounded instance; near a tight scalar benchmark, the first-order sampling variance gives a sharper bound.

Lemma 11.3. *For a product distribution P on $[0,1]^n$, a fixed $\lambda \in [0,1]$ and the product empirical distribution Q from $m \ge 1$ independent samples per distribution,*

$$\mathbb{E}\left[|V(Q;\lambda,1-\lambda) - V(P;\lambda,1-\lambda)|\right] \le 4m^{-1/2}.$$

Proof. Fix $z \ge 0$. Conditional on the arrival times, index the variables in arrival order. Let $\widehat{v}_i$ be the empirical continuation threshold after variable i, and put $\xi_i = (Q_i - P_i)(x - \widehat{v}_i)_+$. This threshold depends only on later empirical distributions, so revealing the sample blocks in reverse order gives centered increments. Refine each increment into its m sample contributions. With $\Xi_* = \max_j |\sum_{i \ge j} \xi_i|$ and $U = \sum_i P_i(x - \widehat{v}_i)_+$, Doob's inequality, followed by averaging over the arrival times, gives $\mathbb{E}[\Xi_*^2] \le 4\mathbb{E}[U]/m$.

Centering gives $\mathbb{E}[U] = \mathbb{E}[\sum_i Q_i(x - \widehat{v}_i)_+]$. Expanding the empirical Bellman recursion through all arrivals makes the elapsed duration payments sum to z and the positive excesses sum to the remaining value. It follows that

$$\mathbb{E}[U] = \mathbb{E}[V(Q;1,z)] - z \le 1,$$

since the selected value and duration are each at most one.

For fixed samples and clocks, a policy's reach probabilities form a decreasing sequence in arrival order. Summation by parts bounds the corresponding weighted sum of the ξ_i by $2\Xi_*$. The Bellman residual is this weighted sum plus nonnegative action losses; those losses vanish for the empirical optimizer. Comparing the true and empirical optimizers and averaging over clocks therefore gives

$$|V(Q;1,z) - V(P;1,z)| \le 2\mathbb{E}_{\text{clocks}}[\Xi_*].$$

Averaging and Cauchy–Schwarz bound the unnormalized error by $4/\sqrt{m}$. Multiplying by λ gives the normalized estimate for $0 < \lambda < 1$. Continuity includes the reward endpoint, and the duration-only value is exactly one. □

Since $mL(Q)$ counts the positive samples and $L(P) \le 2\log m$, independence gives

$$\mathbb{E}[\exp(mL(Q))] \le e^{2(e-1)m\log m}, \qquad \Pr(L(Q) > \bar{L}) \le e^{2(e-1)m\log m - m\bar{L}} \le e^{-2m}. \tag{46}$$

Lemma 11.4. *The implemented target in Construction 11.1 satisfies*

$$\frac{1}{2} \le \widehat{\rho} \le \frac{4}{5} \quad \textit{on every outcome}, \qquad \mathbb{E}[\widehat{\rho}] \ge C_{\text{PS}} - m^{-2}. \tag{47}$$

For every fixed $S_0 < \infty$, when $S(P) \le S_0$ the following bound holds with K also depending on S_0:

$$\mathbb{E}[(\widehat{\rho} - \Gamma(P))_+] \le K\sqrt{\frac{\Gamma(P) - C_{\text{PS}}}{m}} + K/m. \tag{48}$$

Without bounding S, the right side can be replaced by $K/\sqrt{m}$.

Proof. First consider the exact target $\rho_0 = \min(4/5, \Gamma(Q))$ when $\mathbb{E}_Q[X^*] \ge \mu_{\min}/2$, and $\rho_0 = 4/5$ otherwise. It is always at least C_{PS}. Product empirical averaging is unbiased. In the orthogonal product decomposition of X^* from the proof of Lemma 6.4, a component of order r has its variance multiplied by m^{-r}. Since $0 \le X^* \le 1$, this gives

$$\mathbb{E}[\mathbb{E}_Q[X^*]] = \mathbb{E}_P[X^*], \qquad \operatorname{Var}[\mathbb{E}_Q[X^*]] \le \operatorname{Var}_P(X^*)/m \le \mathbb{E}_P[X^*]/m.$$

Consequently,

$$\Pr(\mathbb{E}_Q[X^*] < \mu_{\min}/2) \le \frac{4}{\mu_{\min} m}, \qquad \mathbb{E}[|\mathbb{E}_Q[X^*] - \mathbb{E}_P[X^*]|] \le m^{-1/2}.$$

Choose a deterministic minimizing weight λ_Γ in (41). On the event $\mathbb{E}_Q[X^*] \ge \mu_{\min}/2$, its empirical denominator is at least $\mu_{\min}/2$, and

$$\Gamma(Q) - \Gamma(P) \le \frac{V(Q;\lambda_\Gamma, 1-\lambda_\Gamma) - \Gamma(P)[\lambda_\Gamma \mathbb{E}_Q[X^*] + 1 - \lambda_\Gamma]}{\lambda_\Gamma \mathbb{E}_Q[X^*] + 1 - \lambda_\Gamma}. \tag{49}$$

Lemma 11.3 and the expected-maximum estimate above bound the expected absolute numerator by $K/\sqrt{m}$. Adding the exceptional event proves the coarse bound for ρ_0.

Suppose now $S \le S_0$ and $\Gamma(P) - C_{\text{PS}}$ is below the uniform endpoint slack. The defining equality

$$F_P(\lambda_\Gamma) = (\Gamma(P) - C_{\text{PS}})[\lambda_\Gamma \mathbb{E}_P[X^*] + 1 - \lambda_\Gamma] \le \Gamma(P) - C_{\text{PS}}$$

places λ_Γ in I. Let π be a true optimal policy at this weight. Integrating out its arrival times and randomness, define the function on fixed value vectors $x = (x_1, \dots, x_n)$ by

$$\begin{aligned} f(x) = {} & \mathbb{E}[\lambda_\Gamma X_{\tau_\pi} + (1-\lambda_\Gamma)\tau_\pi \mid V_i = x_i \ (1 \le i \le n)] \\ & - C_{\text{PS}}[\lambda_\Gamma \max_i x_i + 1 - \lambda_\Gamma]. \end{aligned}$$

It has mean $F_P(\lambda_\Gamma)$ and $|f| \le 1$. Set

$$g_i(V_i) = \mathbb{E}[f \mid V_i] - F_P(\lambda_\Gamma), \qquad \sigma_1^2 = \sum_i \mathbb{E}[g_i^2].$$

Tilt every marginal P_i by $1 - tg_i$. The envelope derivative of the scalar gap is $-\sigma_1^2$: large first-order variance would make a nearly zero gap decrease quickly. Since $|g_i| \le 2$, apply Theorem 8.1 to $g_i/2$ and rescale the parameter. Nonnegativity of the gap gives, for sufficiently small positive t,

$$0 \le F_P(\lambda_\Gamma) - t\sigma_1^2 + K(1+S)\sigma_1^2 t^2.$$

Taking t a small fixed multiple of $(1+S)^{-1}$ keeps every tilt nonnegative and proves

$$\sigma_1^2 \le K(1+S)F_P(\lambda_\Gamma).$$

In the orthogonal product decomposition of f, empirical averaging multiplies the first-order variance by m^{-1} and every higher-order variance by at most m^{-2}. The original variances sum to at most $\mathrm{Var}(f) \le 1$. Therefore

$$\mathrm{Var}(Qf) \le K\frac{\Gamma(P) - C_{\mathrm{PS}}}{m} + m^{-2}.$$

Let Δ be the empirical optimal payoff minus the empirical payoff of the fixed true policy π, both at weight λ_Γ. Then $\Delta \ge 0$ and $\mathbb{E}[\Delta] \le K/m$: the fixed policy is unbiased, and Theorem 7.1 bounds the optimism of the empirical optimum. The numerator in (49) is

$$Qf - F_P(\lambda_\Gamma) + \Delta - (\Gamma(P) - C_{\mathrm{PS}})\lambda_\Gamma(\mathbb{E}_Q[X^*] - \mathbb{E}_P[X^*]).$$

Its expected positive part is at most

$$K\sqrt{\frac{\Gamma(P) - C_{\mathrm{PS}}}{m}} + K/m,$$

using $0 \le \Gamma(P) - C_{\mathrm{PS}} \le 1$. Dividing by the empirical denominator and adding the exceptional event proves the sharper bound for ρ_0. If $\Gamma(P) - C_{\mathrm{PS}}$ exceeds the fixed slack, the coarse bound gives the same conclusion with a larger constant.

It remains to compare the computed target with ρ_0. The count-cutoff fallback changes bounded target errors by at most e^{-2m}, by (46). Otherwise a union bound over the $4m+1$ target queries makes all their estimates accurate except with probability m^{-2}. On their success event, (42) is an upper approximation to ρ_0 with error $O(1/m)$. Indeed, waiting gives

$$\frac{V(Q;1,z)}{\mathbb{E}_Q[X^*] + z} \ge 4/5 \qquad \text{for } z \ge 4\mathbb{E}_Q[X^*],$$

so the capped infimum only needs $z \in [0,4]$. For a fixed policy π, since $\mathbb{E}_Q[X_{\tau_\pi}] \le \mathbb{E}_Q[X^*]$ and $\mathbb{E}_Q[\tau_\pi] \le 1$,

$$\left|\frac{d}{dz}\frac{\mathbb{E}_Q[X_{\tau_\pi}] + z\mathbb{E}_Q[\tau_\pi]}{\mathbb{E}_Q[X^*] + z}\right| = \frac{|\mathbb{E}_Q[X^*]\mathbb{E}_Q[\tau_\pi] - \mathbb{E}_Q[X_{\tau_\pi}]|}{(\mathbb{E}_Q[X^*] + z)^2} \le \frac{1}{\mathbb{E}_Q[X^*]}.$$

The optimal ratio, being the supremum of these functions, has the same Lipschitz bound. The grid and value errors sum to at most $3/[m\mathbb{E}_Q[X^*]] \le 6/(\mu_{\min} m)$. Clipping below at $1/2$ does not change a successful upper approximation, since $\rho_0 \ge C_{\mathrm{PS}} \ge 1/2$.

These estimates prove (48) and its coarse version for $\widehat{\rho}$. Except on a failed target query, $\widehat{\rho} \ge C_{\mathrm{PS}}$; its range is always $[1/2, 4/5]$. This proves (47). □

The cutoff also bounds the time slope of an empirical continuation from below. This makes small numerical errors harmless even for discrete value distributions.

Lemma 11.5. *Conditional on $\mathcal{T}$, the numerical policy at any grid coefficient can be coupled to the cap-modified optimal empirical threshold policy, with the same count-cutoff fallback, so that their actions differ with probability at most $3/m$. The estimate is uniform after validation selects the coefficient. In particular,*

$$q_i(\widetilde{\pi}_j) \geq \mathbb{E}[\tau_{\widetilde{\pi}_j}] - 6/m \qquad \text{for every } i, j. \tag{50}$$

Proof. The two policies coincide in a cap-only branch. In the other branch, $L(Q) \leq \bar{L}$. A zero-rejecting policy cannot stop before the first positive arrival. If N_+ is the number of positive values in an empirical remaining instance, that first time has conditional mean $1/(N_+ + 1)$ on the unit horizon, with time 1 when $N_+ = 0$. Jensen gives normalized expected duration at least $1/(1 + \bar{L})$. Reusing a scalar-optimal normalized-time policy at a larger horizon gives

$$v_A^Q(u_2; 1, z) - v_A^Q(u_1; 1, z) \geq \frac{z(u_2 - u_1)}{1 + \bar{L}} \qquad (0 \leq u_1 \leq u_2 \leq 1). \tag{51}$$

Every actual query state is determined before its fresh random seeds are drawn. The pointwise confidence guarantee in Theorem 12.1 thus applies conditionally at each query. There are at most $n(J + 1)$ calls on an online trajectory, so their total failure probability is at most m^{-2}. On accurate calls, (43) differs from the exact continuation by at most h: true continuations are nondecreasing in z and at least zu.

For a first different action, the positive candidate value must therefore be within h of the exact threshold. Fix its value and all other clocks. As its own clock increases, this threshold decreases at speed at least $z_-/(1 + \bar{L})$ between other arrivals, by (51), and jumps only downward when an unseen identity is deleted. The times within h of the candidate value have total length at most $2h(1 + \bar{L})/z_-$. Both policies reject zero and agree on caps. Summing over positive candidates multiplies this length by at most $L(P) \leq \bar{L}$. Including query failures, the disagreement probability is at most

$$\frac{2h(1 + \bar{L})\bar{L}}{z_-} + m^{-2} \leq 2/m + m^{-2} \leq 3/m.$$

The comparison uses the conditional failure bounds for queries made along the actual trajectory.

If actions agree, reward, duration and each reach indicator agree. Each changes by at most one otherwise. The exact comparison policy has reach at least its duration by Lemma 3.2, including the cap-only fallback. Comparing its reach and duration separately proves (50). Online values, clocks and planner seeds are fresh after coefficient selection, so these bounds hold conditional on that selection as well. □

Let r be the selected output policy's true scalar regret conditional on $\mathcal{T}$ and Z, using weights $(b, 1 - b)$ on the left extension and $(1, Z)$ elsewhere. Then

$$\mathbb{E}[r] \leq K \frac{1 + S(1 + S)}{m}. \tag{52}$$

To prove this, first use the exact empirical optimizer at a selected grid endpoint. Its coefficient may depend on Q, validation and all auxiliary seeds, so Theorem 7.1 applies even though the target uses the same Q. The cap modification in Lemma 6.4 costs $O(1/m)$. Changing an endpoint's coefficient to the middle-segment coefficient Z costs at most $1/m$, since both optimal and fixed-policy values have duration slopes in $[0, 1]$. Apply this to each endpoint of the selected mixture; on the left, multiply the unnormalized regret by b. The count-cutoff fallback costs $O(e^{-2m})$ by (46), and Lemma 11.5 changes scalar payoff by $O(1/m)$. Together these estimates give (52).

11.3 Reward, reach, and work

Proof of Theorem 4.2. First suppose $S \leq S_0$ for a fixed bound. Choose a deterministic minimizer $\lambda_* \in \arg\min_{\nu \in [a,1]} F_P(\nu)$ and set $z_* = (1 - \lambda_*)/\lambda_*$. Since the selected λ lies in $[a, 1]$, its gap is at least $F_P(\lambda_*)$.

The map from λ to z is Lipschitz on this interval, so Theorem 10.1 gives

$$\mathbb{E}[(Z - z_*)^2] \le K\mathbb{E}[F_P(\lambda) + F_P(\lambda_*)] \le K\mathbb{E}[F_P(\lambda)].$$

The reference $z_* \in [0, z_+]$ is deterministic. Thus (45) yields

$$\begin{aligned} \mathbb{E}[Zg] &= z_*\mathbb{E}[g] + \mathbb{E}[(Z - z_*)g] \\ &\le \frac{z_+}{m+1} + \sqrt{\mathbb{E}[(Z - z_*)^2]\mathbb{E}[g^2]} \\ &\le K\sqrt{\frac{\mathbb{E}[F_P(\lambda)]}{m}} + K/m. \end{aligned} \tag{53}$$

The ratio definition gives

$$F_P(\lambda) \ge (\Gamma(P) - C_{\mathrm{PS}})[\lambda\mathbb{E}_P[X^*] + 1 - \lambda].$$

Consequently, since $\lambda = (1 + Z)^{-1}$ and the bracketed denominator is at most one,

$$\begin{aligned} (1 + Z)F_P(\lambda) - Z(\Gamma(P) - C_{\mathrm{PS}}) &\ge \frac{\mathbb{E}_P[X^*]}{\lambda\mathbb{E}_P[X^*] + 1 - \lambda} F_P(\lambda) \\ &\ge \mu_{\min} F_P(\lambda). \end{aligned}$$

Let $T = \mathbb{E}[\tau \mid \mathcal{T}, Z]$ be the output policy's conditional expected duration. In a middle segment $T = \widetilde{T}$; on the right $T \le \widetilde{T}$. On either type of segment, the scalar-regret identity therefore gives

$$\begin{aligned} \mathbb{E}[X_\tau \mid \mathcal{T}, Z] - C_{\mathrm{PS}}\mathbb{E}_P[X^*] &\ge (1 + Z)F_P(\lambda) - r - Z(g + \widehat{\rho} - C_{\mathrm{PS}}) \\ &\ge \mu_{\min} F_P(\lambda) - r - Zg - z_+(\widehat{\rho} - \Gamma(P))_+. \end{aligned}$$

On the left, $Z = 0$, $\lambda = 1$, and Corollary 6.3 gives conditional reward excess at least $F_P(1)/40 - Kr$. Since $\mu_{\min} < 1/80$, every selected point satisfies

$$\mathbb{E}[X_\tau \mid \mathcal{T}, Z] - C_{\mathrm{PS}}\mathbb{E}_P[X^*] \ge \mu_{\min} F_P(\lambda) - Kr - Zg - z_+(\widehat{\rho} - \Gamma(P))_+. \tag{54}$$

Each denominator $\lambda\mathbb{E}_P[X^*] + 1 - \lambda$ is at least $\mu_{\min}$, so

$$\Gamma(P) - C_{\mathrm{PS}} \le \frac{\mathbb{E}[F_P(\lambda)]}{\mu_{\min}}.$$

Taking expectations in (54) and applying (48), (52) and (53) gives

$$\mathbb{E}[X_\tau] - C_{\mathrm{PS}}\mathbb{E}_P[X^*] \ge \mu_{\min}\mathbb{E}[F_P(\lambda)] - K\sqrt{\frac{\mathbb{E}[F_P(\lambda)]}{m}} - K/m \ge -K'/m.$$

The last step completes the square in $\sqrt{\mathbb{E}[F_P(\lambda)]}$. The scalar gap absorbs the calibration fluctuations.

The same algorithm handles large first moment. By Theorem 9.1 and the endpoint slack, there are $S_*, \kappa > 0$ such that

$$S \ge S_* \quad \Longrightarrow \quad F_P(\nu) \ge \frac{\kappa}{(1 + L)^4} \qquad \text{for every } \nu \in [0, 1].$$

Indeed, take S_* above the first-moment bound in that theorem for weights in I. Its contrapositive bounds the gap below by a constant times $(1 + L)^{-4}$ there; outside I, use the fixed endpoint slack. Choose $S_0 \ge S_*$ in the bounded-moment argument. If $S \ge S_*$, the coarse target estimate and (45) give

$$\mathbb{E}[(\widehat{\rho} - \Gamma(P))_+] \le K/\sqrt{m}, \qquad |\mathbb{E}[Zg]| \le K/\sqrt{m}.$$

Since $S \le L \le 2\log m$, (54) now yields

$$\mathbb{E}[X_\tau] - C_{\mathrm{PS}}\mathbb{E}_P[X^*] \ge \frac{\mu_{\min}\kappa}{(1+2\log m)^4} - K\frac{1+\log^2 m}{m} - \frac{K}{\sqrt{m}} \ge 0$$

for all sufficiently large m. The finitely many smaller values are covered by increasing K.

For reach, the last policy in the numerical family satisfies

$$X_{\tau_{\widetilde{\pi}_J}} \ge z_+(1-\tau_{\widetilde{\pi}_J}) \quad \text{pathwise}, \qquad \mathbb{E}[\tau_{\widetilde{\pi}_J} \mid \mathcal{T}] \ge 1 - \mathbb{E}_P[X^*]/z_+ \ge 19/20.$$

The first inequality follows from the threshold floor in (43), including cap acceptance and the cap-only fallback. The second uses reward at most the realized maximum. A right extension is selected only if its validation endpoint mean is at most $\widehat{\rho} \le 4/5$. Conditional Hoeffding therefore gives

$$\Pr(\text{right extension} \mid \mathcal{T}) \le e^{-2m(19/20-4/5)^2} = e^{-9m/200}.$$

Actual duration dominates the selected surrogate on the left, agrees in the middle, and can lose at most one on the right. By (47), (45) and (50),

$$q_i \ge \mathbb{E}[\tau] - 6/m \ge \mathbb{E}[\widehat{\rho}] + \mathbb{E}[g] - e^{-9m/200} - 6/m \ge C_{\mathrm{PS}} - K/m$$

for every i. Mixtures preserve the uniform reach comparison.

The output policy prescribes rejecting zero and accepting every reached cap. The learner uses exactly $2m$ samples per distribution. There are $O(m)$ target queries and at most $n(J+1)$ continuation calls on each of m validation trajectories and on the online trajectory, hence $O(nm^2)$ calls in total. Every invoked empirical distribution has $L(Q) \le \bar{L} = O(\log m)$, all inverse accuracies are polynomial in m, and logarithmic inverse failure probabilities are $O(\log n + \log m)$. The hard work bound in Theorem 12.1 is therefore polynomial in n, m on every outcome. Root finding and empirical CDF computations are polynomial as well. Appendix G proves the rational-input implementation with the same statistical order. □

12 Computing the scalar policies

The calibrated learner queries continuation values at the states it encounters. Computing each value by enumerating all remaining subsets would take exponential time. We instead write the value as an expectation over smaller states and approximate that expectation recursively. A multiplicative weight makes this recursion contract by a fixed factor. Its size depends on the sum of the positive probabilities, which the learner bounds before making a query.

Theorem 12.1 (Scalar planning)**.** *Let $Q = \bigotimes_{i=1}^n Q_i$, where each Q_i is an empirical distribution on $[0,1]$ represented by at most m observations. Given $A \subseteq [n]$, $u \in [0,1]$, $z \in [0,4]$, an error tolerance $0 < h < 1$ and a failure probability $0 < p < 1/2$, a randomized procedure returns $\widehat{v}$ satisfying*

$$\Pr\bigl(|\widehat{v} - v_A^Q(u;1,z)| > h\bigr) \le p.$$

Its number of arithmetic operations, comparisons, and internal random draws is at most

$$e^{O(L(Q))}\,\mathrm{poly}\bigl(n, m, h^{-1}, \log(1/p)\bigr) \tag{55}$$

on every outcome. For rational inputs, the same bound holds for bit operations, with an additional polynomial dependence on the input representation length.

The procedure below proves the arithmetic bound. The finite-bit implementation is given in Appendix G. In particular, when every queried empirical distribution has $L(Q) = O(\log m)$, inverse-polynomial accuracy costs polynomial time.

12.1 An all-arrivals recursion and a weighted contraction

Fix the empirical distribution Q. Write

$$p_i = Q_i(V_i > 0), \qquad h_i(y) = \mathbb{E}_{Q_i}\big[(V_i - y)_+ \mid V_i > 0\big].$$

We omit empirically zero variables from continuation computations. Their arrivals carry no information about the other independent values, and their clocks can be simulated as auxiliary randomness. The online phase still tests a positive value from such an identity against the continuation threshold. Thus every active identity has $p_i > 0$. For an empty active set, $v^Q_\varnothing(u; 1, z) = zu$, which is computed directly. Each h_i takes values in $[0, 1]$ on nonnegative arguments and is one-Lipschitz. Its empirical average costs polynomial time.

Expanding the Bellman recursion through all arrivals gives the waiting payoff plus the expected excess above each future continuation value:

$$v^Q_A(u; 1, z) = zu + \sum_{i\in A} p_i \int_0^1 \mathbb{E}_{B\sim \mathrm{Bern}(s)^{A\setminus\{i\}}}\big[h_i(v^Q_B(us; 1, z))\big]\, ds. \tag{56}$$

Here each identity in $A \setminus \{i\}$ belongs to B independently with probability s. To verify the identity, let $r = |A|$ and condition on the first arrival. Its remaining-time fraction has density rs^{r-1}, and its identity is uniform in A. The Bellman equation is therefore

$$v^Q_A(u; 1, z) = \frac{zu}{r+1} + \sum_{i\in A} \int_0^1 s^{r-1}\big\{v^Q_{A\setminus\{i\}}(us; 1, z) + p_i h_i(v^Q_{A\setminus\{i\}}(us; 1, z))\big\}\, ds.$$

This uses $\max(x, v) = v + (x - v)_+$. Expand each child value in the same way. After at most r expansions along a branch, the elapsed duration payments and the final empty-state payment sum to zu. The remaining terms sum the positive excess at every arrival. For a fixed identity i, its remaining-time fraction s is uniform, and each other identity is still unseen independently with probability s. This gives (56), with no reach factors. At $u = 0$, continuity in the duration coefficient gives the reward-only identity; $v^Q_A(0; 1, z)$ need not vanish.

Errors in child values can add up in the unweighted recursion. We divide by a product weight so that their total coefficient is at most $1/4$. Define

$$\beta_A = \prod_{i\in A}(1 + 4p_i), \qquad \bar v_A(u; z) = v^Q_A(u; 1, z)/\beta_A, \qquad \kappa_A = \frac{\beta_A - 1}{4\beta_A} \le \frac{1}{4}.$$

Set $\beta_\varnothing = 1$ and $\kappa_\varnothing = 0$. Differentiating $\prod_{i\in A}(1 + 4sp_i)$ and integrating gives

$$\sum_{i\in A} p_i \int_0^1 \mathbb{E}[\beta_B]\, ds = \frac{\beta_A - 1}{4}.$$

For nonempty A, define a probability kernel on (i, s, B) by

$$K_A(i, ds, B) = \frac{4p_i\beta_B}{\beta_A - 1} s^{|B|}(1 - s)^{|A|-1-|B|}\, ds, \qquad B \subseteq A \setminus \{i\}.$$

Put $f_{i,B}(y) = h_i(\beta_B y)/\beta_B$. This function is one-Lipschitz and belongs to $[0, 1]$ for $y \ge 0$. Dividing (56) by β_A yields

$$\bar v_A(u; z) = \frac{zu}{\beta_A} + \kappa_A \mathbb{E}_{(i,s,B)\sim K_A}\big[f_{i,B}(\bar v_B(us; z))\big]. \tag{57}$$

The right side contracts in the supremum norm by at most $1/4$. Its solution lies in $[0, 5]$, since $v^Q_A(u; 1, z) \le 1 + z \le 5$, and

$$1 \le \beta_A \le e^{4L(Q)}. \tag{58}$$

Thus the weight that produces a constant contraction costs only an exponential factor in $L(Q)$.

Lemma 12.2. *For every nonempty active set A, the kernel K_A defined above can be sampled with $O(|A|)$ random draws and polynomially many arithmetic operations, without subset enumeration or rejection sampling.*

Proof. Draw a nonempty set $D \subseteq A$ with probability

$$\Pr(D) = \frac{4^{|D|} \prod_{j \in D} p_j}{\beta_A - 1}.$$

Choose i uniformly from D and let s be the maximum of $|D|$ independent uniforms. Include $D \setminus \{i\}$ in B, then include each identity in $A \setminus D$ independently with probability s. For fixed i, s, B, a contributing set has $D \setminus \{i\} \subseteq B$. The factor $1/|D|$ for choosing i cancels the factor $|D|$ in the maximum's density. Summing over these sets gives

$$\frac{4p_i}{\beta_A - 1} s^{|B|} (1-s)^{|A|-1-|B|} \prod_{j \in B} (1 + 4p_j),$$

which is the density of K_A.

To draw D, use Bernoulli probabilities $r_j = 4p_j/(1+4p_j)$ conditioned on at least one success. In a fixed ordering of A, draw the first success at i with probability

$$\frac{r_i \prod_{j<i}(1-r_j)}{1 - \prod_{j \in A}(1-r_j)},$$

then draw the later Bernoullis independently. A single uniform draw and the cumulative probabilities choose the first success. All subsequent draws have fixed bounds on their number. The empty state is evaluated directly. □

12.2 Multilevel approximation

A direct nested Monte Carlo computation would repeatedly pay for accurate child values. We use a recursive multilevel Monte Carlo estimator [19, 2] to estimate differences between successive levels. At higher levels these differences are smaller, so they need fewer samples.

Write $x = (A, u, z)$ for a query state, $Y = (i, s, B)$ for a transition drawn from K_A, and $x_Y = (B, us, z)$ for its child state. In the estimator below, $f_Y = f_{i,B}$ and $\bar{v}(x) = \bar{v}_A(u; z)$. Define $W_0(x) = 0$. For an integer $r \geq 1$, form

$$\mathcal{W}_r(x) = \frac{zu}{\beta_A} + \kappa_A \Bigg[\frac{1}{4^r} \sum_{j=1}^{4^r} f_{Y_{0j}}(0) + \sum_{\ell=1}^{r-1} \frac{1}{4^{r-\ell}} \sum_{j=1}^{4^{r-\ell}} \left\{ f_{Y_{\ell j}}(W_\ell^+(x_{Y_{\ell j}})) - f_{Y_{\ell j}}(W_{\ell-1}^-(x_{Y_{\ell j}})) \right\} \Bigg],$$

and set $W_r(x) = \mathrm{clip}_{[0,5]} \mathcal{W}_r(x)$, the projection of the raw estimate onto $[0, 5]$. The transitions in distinct summands are independent. Within a difference, the superscripts + and − distinguish recursive copies that share Y and its child state, but whose recursive random trees are independent conditional on Y. Sharing this transition is what makes their difference small. For $r \geq 1$, an empty state returns zu exactly; the level-zero value remains $W_0 = 0$ on every state.

For the fixed empirical instance, define the worst-case mean-square error by

$$e_r^2 = \sup_{A,u,z} \mathbb{E}\left[|W_r(A, u, z) - \bar{v}_A(u; z)|^2 \right].$$

The mean of the raw estimator telescopes:

$$\mathbb{E}[\mathcal{W}_r(x)] = \frac{zu}{\beta_A} + \kappa_A \mathbb{E}\big[f_Y(W_{r-1}(x_Y))\big].$$

Comparison with (57), Lipschitz continuity, and Jensen give squared bias at most $e_{r-1}^2/16$. The base average has variance at most $4^{-r}/16$. In a paired difference, subtract and add $f_Y(\bar{v}(x_Y))$. Its second moment is at most $2(e_\ell^2 + e_{\ell-1}^2)$. Independence across summands and the fact that clipping cannot increase distance to $\bar{v} \in [0,5]$ now give

$$e_r^2 \le \frac{1}{16}\left[e_{r-1}^2 + 4^{-r} + 2\sum_{\ell=1}^{r-1} 4^{-(r-\ell)}(e_\ell^2 + e_{\ell-1}^2)\right]. \tag{59}$$

Since $e_0^2 \le 25$, induction proves

$$e_r^2 \le 25\,2^{-r}. \tag{60}$$

Indeed, inside the brackets in (59), the inductive bias bound is at most $2 \cdot 25\,2^{-r}$ and the base term at most $25\,2^{-r}$. The paired terms are at most

$$6 \cdot 25\,2^{-r} \sum_{j=1}^{r-1} 2^{-j} \le 6 \cdot 25\,2^{-r}.$$

Multiplication by $1/16$ gives the claimed bound.

Let $c_0 = \mathrm{poly}(n,m)$ bound one transition, one empirical average, and the associated bookkeeping. If C_r bounds the work of a level-r call, then $C_0 \le c_0$ and

$$C_r \le c_0 + \sum_{\ell=0}^{r-1} 4^{r-\ell}\big(c_0 + C_\ell + \mathbf{1}_{\{\ell \ge 1\}} C_{\ell-1}\big).$$

A geometric-series induction gives $C_r \le K c_0\, 16^r$. For example, the normalized coefficient of the inductive recursive terms is at most

$$\sum_{j \ge 1} 4^{-j} + \frac{1}{16}\sum_{j \ge 1} 4^{-j} = \frac{17}{48} < 1,$$

and the nonrecursive terms are absorbed by increasing K.

Choose the smallest r for which $25\beta_A^2 2^{-r} \le h^2/64$. By (60), the estimator $\beta_A W_r$ has mean-square error at most $h^2/64$, with work

$$\mathrm{poly}(n,m)\, O\big((\beta_A/h)^8\big).$$

Its error exceeds h with probability at most $1/64$. Take the median of an odd number $N = O(\log(1/p))$ of independent copies. If the median is inaccurate, at least half the copies are inaccurate; the union bound over their subsets gives probability at most $2^N (1/64)^{N/2} \le 4^{-N}$. Choose N to make this at most p. Final clipping to $[0,5]$ can only improve the error. Together with (58), these estimates prove (55).

Part II
The lower bound

Section 13 compares the loss from two hard-to-distinguish distributions with the gain from adapting to samples. Section 14 starts from a limit where the weighted guarantee is tight. A perturbation changes the expected stopping time of a fixed optimal policy, and finite approximations preserve that change.

13 The sample lower bound

Theorem 13.1 (Sample lower bound)**.** *There are constants $a > 0$ and $\varepsilon_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$, any learner that uses an online policy with stopping time τ and guarantees*

$$\mathbb{E}_P[X_\tau] \geq (C_{\mathrm{PS}} - \varepsilon)\mathbb{E}_P[X^*]$$

for every finite product distribution P with $0 < \mathbb{E}_P[X^*] < \infty$ requires a hard budget of at least a/ε samples per distribution in the worst case. The conclusion holds even for adaptive training requests; the reward expectation includes training and online randomness.

A reward guarantee on every instance also constrains the expected stopping time. After we append many deterministic-zero variables, Lemmas 3.4 and 3.5 give the same lower bound on expected duration, up to a vanishing error. We can therefore evaluate the learned policy using positive reward–duration weights correlated with the underlying distribution. These weights are not revealed to the learner.

13.1 A statistical experiment with two statistically close distributions

The experiment pairs two statistically close distributions with scalar weights a fixed distance apart. Their correlation creates a loss of order t when the distributions differ at a small scale $t > 0$. Samples help the learner adapt its policy, and scalar curvature bounds this gain by the squared posterior bias about the hidden sign. With k samples per distribution the expected squared posterior bias is $O(kt^2)$, so balancing the loss and the gain gives the lower bound $k = \Omega(1/t)$. The theorem below applies this comparison uniformly to the nearly critical instances used in the construction.

For integers $m \geq 1$ and θ in a common neighborhood of zero, let $P_{m,\theta}$ be product perturbations of base distributions $P_m = P_{m,0}$ with $\mathbb{E}_{P_m}[X^*] = 1$, all variables bounded by a common $H < \infty$, and weights $w_m \in [\alpha, 1-\alpha]$ for a fixed $\alpha > 0$. Let π_m be scalar-optimal for $(P_m, w_m, 1-w_m)$. For a competing policy π, write $r_m(\pi) = V(P_m; w_m, 1-w_m) - \mathbb{E}_{P_m}[w_m X_{\tau_\pi} + (1-w_m)\tau_\pi]$ for its base regret. Holding π fixed under perturbation, define

$$\begin{aligned} A_{m,\pi}(\theta) &= w_m \mathbb{E}_{P_{m,\theta}}[X_{\tau_\pi}] + (1-w_m)\mathbb{E}_{P_{m,\theta}}[\tau_\pi] - C_{\mathrm{PS}}[w_m \mathbb{E}_{P_{m,\theta}}[X^*] + 1 - w_m], \\ B_{m,\pi}(\theta) &= \mathbb{E}_{P_{m,\theta}}[X_{\tau_\pi}] - \mathbb{E}_{P_{m,\theta}}[\tau_\pi] - C_{\mathrm{PS}}[\mathbb{E}_{P_{m,\theta}}[X^*] - 1]. \end{aligned}$$

The quantity $A_{m,\pi}$ is the fixed policy's excess over the scalar benchmark at the base weight, and $B_{m,\pi}$ is its change per unit increase in that weight while the policy remains fixed.

The criterion below separates the statistical and optimization requirements. Limited information keeps the learner uncertain about the distribution, while curvature and stability bound how much a policy can gain from that information. A negative mixed response to the distribution and evaluation weight supplies the loss that these gains must overcome.

Theorem 13.2. *For the perturbation families and normalized base distributions just defined, suppose $\delta_m = A_{m,\pi_m}(0) \to 0$. Assume that for sufficiently small $t > 0$ the following estimates hold uniformly in m and in all randomized online policies π for constants $I, b > 0$:*

(i) For every integer budget $k \geq 0$ and every adaptive training procedure with at most k requests per distribution, its transcript (requests, observed samples and training randomness) has distributions $D_{m,+t}$ and $D_{m,-t}$ under $\theta = \pm t$ obeying

$$\mathrm{KL}(D_{m,+t} \| D_{m,-t}) + \mathrm{KL}(D_{m,-t} \| D_{m,+t}) \leq Ikt^2.$$

(ii) The functions $A_{m,\pi}$ and $B_{m,\pi}$ have uniform $O(t^2)$ first-order Taylor remainders about $\theta = 0$ at $\theta = \pm t$.

(iii) There is a bounded nondecreasing modulus $\omega : [0,\infty) \to [0,\infty)$, with $\omega(0) = 0$ and $\omega(s) \to 0$ as $s \downarrow 0$, such that

$$|A'_{m,\pi}(0) - A'_{m,\pi_m}(0)| + |B'_{m,\pi}(0) - B'_{m,\pi_m}(0)| \le \omega(r_m(\pi)).$$

(iv) $B'_{m,\pi_m}(0) \le -b < 0$.

Then there are constants $a, \varepsilon_0 > 0$ such that every learner guaranteeing expected reward at least $(C_{\mathrm{PS}} - \varepsilon)\mathbb{E}_P[X^*]$ for every finite product distribution P with positive finite $\mathbb{E}_P[X^*]$ requires a hard budget of at least a/ε samples per distribution for $0 < \varepsilon < \varepsilon_0$, including adaptive training.

Proof. Fix an accuracy $\varepsilon > 0$ and a learner with hard budget k per distribution that guarantees competitive ratio $C_{\mathrm{PS}} - \varepsilon$ on every finite instance. Fix $0 < v < \alpha$, so $w_m \pm v \in (0,1)$ for every m. For an index m and sufficiently small $t > 0$, draw a uniform sign $\sigma \in \{-1, 1\}$, use distribution $P_{m,\sigma t}$ and target weight $w_m + \sigma v$. Conditional on the training transcript D, let $\pi = \pi_D$ be the learner's randomized online policy and put $Z = \mathbb{E}[\sigma \mid D]$. The sign of Z identifies which distribution the samples favor; $|Z|$ measures the strength of this evidence. The policy may still respond to every online observation; all hypotheses apply uniformly to such policies. The conditional expected excess of the learner's scalar payoff over its benchmark is

$$\delta_m - r_m(\pi) + vZB_{m,\pi}(0) + tZA'_{m,\pi}(0) + vtB'_{m,\pi}(0) + O(t^2).$$

The terms involving the posterior bias measure the possible benefit of training. The mixed term has no posterior factor: the two hidden signs multiply to one. The negative derivative at the base optimizer therefore supplies a loss even when the samples reveal no information.

Figure 4 shows which observations the learner receives and which hidden weight is used to evaluate it. The terms $vZB_{m,\pi_m}(0)$ and $tZA'_{m,\pi_m}(0)$ vanish after averaging because $\mathbb{E}[Z] = 0$. Write $r = r_m(\pi)$.

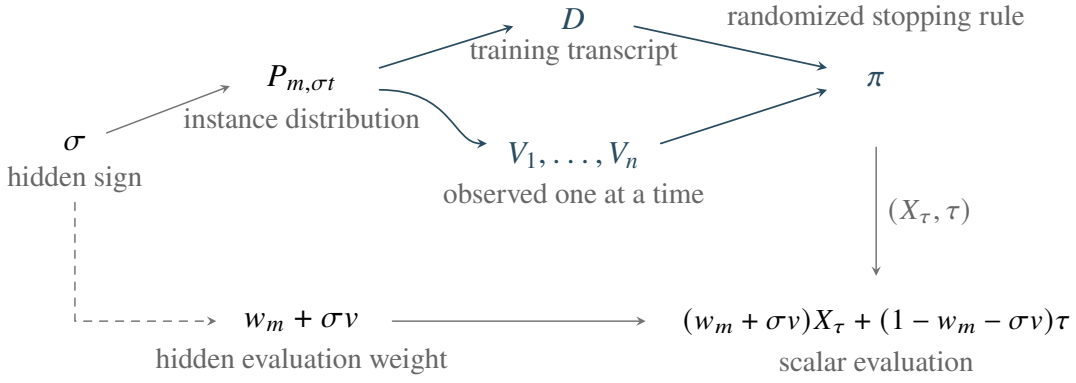


Figure 4: The learner uses training data to choose a policy. The proof evaluates it at a hidden weight.

Corollary 3.7 and derivative stability bound the remaining expression by

$$\delta_m - r + Kv|Z|\sqrt{r} - vbt + (1+v)t\omega(r) + O(t^2).$$

The regret r measures the payoff lost at the base instance. Curvature bounds the change in reward–duration slope by $K\sqrt{r}$, so Young's inequality absorbs half of r and leaves a multiple of Z^2. Thus the base regret pays for changing the policy, and the remaining benefit is quadratic in the evidence supplied by training.

For the resulting constant K', splitting at $r = 2K't\|\omega\|_\infty$ bounds the remaining modulus term uniformly:

$$0 \le \sup_{r \ge 0}\{-r/2 + K't\omega(r)\} \le K't\omega(2K't\|\omega\|_\infty) = o(t).$$

Above the threshold the expression is nonpositive; below it use monotonicity. The expected excess is at most $\delta_m - vbt + K_1\mathbb{E}[Z^2] + o(t)$ for a uniform constant K_1.

We now bound the benefit of adapting to the samples through $\mathbb{E}[Z^2]$. For fixed m, t, abbreviate the transcript distributions $D_{m,+t}, D_{m,-t}$ as D_+, D_-, and write p_+, p_- for their densities with respect to a common dominating measure. Then

$$\mathbb{E}[Z^2] = \frac{1}{2}\int \frac{(p_+ - p_-)^2}{p_+ + p_-} \le \frac{1}{4}\big[\mathrm{KL}(D_+\|D_-) + \mathrm{KL}(D_-\|D_+)\big].$$

The inequality uses $(a-b)\log(a/b) \ge 2(a-b)^2/(a+b)$ for $a, b > 0$. The information bound gives expected excess at most $\delta_m - vbt + K_2kt^2 + o(t)$ for a uniform constant K_2. The first-order loss now competes directly with the information gained from samples. Choose $m = m(t)$ with $\delta_m = o(t)$ and then a fixed $a > 0$ small enough that $k \le a/t$ makes the excess at most $-vbt/2$.

To compare this negative scalar excess with the reward guarantee, append N deterministic-zero variables. The learner must also satisfy its guarantee on these augmented instances, whose expected maxima are unchanged. The reward guarantee and Lemmas 3.4 and 3.5 force the scalar excess at either target weight to be at least $-O(\varepsilon) - O(N^{-1/2})$. Their training reveals nothing about σ, and their arrival schedule is independent auxiliary randomness. By Lemma 3.5, replace any zero selection by waiting to time 1, obtaining a base policy π' with $X_{\tau_{\pi'}} = X_{\tau_\pi}$ and $\tau_{\pi'} \ge \tau_\pi$ pathwise. Its scalar excess is no smaller, and the preceding upper estimates apply to it.

Take N sufficiently large and t a sufficiently large fixed multiple of ε. The two excess bounds contradict each other when $k \le a'/\varepsilon$ for a sufficiently small constant $a' > 0$. □

14 Finite instances for the lower bound

The statistical experiment needs instances whose scalar ratios approach C_{PS} and a perturbation that changes the fixed optimal policy's expected duration to first order. Bounded normalization and compactness give a critical limiting instance; nondegeneracy identifies the perturbation. To use it in the sampling problem, we approximate the limiting instance by finite ones while preserving the duration response and the uniform information and stability bounds. Section 14.3 combines these steps to prove the following proposition.

Proposition 14.1. *There are $0 < \alpha < 1/2$, $H < \infty$, finite product perturbations $P_{m,\theta}$ with all values in $[0, H]$, indexed by integers $m \ge 1$ and θ in a common neighborhood of zero, and weights $w_m \in [\alpha, 1-\alpha]$ with normalized base distributions $P_m = P_{m,0}$ such that $\mathbb{E}_{P_m}[X^*] = 1$ and $F_{P_m}(w_m) \to 0$. For their base scalar optimizers π_m, the information, Taylor-remainder and derivative-stability hypotheses of Theorem 13.2 hold uniformly in m and the competing policy, and the target-difference functions B_{m,π_m} defined there satisfy $B'_{m,\pi_m}(0) \le -b$ for a fixed $b > 0$. The likelihoods also have uniform L^1 second-order remainders.*

The finite perturbation families must approximate both values and derivatives. We first construct finite approximations and compare policies on them in Lemmas 14.3 and 14.4. We then construct likelihood scores, show that their expected products with reward, stopping time, and the maximum converge, and bound the information revealed by adaptive training in Lemmas 14.6 and 14.7. The final steps identify the required target derivative at criticality and check the hypotheses of Theorem 13.2.

14.1 Finite approximations

The limiting instance may contain a Poisson process, whereas the learner samples a finite list of distributions. We replace the process by many rare variables, each producing at most one positive arrival. The approximation must preserve both optimal values and the positive-level tail bounds needed for policy stability.

Construction 14.2. *For a limiting instance with cap H, Poisson mark measure μ and zero terminal reward, include an increasing finite initial set of individual variables and choose positive cutoffs $\zeta_m \downarrow 0$, indexed by integers $m \geq 1$. Let μ_m be μ restricted to $(\zeta_m, H]$. Replace that Poisson process by a positive integer N_m of independent variables, each with positive subprobability measure μ_m/N_m and its remaining mass at zero, where $N_m \geq \mu_m((0,H])$. Give the included individual variables and all new variables independent uniform arrival times. Denote the resulting finite product distribution by P_m. Throughout this section, write n_m for its number of variables, P_{mi} for their marginal distributions and V_{mi} for their values, so $P_m = \bigotimes_{i=1}^{n_m} P_{mi}$.*

Lemma 14.3. *In Construction 14.2, one can choose the number of individual variables to include in each approximant and the numbers N_m of new variables so that expected maxima and scalar values converge to those of the limiting instance and, for every $b > 0$,*

$$\sup_m \sum_i \Pr_{P_m}(V_{mi} > b) < \infty.$$

Convergence holds also after a bounded continuous likelihood tilt of a fixed individual variable, or after multiplying the Poisson intensity on a fixed positive continuity band by $1 + \theta$, for each fixed sufficiently small real θ.

Proof. Fix scalar weights $w, c \geq 0$ and a cutoff $\delta > 0$. Discard values at most δ by waiting instead of selecting them; reward falls by at most δ and duration increases. Above fixed δ, include enough individual variables that the expected maximum of the omitted variables vanishes, and take $N_m \gg \mu_m((0,H])^2$. Lemma 3.10 bounds the Poisson-approximation error by $(\pi/2)wH\mu_m((0,H])/\sqrt{N_m}$. Rare-point coupling or the product cumulative distribution function gives the corresponding maximum error.

For every fixed positive b, the sum of the approximating variables' probabilities above b is at most $\mu((b,H])$, and the sum over individual variables is bounded by the corresponding summable exceedance probabilities of the limiting individual variables. Bounded tilts preserve these estimates uniformly for small θ. □

Value convergence alone does not compare the policies on the two observation spaces. We first follow one fixed policy that uses only finitely many variables and marks above a positive cutoff. This common policy will later connect the finite and limiting optimizers through their small regrets.

Lemma 14.4. *In Construction 14.2, fix positive scalar weights w, c, finitely many individual variables and a positive continuity level b of the Poisson intensity measure. An optimal threshold policy for those variables and the Poisson process above b, run on the direct approximants by ignoring all other observations, has convergent reward and duration expectations. Joint expectations with either a bounded continuous score of an included individual variable or the centered total count in a fixed positive continuity band also converge.*

Proof. On a fixed positive band, the point process of the approximating variables converges to the finite-intensity marked Poisson process. For a continuous nonnegative test function $f(t,x)$ supported on that band, the expectation of $\exp[-\sum f(t,x)]$, with the sum over its arrivals, is

$$\left(1 - \frac{1}{N_m}\int_0^1 \int \left(1 - e^{-f(t,x)}\right)\mu_m(dx)\,dt\right)^{N_m},$$

which converges to the same expectation for the Poisson process. Couple corresponding individual variables directly; on each fixed positive band, counts have uniformly bounded moments of every fixed order.

To pass point-process convergence through the stopping rule, we must control the configurations at which a small change can alter a decision. Between the finitely many arrivals of the included individual variables, positive continuation strictly decreases in time: a longer horizon increases duration payoff, and the extra

Poisson opportunities can only increase the continuation value. Conditional on other arrivals, the candidate's continuous uniform arrival time therefore makes ties between positive acceptance and continuation values null. Simultaneous arrival times are also null, so reward and duration are almost surely continuous at the limiting marked configuration. Truncate the Poisson point count and use coupled bounded convergence; uniform count moments allow removal of the truncation and give joint convergence with scores. □

14.2 Likelihood scores and training information

For a finite variable, increasing the probability of a positive band must remove the same mass elsewhere. Taking it from the zero atom simulates an intensity tilt of the Poisson process. The sum of the finite likelihood scores then approaches the centered Poisson count.

Construction 14.5. *Let P_m be the direct approximants in Construction 14.2, with marginal distributions P_{mi} and values V_{mi}. For a fixed individual variable i, use a bounded continuous score recentered under P_{mi}, and use score zero on all other variables. For the nondegeneracy direction, the score $(x - \mathbb{E}_{P_{mi}}[V_{mi}])/H$ has absolute value at most one. Alternatively, let B be a continuity band of μ bounded away from zero, multiply each approximating variable's mass in B by $1 + \theta$ and subtract the added mass from its zero atom. Write $p_{mB} = \mu(B)/N_m$ for its band probability and p_{m0} for its zero mass. Its score is*

$$h_m(x) = \mathbf{1}_B(x) - \frac{p_{mB}}{p_{m0}}\mathbf{1}_{\{0\}}(x).$$

Start the sequence after $\zeta_m < \inf B$ and choose N_m large enough that $p_{m0} \geq 1/2$; use score zero on the included individual variables. Denote these likelihood scores by h_{mi} and their sum on the complete vector of values by $Z_m = \sum_i h_{mi}(V_{mi})$. The perturbed product distribution $P_{m,\theta} = \bigotimes_i P_{mi,\theta}$ has marginals $dP_{mi,\theta} = (1 + \theta h_{mi})\, dP_{mi}$ on a fixed sufficiently small interval around zero.

Products of likelihood scores with policy outcomes are the derivatives of fixed-policy expectations. Their convergence transports the duration response from the limiting instance to finite experiments, while stability makes the transport uniform over policies with small regret.

Lemma 14.6. *Let P be a critical limiting instance with zero terminal reward, and let P_m be its direct approximants from Construction 14.2. Fix positive weights (w, c) at which P is critical. Let π_m be the optimal threshold policy for P_m at these weights, and let π_* be the corresponding limiting optimizer. Let Z be either a centered bounded continuous individual-variable score or a centered count score in a fixed positive continuity band B. The scores Z_m of Construction 14.5 have uniformly bounded second moments and satisfy*

$$\mathbb{E}_{P_m}[Z_m X_{\tau_{\pi_m}}] \longrightarrow \mathbb{E}_P[Z X_{\tau_{\pi_*}}], \qquad \mathbb{E}_{P_m}[Z_m \tau_{\pi_m}] \longrightarrow \mathbb{E}_P[Z \tau_{\pi_*}].$$

For the finite maximum $X_m^* = \max_i V_{mi}$ and the limiting maximum X^*, also $\mathbb{E}_{P_m}[Z_m X_m^*] \to \mathbb{E}_P[Z X^*]$. Moreover, if π is any competing policy on P_m, the differences between its two score expectations and those of π_m tend to zero uniformly as its scalar regret tends to zero.

Proof. The bounded continuous individual-variable score converges with its variable and has bounded second moment. For a band of Poisson marks, the mass is exact once $\zeta_m < \inf B$, and the independent scores of the approximating variables have total variance at most $2\mu(B)$. Writing N_{mB}, N_{m0} for band and zero counts in the online instance,

$$Z_m = N_{mB} - \frac{p_{mB}}{p_{m0}} N_{m0} = N_{mB} - \mu(B) + o_{L^2}(1).$$

This representation and Lemma 14.4 give joint score convergence for each fixed policy for the truncated instance. The score is evaluated on the complete value vector after the policy's execution.

The finite and limiting optimizers may use different observations. A fixed policy for the truncated instance provides a common comparison, and small regret controls the error when it is replaced by either optimizer.

As the initial set of individual variables grows and the cutoff decreases, their limiting-instance regret vanishes; scalar value convergence and Lemma 14.4 give the same limiting bound on P_m. The uniform tail counts allow us to apply (3); Cauchy–Schwarz with the score variance bound then compares the policies for the truncated instances to π_m in score expectation. Lemma 3.13 gives the comparison with π_*. Let $m \to \infty$ first, then refine the policy for the truncated instance. The uniform regret assertion follows from the same stability bound.

For maxima, boundedness by H turns vanishing expected truncation error into vanishing squared error; Cauchy–Schwarz gives the maximum-score limit. □

Lemma 14.7. *The product tilts in Construction 14.5 have uniformly bounded variance of the likelihood score of the complete vector of values and L^1 second-order likelihood remainders $O(\theta^2)$. Consequently all bounded policy payoffs and maxima have uniform first-order Taylor remainders. For each integer budget $k \geq 0$ and sufficiently small $t > 0$, adaptive training with at most k requests per distribution gives a sum of the two relative entropies between the transcript distributions at $\theta = \pm t$ of $O(kt^2)$.*

Proof. The tilt of an individual variable has one bounded score; the finite approximation of the Poisson intensity tilt has independent scores bounded by one with uniformly bounded total variance. For fixed m, abbreviate the random scores $h_{mi}(V_{mi})$ as h_i. On a fixed sufficiently small interval around zero, the likelihood ratio relative to P_m is $L_\theta = \prod_i (1 + \theta h_i)$. The second derivative has uniformly bounded L^1 norm: under the tilted distribution, both the centered score sum's variance and the second log derivative's expected absolute value are bounded by a constant times $\sum_i \mathbb{E}_{P_m}[h_i^2]$. Every bounded policy payoff and the maximum therefore have an $O(\theta^2)$ Taylor remainder, uniformly in the approximant and policy.

Per-variable relative entropy between the $+\theta$ and $-\theta$ distributions is at most $K\theta^2 \mathbb{E}_{P_m}[h_i^2]$. An adaptive request contributes the divergence of the distribution chosen at that step; the hard budget limits how often each distribution can contribute. The chain rule therefore bounds the sum of the two training relative entropies by $Kk\theta^2$ for every adaptive experiment with at most k draws per distribution. Extra deterministic zero variables carry no information and do not change any bound. □

For approximants of a limiting instance with expected maximum 1, we have $\mathbb{E}_{P_m}[X^*] \to 1$ and take normalized limiting weights $w, c > 0$ with $w + c = 1$. Discard finitely many terms so $\inf_m \mathbb{E}_{P_m}[X^*] > 0$. To normalize the finite expected maxima without changing the optimizer, divide the rewards by $\mathbb{E}_{P_m}[X^*]$ and use weights

$$w_m = \frac{w\mathbb{E}_{P_m}[X^*]}{w\mathbb{E}_{P_m}[X^*] + c}, \qquad c_m = \frac{c}{w\mathbb{E}_{P_m}[X^*] + c}. \tag{61}$$

The payoff and gap are divided by $w\mathbb{E}_{P_m}[X^*] + c$, preserving the optimal threshold policy. Keep the reward scaling and these weights fixed throughout each θ perturbation, so differentiating the distribution introduces no extra normalization terms. Scores are unchanged, and reward and maximum score expectations are divided by $\mathbb{E}_{P_m}[X^*]$; all limits and uniform bounds still hold. Continue to write P_m for the normalized base distribution.

14.3 Completing the lower bound

The statistical criterion requires a negative mixed response to the distribution and evaluation weight. At criticality, stationarity relates this response to the change in expected stopping time. We establish that relation first, then apply the finite approximation results.

Fix a critical limiting instance P with cap H, Poisson mark measure μ, $\mathbb{E}_P[X^*] = 1$ and weights $(w, 1-w)$, where $0 < w < 1$, and let π_0 be its optimal threshold policy. Consider a distributional perturbation P^θ with $P^0 = P$, keeping π_0 fixed as θ varies near zero.

At an interior critical weight, the nonnegative scalar gap and its derivative with respect to the weight both vanish. Since $\mathbb{E}_P[X^*] = 1$, the scalar benchmark is independent of the weight. Thus

$$w\mathbb{E}_P[X_{\tau_{\pi_0}}] + (1-w)\mathbb{E}_P[\tau_{\pi_0}] = C_{\mathrm{PS}}, \qquad \mathbb{E}_P[X_{\tau_{\pi_0}}] - \mathbb{E}_P[\tau_{\pi_0}] = 0,$$

and $\mathbb{E}_P[X_{\tau_{\pi_0}}] = \mathbb{E}_P[\tau_{\pi_0}] = C_{\mathrm{PS}}$. Lemma 3.14 and nonnegativity of the gap give stationarity under distributional perturbations:

$$\frac{d}{d\theta}\left\{w\big(\mathbb{E}_{P^\theta}[X_{\tau_{\pi_0}}] - C_{\mathrm{PS}}\mathbb{E}_{P^\theta}[X^*]\big) + (1-w)\mathbb{E}_{P^\theta}[\tau_{\pi_0}]\right\}\Big|_{\theta=0} = 0.$$

For the target difference

$$B(\theta) = \mathbb{E}_{P^\theta}[X_{\tau_{\pi_0}}] - \mathbb{E}_{P^\theta}[\tau_{\pi_0}] - C_{\mathrm{PS}}[\mathbb{E}_{P^\theta}[X^*] - 1],$$

this identity yields

$$B'(0) = -\frac{1}{w}\,\frac{d}{d\theta}\mathbb{E}_{P^\theta}[\tau_{\pi_0}]\Big|_{\theta=0}.$$

Lemma 3.15 therefore gives the nonzero cross derivative needed by Theorem 13.2, after reversing the direction if necessary.

Use Construction 14.5 on the direct approximants of Construction 14.2. For an individual variable V_i with distribution Q_i, conditional duration is monotone in its value, so a nonzero response is detected by the linear score $h(x) = (x - \mathbb{E}_{Q_i}[V_i])/H$; recenter it under each approximating distribution. For a Poisson band $B = [\delta, H]$ with a positive continuity endpoint δ, the scores of the approximating variables in that construction have total variance at most $2\mu(B)$, and their sum converges to $N_B - \mu(B)$, where N_B is the limiting band's point count.

Lemma 14.7 gives the uniform L^1 second-order likelihood expansion and $O(kt^2)$ training information, including adaptive requests. Lemma 14.6 gives convergence of the reward, duration and maximum score expectations and their uniform stability with regret. These are the derivative and information conditions of Theorem 13.2.

Proof of Proposition 14.1 and Theorem 13.1. Bounded finite instances suffice in the definition of C_{PS}: truncate all values at a common cap and use integrability of the maximum to make both prophet and online reward errors tend to zero. Choose a sequence whose reward ratios tend to C_{PS}, and apply Lemma 3.3, starting from duration weight zero. The clipped scalar ratios still tend to C_{PS} by Lemma 3.1; their expected maxima are one, their caps are uniformly bounded, and their weights stay in a fixed compact interior interval. Compact extraction gives a critical limiting instance whose terminal reward is zero by Lemma 3.12.

The limiting optimal threshold policy and the envelope statements of Lemma 3.14 give stationarity under perturbations of values and permit Lemma 3.15. Choose the resulting score direction on one individual variable or one finite band, reversing its sign as necessary. Direct finite approximants have scalar gaps tending to zero. Apply the optimizer-preserving normalization in (61), keeping it fixed throughout each perturbation family. The caps and positive-level tail counts remain uniformly bounded. Optimal reward, duration, and maximum score expectations converge, so the finite target-difference derivatives are eventually bounded above by a negative constant. The Taylor, information, and uniform derivative-stability bounds are precisely those in Theorem 13.2. Theorem 13.2 gives the inverse-linear lower bound.

□

Proof of Theorem 1.1. Theorem 13.1 gives the lower bound, including adaptive training. Let $K_{\text{out}} > 0$ be the universal sample constant supplied by Theorem 4.1. For accuracy $0 < \varepsilon < 1$, a budget $k = \lceil K_{\text{out}}/\varepsilon \rceil$ samples per distribution gives competitive ratio at least $C_{\text{PS}} - \varepsilon$. This budget is at most $(K_{\text{out}} + 1)/\varepsilon$, and the running time is polynomial in $n, 1/\varepsilon$. Both constants are independent of the number of variables and their distributions. □

Discussion and formalization

15 Discussion

The sharp sample bound is achieved by a polynomial-time algorithm. For every number of variables, $O(1/\varepsilon)$ samples per distribution suffice to achieve competitive ratio $C_{\text{PS}} - \varepsilon$. The matching worst-case lower bound holds even for adaptive sample requests. The learner uses its samples to control both reward and the probability of reaching each variable before stopping. This protects the expected reward from rare, large values absent from the samples.

The same reduction handles rare, large values with $O(1/\varepsilon)$ samples per distribution and enables polynomial-time computation. The planner's cost is exponential in the expected number of positive values, so we call it only when the sample estimate is logarithmic in the budget. Threshold errors change decisions only in short arrival intervals, where mistakes cost little. This makes the expected loss quadratic. At a limit where the weighted benchmark is tight, a distributional perturbation shifts a fixed optimal policy's expected stopping time, helping control calibration error in the upper bound and produce a first-order loss in the lower bound.

Duration calibration replaces the minimax policy selection of Cristi and Ziliotto [12]. For optimal threshold rules, expected stopping time bounds each reach probability from below. This lets validation choose among ordered rules using a single duration target. The planner approximates their continuation values with a weighted Bellman recursion and recursive multilevel Monte Carlo, related to multilevel fixed-point methods [2]. A smaller policy family could reduce the number of queries, and a better estimator could lower the exponent of inverse accuracy in the cost per query. The aim is to keep the same sample budget and competitive ratio. Calibration may extend to other selection problems where one statistic orders the policies and controls several guarantees. The planner could apply to other undiscounted stopping problems if reweighting makes errors shrink uniformly and differences between successive approximations can be sampled efficiently with low variance.

Another direction is the sharp sample complexity of free-order selection. Here the learner chooses the inspection order from its samples before seeing the online values. Cristi and Ziliotto [12] show that $O(\varepsilon^{-5})$ samples per distribution suffice to approach the optimal full-information competitive ratio in this model within ε, independently of the number of variables. Determining the optimal dependence on accuracy would explain how control of the order changes the information needed for selection. Independent uniform arrival times play two roles in our argument: they turn threshold errors into quadratic loss and yield the recursion used to compute continuation values. Extending this approach requires control of both threshold estimation and the choice of order.

16 Formalization

We formalize the main theorem in Lean using mathlib. The development proves the matching sample bounds, including the lower bound for adaptive requests, and verifies the learner's polynomial work bound in the sampling and real-arithmetic model. For rational observations and a rational accuracy parameter, it also verifies the finite-bit learner's reward guarantee and bounds on sample requests, bit operations, and fair random bits. The GitHub repository contains the Lean sources, build instructions, and a guide linking the paper's results to Lean declarations and explaining differences in proof methods.

Declaration of AI use

Our early work supplied the ideas, arguments, and partial results for the sample-complexity analysis. We guided successive mathematical discussions with GPT-6 Astra Ultra, which helped us sharpen the bounds to the $\Theta(1/\varepsilon)$ rate. We supplied ideas for turning our minimax existence argument into an explicit learner. GPT-6 Pro developed those ideas into the polynomial-time construction through mathematical discussions with us. The learner uses duration calibration to choose a stopping policy and a weighted Bellman recursion with multilevel Monte Carlo to approximate continuation values. The learner and planner are described in Sections 11 and 12, with the bit analysis in Appendix G. GPT-6 and Codex assisted with organizing, drafting, and revising the manuscript. Codex also helped draw several figures and assisted with developing and checking the Lean formalization. We take responsibility for the correctness of this work.

A Reward and reach probabilities

A hidden rare jackpot turns a universal reward guarantee into a constraint on reaching its identity. Independent zero arrivals then estimate elapsed time. These arguments also justify treating duration as a reward component.

Proof of Lemma 3.4. We make the jackpot unlikely to appear in training while letting it dominate the prophet's expectation.

Fix a bounded base instance, a variable i, and a learner with competitive guarantee ρ and hard budget k per distribution. For $0 < p < 1$, replace variable i by its base distribution with probability $1 - p$ and a jackpot $A = p^{-2}$ otherwise. Couple each training request to a base sample and an independent jackpot indicator. With at most k requests to this distribution, the training data agree with the base experiment with probability at least $1 - kp$, even under adaptive requests. Reach is decided before the online value is observed, so conditional on a jackpot occurring online its probability is $q_i + o(1)$ as $p \downarrow 0$, where q_i is its base reach probability. The expected maximum is $pA + O(1)$ and the reward is at most $pA(q_i + o(1)) + O(1)$, forcing $q_i \geq \rho$. □

Proof of Lemma 3.5. The fraction of zero arrivals reached estimates elapsed time uniformly over stopping rules.

Let P be the bounded base distribution, append $N \geq 1$ deterministic-zero variables, and fix an augmented-instance policy with stopping time τ. Let F_N be the empirical distribution function of the zero variables' arrival times. Pathwise, the fraction reached differs from τ by at most $\sup_{0\leq t\leq 1} |F_N(t) - t| + N^{-1}$, including the stopping variable. If each zero variable is reached with probability at least ρ, integrating the Dvoretzky–Kiefer–Wolfowitz tail bound [23] gives $\mathbb{E}[\tau] \geq \rho - O(N^{-1/2})$. Simulate these independent arrival times as policy randomness and replace selection of a zero variable by waiting to time 1. The resulting base policy has the same reward and no smaller duration, so the augmented policy satisfies $\mathbb{E}[wX_\tau + c\tau] \leq V(P; w, c)$ for $w, c \geq 0$. □

Proof of Lemma 3.1. Replacing the zero values by vanishingly rare jackpots converts duration payoff into reward.

Fix a bounded product distribution P and weights $w, c \geq 0$. The case $w = 0$ follows by waiting to time 1. For $w > 0$, an integer $N \geq 1$ and $0 < p < 1$, add N new variables, each zero except for a jackpot of probability p and height $c/(wNp)$. First let $p \downarrow 0$ with N fixed. Simultaneous jackpots contribute an error tending to zero. The prophet expectation tends to $\mathbb{E}_P[X^*] + c/w$, while the full-information optimum is at most the supremum over base policies π of

$$\mathbb{E}[X_{\tau_\pi}] + \frac{c}{wN} \sum_{j=1}^{N} q_j(\pi) + o(1).$$

Here $q_j(\pi)$ is the probability that π reaches new variable j. These base policies observe the identities and arrival times of all zero-valued variables. The empirical-distribution bound used in Lemma 3.5 gives $N^{-1}\sum_j q_j(\pi) \le \mathbb{E}[\tau_\pi] + O(N^{-1/2})$, bounding the supremum by $V(P;1,c/w) + O(c/(w\sqrt{N})) + o(1)$. The competitive guarantee bounds the augmented optimum below by $C_{\mathrm{PS}}[\mathbb{E}_P[X^*] + c/w]$ in the limit. Let $N \to \infty$ and multiply by w. This proves the scalar inequality; taking $c = 0$ shows that its constant is sharp. □

Proof of Lemma 3.2. The upper bound needs a reach guarantee from an expected-duration bound. Threshold monotonicity provides this reverse comparison.

Fix a policy with stopping time τ, expected duration $T = \mathbb{E}[\tau]$, and reach and selection probabilities q_i, p_i^{sel} at variable i. Run the policy with i kept pending in the unseen set, and let σ be this run's stopping time, with $\sigma = 1$ if it selects nothing. Then σ is independent of (U_i, V_i); the actual run reaches i exactly when $U_i \le \sigma$ and stops no earlier than $\min(U_i, \sigma)$. Hence

$$q_i = \mathbb{E}[\sigma], \qquad T \ge \mathbb{E}[\min(U_i,\sigma)] = \mathbb{E}[\sigma - \sigma^2/2] \ge q_i/2.$$

Under the stated threshold monotonicity, $\tau \le \sigma$: after rejecting i, the actual run has no larger thresholds than the pending run, and selecting i gives $\tau = U_i$. Therefore

$$q_i - T \ge \mathbb{E}\left[\mathbf{1}_{\{i \text{ selected}\}}(\sigma - U_i)\right].$$

Condition on all randomness except (U_i, V_i), and let $a(t)$ be the conditional selection probability given $U_i = t$. Write $A = \int_0^\sigma a(t)\,dt$. Since $0 \le a \le 1$, the integral $\int_0^\sigma (\sigma - t)a(t)\,dt$ is minimized by placing mass A nearest σ, and is at least $A^2/2$. Since $\mathbb{E}[A] = p_i^{\mathrm{sel}}$, averaging and Jensen give (1).

Extra unseen options cannot lower optimal continuation values, giving the required threshold monotonicity. For an individual variable of a limiting instance, keep i unseen and apply the same threshold rule at all other arrivals, using their observed identities and values. Define σ as the infimum of the times at which this rule prescribes acceptance, with empty infimum 1. This measurable quantity depends only on the other variables, the Poisson process and the policy randomness, so it is independent of (U_i, V_i).

Admissibility under the evaluation distribution makes the actual acceptance infimum equal to the realized stopping time. Before U_i, the actual and pending histories agree. Since U_i is continuous and independent of σ, this gives the same reach event $\{U_i \le \sigma\}$ up to a null event, and the lower bound $\tau \ge \min(U_i, \sigma)$. At each pending acceptance time, the actual rule has already stopped or also accepts, since removing i can only lower its threshold. Taking infima gives $\tau \le \sigma$, even when the pending infimum is not attained. The preceding integration and Jensen argument applies under the original product distribution. Rescaling time gives the remaining-horizon bounds. □

B Bounded normalization

The proof compares the original optimum with a capped problem at the same value. At this value, the optimal threshold policy accepts every reached cap. Running that policy on the original distributions recovers tail reward with the corresponding reach probabilities. Since reach is at least expected duration, this recovered reward pays for increasing the duration coefficient by the reward coefficient times the total clipped tail expectation. The resulting instance has no larger optimal value.

Proof of Lemma 3.3. Use the distribution P, weights w, c, cap h, capped distribution P^h, duration coefficient c_h and ratio r_0 from Lemma 3.3. The original optimal value is wh. Write $e_i = \mathbb{E}[(V_i - h)_+]$. For $d \ge 0$, set $f(d) = V(P^h; w, d)$. Capping gives $f(0) \le wh$. If $f(c_h) > wh$, continuity gives $0 \le d < c_h$ with $f(d) = wh$.

An optimal threshold policy π_h for $f(d)$ accepts every reached cap, including ties: after any arrival, its additional continuation value is at most $f(d) = wh$, since both the unseen set and the remaining duration

coefficient have decreased. Write $T = \mathbb{E}_{P^h}[\tau_{\pi_h}]$ and $q_i = q_i(P^h, \pi_h)$ for its duration and reach probabilities under the capped distribution. Run it on the original instance after capping observations. Every reached cap is accepted, and reach is independent of the variable's value, so the expected reward gain is $\sum_i q_i e_i$. Its expected reward under the original distribution is therefore $\mathbb{E}_{P^h}[X_{\tau_{\pi_h}}] + \sum_i q_i e_i$. By Lemma 3.2, its scalar payoff satisfies

$$w\mathbb{E}_{P^h}[X_{\tau_{\pi_h}}] + cT + w\sum_i q_i e_i \geq w\mathbb{E}_{P^h}[X_{\tau_{\pi_h}}] + c_h T = f(d) + (c_h - d)T > wh.$$

The final inequality uses $T > 0$, since the instance has finitely many uniform arrival times. This contradiction proves $V(P^h; w, c_h) \leq wh$.

It remains to normalize the expected maximum and keep both weights away from zero. Write $D = w\mathbb{E}_{P^h}[X^*] + c_h$ for the capped benchmark. The pathwise bound on the maximum gives $D \geq (w\mathbb{E}_P[X^*] + c)$, while waiting and $\mathbb{E}_{P^h}[X^*] \leq h$ give $c_h \leq wh$ and $w\mathbb{E}_{P^h}[X^*] \leq wh$. In particular, $w\mathbb{E}_{P^h}[X^*] = D - c_h \geq (w\mathbb{E}_P[X^*] + c) - wh > 0$. Scale the variables by $1/\mathbb{E}_{P^h}[X^*]$ and use weights $\bar{w} = w\mathbb{E}_{P^h}[X^*]/D$ and $\bar{c} = c_h/D$. Their sum is one, each is at most $wh/D \leq r_0$, and the new ratio is at most r_0. Since the weights sum to one, the same upper bound also bounds each weight away from zero. The normalized cap satisfies

$$\frac{h}{\mathbb{E}_{P^h}[X^*]} = \frac{wh}{w\mathbb{E}_{P^h}[X^*]} \leq \frac{wh}{(w\mathbb{E}_P[X^*] + c) - wh} = \frac{r_0}{1 - r_0},$$

which proves the normalization bounds. □

C Scalar curvature

Proof of Theorem 3.6. Fix a product distribution $P = \bigotimes_{i=1}^n P_i$ with values in $[0, H]$, and abbreviate $V(P; \lambda, 1-\lambda)$ as $V(\lambda)$ for $\lambda \in (0, 1)$. Changing the weight moves the acceptance thresholds. Integrating over a candidate's random arrival time turns this motion into a curvature contribution at each threshold crossing. We charge these contributions to later selections; since at most one variable is selected, the final bound is independent of the number of variables.

State values and regularity. For an unseen set $A \subseteq \{1, \ldots, n\}$ and horizon $u \in (0, 1]$, apply the payoff-process notation with time measured from this state's start. Let $\tau_A \in [0, u]$ be the additional stopping time of the optimal threshold policy defining $v_A^P(u; \lambda, 1-\lambda)$, and put $T_A = \mathbb{E}[\tau_A]/u$ for its normalized additional duration. Suppressing the arguments of v_A, at a point of differentiability,

$$v_A = \lambda\mathbb{E}[X_{\tau_A}] + (1-\lambda)uT_A, \qquad \partial_\lambda v_A = \mathbb{E}[X_{\tau_A}] - uT_A, \qquad \partial_u v_A = (1-\lambda)T_A. \tag{62}$$

A candidate value $x \in [0, H]$ whose rejection leaves continuation state (A, u) is accepted when $\lambda x \geq v_A^P(u; \lambda, 1-\lambda)$. The optimizer rejects zero except at null terminal events. Along an arrival path with the candidate excluded, followed without stopping, continuation decreases continuously between arrivals and jumps downward when variables are removed.

For now, constants may depend on n, H and a compact scalar interval in $(0, 1)$. A zero-rejecting policy waits at least until the first positive arrival, so every state's normalized duration is at least $1/[4(n+1)]$. Its continuation has derivative with respect to the remaining horizon at least $(1-\lambda)/[4(n+1)]$.

Changing λ to $\lambda + h \in (0, 1)$ changes continuation by at most $(H+1)|h|$. Actions can therefore disagree only within $O_H(|h|)$ of the old acceptance boundary. This derivative bound limits this interval to length $O_{n,H}(|h|)$, with Bellman loss $O_H(|h|)$. Write π_λ for the optimal threshold policy at weight λ. Bounding reach probabilities by one and summing candidates yields

$$0 \leq V(\lambda + h) - V(\lambda) - h\mathbb{E}[X_{\tau_{\pi_\lambda}} - \tau_{\pi_\lambda}] \leq K_{n,H}|h|^2$$

locally uniformly in λ, proving local $C^{1,1}$ regularity at every continuation state. Conditional on other arrival times and the candidate value, a continuous threshold equality occurs at at most one arrival time, so optimal outcomes are almost surely unique. Bounded convergence gives (62) and continuity of its derivatives at positive horizons.

The boundary Hessian formula. Only candidates whose acceptance decisions change contribute to curvature. We locate their threshold crossings and compute how fast the acceptance boundary moves when the weight changes.

Hold variable i pending and follow the other arrival times without stopping. For a candidate value $x > 0$ and time $t \in [0, 1]$, write $A_i(t)$ for the other unseen variables and $g_i(t, x; \lambda) = \lambda x - v^P_{A_i(t)}(1-t; \lambda, 1-\lambda)$. This gap increases, with upward jumps. Let $\sigma_i(x)$ be the first time at which g_i is nonnegative, $u_i(x) = 1 - \sigma_i(x)$, and $K_i(x)$ the event that this first crossing is continuous and in $(0, 1)$. Let $T_i(x)$ be the continuation policy's normalized duration there and $B_i(t)$ the indicator that the optimal threshold policy with i pending has not stopped before t. These crossing quantities are measurable from the history of the other arrivals and observed values; in particular, $K_i(x)$ is determined by the history at the crossing.

At a continuous crossing, put $A = A_i(\sigma_i(x))$ and $u = u_i(x)$. Then $v^P_A(u; \lambda, 1-\lambda) = \lambda x$, and (62) gives the exact identities

$$\partial_\lambda g_i = x - (\mathbb{E}[X_{\tau_A}] - uT_A) = \frac{uT_A}{\lambda}, \qquad \partial_t g_i = (1-\lambda)T_A. \tag{63}$$

These identities give, for almost every λ,

$$V''(\lambda) = \sum_{i=1}^{n} \int \mathbb{E}\left[\mathbf{1}_{K_i(x)} B_i(\sigma_i(x)) \frac{u_i(x)^2 T_i(x)}{\lambda^2(1-\lambda)}\right] P_i(dx). \tag{64}$$

Here and below zero values contribute zero.

To justify the formula for arbitrary distributions of values, decompose the old optimizer's regret under the new objective into reached Bellman losses. Conditional on the other arrival times and x, its reach indicator is constant between consecutive other arrivals. Near a continuous crossing, local $C^{1,1}$ regularity gives the new gap $g_i + h\partial_\lambda g_i + \rho_h$, with $|\rho_h| \le K_{n,H} h^2$ uniformly over states and times. The disagreement interval and the loss on it are both $O(|h|)$. Integrating this triangular loss gives

$$\frac{h^2}{2} B_i(\sigma_i(x)) \frac{(\partial_\lambda g_i)^2}{\partial_t g_i} + o(h^2).$$

The positive lower bound on the derivative confines disagreement to an $O_{n,H}(|h|)$ interval with $O_H(|h|)$ loss. The remainder is bounded by $K_{n,H} h^2$ times the sum of $|h|$ and the oscillations of $\partial_t g_i$ and $\partial_\lambda g_i$ on this interval. Their continuity at positive horizons makes it $o(h^2)$ pointwise. The identities in (63) also bound the quotient uniformly on the compact scalar interval. Dominated convergence thus permits summing variables and integrating arbitrary P_i.

Strict jump crossings contribute nothing. Equality at another arrival has probability zero because, conditional on the remaining arrival times, continuation is strictly monotone in that continuously distributed arrival time. At time zero, $v^P_A(1; \lambda, 1-\lambda)/\lambda$ strictly decreases with λ, since every policy has positive normalized duration. Each atom of P_i can therefore create an endpoint root at only one scalar; the nonatomic part gives each root zero mass. Endpoint events do not affect the almost-everywhere formula. Local $C^{1,1}$ regularity now proves (64).

Charging crossings and taking limits. A threshold crossing makes a later arrival of the candidate acceptable. Comparing survival in the pending run and the continuation run lets us charge its curvature contribution to that selection probability.

At a crossing in $K_i(x)$, couple the surviving pending-i policy to the optimal continuation policy with i removed. The former has higher thresholds and survives whenever the latter does. Conditional on the history at the crossing, the remaining arrival times are independent uniforms on the remaining interval. Hence

$$\mathbb{E}\left[\int_{\sigma_i(x)}^{1} B_i(t)\,dt \,\middle|\, \text{history at the crossing}\right] \geq B_i(\sigma_i(x))\,u_i(x)T_i(x). \tag{65}$$

After crossing, a reached value x is accepted. Its independent arrival time has density one, so its conditional selection probability $a_i(x)$ is at least $\mathbb{E}[\mathbf{1}_{K_i(x)} \int_{\sigma_i(x)}^{1} B_i(t)\,dt]$. By (65) and $u_i \leq 1$,

$$a_i(x) \geq \mathbb{E}\left[\mathbf{1}_{K_i(x)} B_i(\sigma_i(x)) u_i(x)^2 T_i(x)\right]. \tag{66}$$

There is at most one selection, so $\sum_i \int a_i(x) P_i(dx) \leq 1$. Substituting (66) into (64) now gives (2).

For a compact interval $I \subset (0,1)$, set $K_I = \sup_{\lambda \in I} 1/[\lambda^2(1-\lambda)]$. Integrating the curvature bound gives

$$0 \leq V(\beta) - V(\alpha) - V'(\alpha)(\beta - \alpha) \leq \frac{K_I}{2}(\beta - \alpha)^2, \qquad \alpha, \beta \in I.$$

The symmetric second-difference version passes to every pointwise limit of these value functions on I, and convexity then gives the same local $C^{1,1}$ bound for the limit. □

Proof of Corollary 3.7. Use the distribution P, weight $\lambda \in I$, optimizer π_* and competing policy π with regret r from Corollary 3.7. Write $B_\pi = \mathbb{E}[X_{\tau_\pi}] - \mathbb{E}[\tau_\pi]$ for a policy's reward–duration slope. For a sufficiently small $h > 0$, compare the affine payoff of π with the optimum at weights $\lambda + h$ and $\lambda - h$, in a slightly larger compact interval in $(0,1)$. Theorem 3.6 gives $|B_\pi - B_{\pi_*}| \leq r/h + Lh/2$ for a uniform curvature constant L. Choose h of order $\sqrt{r}$ for small positive regret, pass to the limit for zero regret, and use the bounded reward and duration for larger regret. □

D Stability of policy outcomes

Proof of Theorem 3.8. Use the family, positive weights w, c, coupled policies π, π^* and scalar regret r from Theorem 3.8. For each $d > 0$, write $\overline{L}_d = \sup_P L_d(P) < \infty$ for the family's uniform expected tail count. Before a short terminal window, disagreement either incurs Bellman loss or lies near an acceptance boundary. Positive-level tail counts control these boundary events and the reward in the terminal window uniformly as the number of variables grows.

Stopping-time stability. We first restrict attention to positive candidate values bounded away from zero. Near their acceptance thresholds, continuation has positive expected duration and therefore moves with time at a uniformly positive speed.

Fix $b > 0$ small enough that $u_0 := 2wb/c < 1$, and take $0 < e \leq wb/4$. At a continuation state with horizon $u \in (0,1]$, apply the payoff-process notation with time measured from the state's start. Let $\tau_* \in [0,u]$ be the optimal threshold policy's additional stopping time and put $T = \mathbb{E}[\tau_*]/u$. Write v for its continuation value and q_j for its reach probability at a remaining variable j. Its optimal expected reward satisfies

$$\mathbb{E}[X_{\tau_*}] \leq b/4 + H \sum_j q_j P_j(V_j > b/4) \leq b/4 + 2TH\overline{L}_{b/4}.$$

Here $q_j \leq 2T$ by Lemma 3.2, so $v \leq wb/4 + T(2wH\overline{L}_{b/4} + cu)$. If a candidate value $x \geq b$ satisfies $|wx - v| \leq e$, then $v \geq 3wb/4$ and

$$T \geq \frac{wb}{2(2wH\overline{L}_{b/4} + c)}.$$

Thus the derivative cT with respect to the remaining horizon is uniformly positive on this band. Continuation jumps downward upon variable removal, so an arrival path with the candidate excluded spends at most $K_b e$ time within gap e of acceptance, for a constant K_b uniform over the family. Summing over values with $\sum_i P_i(V_i \geq b) \leq \overline{L}_{b/2}$ gives

$$\Pr(\text{some arrival with } x \geq b \text{ and } |wx - v| \leq e) \leq K_b e. \tag{67}$$

The union bound averages over each candidate's full arrival-time distribution, so it also covers policies using zero-valued arrivals or extra randomness.

The policies share an observed history until their first disagreement. At an arrival before $1 - u_0$, the remaining horizon is $u > u_0$, and accepting a value $x < b$ loses at least wb, since continuation is at least $cu > 2wb$. A larger value either lies in the boundary event above or incurs loss at least e. The performance-difference identity bounds the probability of loss at least e by r/e. Thus (67) gives

$$\Pr(\text{first disagreement before } 1 - u_0) \leq r/e + K_b e.$$

On a later first disagreement, the stopping times differ by at most u_0. Therefore

$$\mathbb{E}[|\tau_\pi - \tau_{\pi^*}|] \leq u_0 + r/e + K_b e. \tag{68}$$

Reward stability. For reward stability, we also bound how much value can arrive during the short terminal window. If the first disagreement is late, both rewards come from the terminal window of length u_0. Their absolute difference is at most the largest value arriving in that window. For any $d > 0$,

$$\mathbb{E}\left[\max_{i:\, U_i > 1-u_0} V_i\right] \leq d + H \sum_i \Pr(V_i > d,\ U_i > 1 - u_0) \leq d + H u_0 \overline{L}_d.$$

Together with the early-disagreement bound, this gives

$$\mathbb{E}[|X_{\tau_\pi} - X_{\tau_{\pi^*}}|] \leq H(r/e + K_b e) + d + H u_0 \overline{L}_d. \tag{69}$$

For fixed b, d, let $r, e \to 0$ with $r/e \to 0$, then let $b \to 0$, and finally $d \to 0$. (68) and (69) prove the uniform modulus in (3); bounded payoffs handle larger regret. □

Proof of Corollary 3.9. The bound follows from Cauchy–Schwarz, Theorem 3.8, and $|X_{\tau_\pi} - X_{\tau_{\pi^*}}| + |\tau_\pi - \tau_{\pi^*}| \leq H + 1$. □

E Limits of bounded instances

The compactness argument must allow the number of variables to grow while accounting for what a policy observes. Rare independent contributions become Poisson arrivals, and values available arbitrarily late give a terminal reward. We first justify truncation and the Poisson comparison, which give extraction of a limit and convergence of optimal values. We then check closure under the perturbations used by both bounds, rule out a terminal reward at criticality, and construct an optimal threshold policy. We use the convergence from Preliminaries; each argument specifies its reward cap and admissible weights.

Policies on the full limiting instance. We first check that the truncation convention captures all admissible policies, including those that use identities omitted by the truncation.

Fix a limiting instance with data $(a, (Q_i), \mu)$ as in Definition 2.1, and let V_i denote its individual variables with distributions Q_i. Write $V_{\text{full}}(w, c)$ for the supremum over all admissible policies of the full instance, and $V_{N,b}(w, c)$ for the value of the truncated instance consisting of the first N individual variables, the Poisson marks above $b > a$, and terminal reward a. Every truncated policy extends to the full instance by ignoring

omitted observations. Conversely, simulate the omitted variables and low Poisson points independently as auxiliary randomness. Feed a full-instance policy only the reconstructed past, including zero-valued arrivals and their identities. If it selects an omitted value, wait until time 1 and collect a. This simulation is causal, its duration is no smaller, and its reward loss is at most the omitted excess maximum plus $b - a$. Hence, for $w, c \geq 0$,

$$0 \leq V_{\mathrm{full}}(w,c) - V_{N,b}(w,c) \leq w\left(\mathbb{E}\left[\sup_{i>N}(V_i - a)_+\right] + b - a\right).$$

For every $\eta > 0$, the expectation on the right is at most $\eta + H\sum_{i>N}\Pr(V_i > a + \eta)$. Summability makes it vanish as $N \to \infty$ and then $\eta \downarrow 0$. Letting $b \downarrow a$ identifies the truncation limit with the full policy supremum.

Proof of Lemma 3.10. We compare the Bellman values through the arrival intensities seen from the observed history. This lets the comparison account for the information revealed by zero arrivals.

Fix scalar weights $w, c \geq 0$, and use the positive-value measures ν_i and probabilities p_i from Lemma 3.10. Conditional on the finite arrivals observed by time $t \in [0,1)$, the variables being replaced have predictable positive-value measure $\nu_t = (1-t)^{-1}\sum_{i:U_i>t}\nu_i$. Set $\nu = \sum_i \nu_i$. Independence of the arrival times gives, for every $x \in [0,H]$,

$$\mathbb{E}\left[|\nu_t((x,H]) - \nu((x,H])|\right] \leq \left(\frac{t}{1-t}\sum_i \nu_i((x,H])^2\right)^{1/2}.$$

Consequently

$$\mathbb{E}\left[\int_0^1\int_0^H |\nu_t((x,H]) - \nu((x,H])|\,dx\,dt\right] \leq \frac{\pi}{2}H\Big(\sum_i p_i^2\Big)^{1/2}. \tag{70}$$

Use the Poisson Bellman value, indexed by the unseen subset of individual variables, as a potential on the finite process. Write v for the applicable continuation value. Waiting guarantees the terminal reward, so the gain $(wx - v)_+$ vanishes at and below that reward. This lets us use $\nu_t - \nu$ in the gain integral. The gain is w-Lipschitz in x, and integration by parts bounds the integral by w times the inner integral in (70). The generator terms for the individual variables left unchanged and for running rewards agree in the two models.

For $0 < \delta < 1$, up to time $1 - \delta$ the stopped drift is this residual minus action losses. Bound the residual by its absolute value, then drop the stopping indicator. This bound applies even when the policy uses the observed arrival history. Let $\delta \downarrow 0$ using (70) and bounded terminal payoffs. Every finite policy has expected scalar payoff at most the Poisson value plus the error; running the Poisson threshold policy removes action losses and proves the reverse bound. □

Proof of Lemma 3.11. We separate variables that persist individually from rare variables that contribute collectively. The level at which arrivals become abundant determines the terminal reward.

Let P_n be the given sequence with common cap H, and write P_{ni} and V_{ni} for its marginal distributions and values. Fix weights $w, c \geq 0$ and put $v_n = V(P_n; w, c)$. For nonnegative tail levels x, set $L_n(x) = \sum_i \Pr(V_{ni} > x)$ and extract extended limits on a countable dense subset of $[0,H]$. Let $a \in [0,H]$ separate divergence from boundedness. Intermediate dense-set levels give $L_n(x) \to \infty$ below a and $\sup_n L_n(x) < \infty$ above it. At each fixed level below a, diverging tail counts make a value above that level appear with probability tending to one even in a short terminal interval; this is the source of the terminal reward. A deterministic individual variable can give the maximum a positive lower endpoint while contributing no terminal reward.

The next ordering makes one initial set capture the variables with substantial exceedance probabilities simultaneously across positive levels. Choose levels $b_\ell > a$ decreasing to a, bounds $B_\ell \geq \max\{\sup_n L_n(b_\ell), 1\}$, and positive coefficients α_ℓ summing to one. Order variables by decreasing $s_{ni} = \sum_\ell(\alpha_\ell/B_\ell)\Pr(V_{ni} > b_\ell)$. Since $\sum_i s_{ni} \leq 1$, we have $s_{ni} \leq 1/i$ and $\Pr(V_{ni} > b_\ell) \leq B_\ell/(\alpha_\ell i)$. Pad with zeros and extract weak limits $P_{ni} \Rightarrow Q_i$ variable by variable and a vague aggregate limit on $(a,H]$. Let V_i have distribution Q_i. Subtracting

the individual-variable measures from the aggregate limit gives a nonnegative residual measure μ: every finite partial sum over individual variables is dominated by the aggregate limit. All these measures are locally finite above a.

Choose truncation levels $b > a$ avoiding the at most countably many atoms of the aggregate measure and the distributions of the individual variables. For any $b_\ell < b$,

$$\sum_{i>N} \Pr(V_{ni} > b)^2 \le \frac{B_\ell^2}{\alpha_\ell (N+1)}.$$

Lemma 3.10 therefore replaces all residual variables above b by a Poisson process with an error tending to zero uniformly in n as N grows.

For fixed N, b, couple the converging distributions of the individual variables in L^1 and the finite Poisson measures, first truncating the Poisson count. Conditional simulation of coupled values at arrival lets us run a policy from either model in the other without revealing future information: duration is unchanged and summed differences in values bound the reward error. Remove the count truncation to obtain value continuity. Summability of the exceedance probabilities of the individual variables above b makes the omitted maximum vanish as $N \to \infty$.

A short fallback near the end realizes the terminal reward through values available near the end. Let v_n^b be the scalar value at weights w, c after setting values at most b to zero and adding terminal reward b. Keep each arrival's identity and clock. A policy can simulate original values conditional on these observations and replace acceptance of a value at most b by waiting for the terminal reward. Thus $v_n^b \ge v_n$. If $a > 0$, choose $\delta \in (0, a)$ and $\eta \in (0, 1)$. Follow a v_n^b optimizer until $1 - \eta$, then take the first value above $a - \delta$ if still active. This gives

$$\begin{aligned} v_n \ge v_n^b &- w(b - a + \delta) - wH\eta L_n(b) \\ &- wH \exp[-\eta L_n(a - \delta)] - c\eta. \end{aligned}$$

The probability that a value above b arrives after $1 - \eta$ is at most $\eta L_n(b)$; the exponential bounds the unconditional probability that no fallback value arrives. Send $n \to \infty$, then $\eta, \delta \downarrow 0$, then $b \downarrow a$. For $a = 0$, mimicking the above-b policy with terminal reward zero loses at most wb.

For another cutoff $b' > a$, changing both cutoff and terminal reward from b to b' changes value by at most $w|b - b'|$, yielding the instance with terminal reward a as the cutoffs decrease. The same truncation proves convergence of maxima: the limiting maximum's cumulative distribution function above a is $F(x) = \prod_i Q_i([0, x]) \exp[-\mu((x, H])]$. If $a = H$, the fallback argument gives value $wH + c$ and maximum H directly. Bounded payoffs give uniform Lipschitz continuity in the weights, so dense-set convergence is uniform on compact weight sets. □

The perturbations below change a limiting instance and then apply the global scalar inequality again. We therefore need the changed instance to remain a limit of finite instances. The next lemma establishes this closure, including the contractions and deletions used in Appendix F.

Lemma E.1. *Let finite instances P_n on $[0, H]$ converge, in the sense of Definition 2.2, to a limiting instance $\mathfrak{B}$ with terminal reward a, individual variables V_i with distributions Q_i, and Poisson intensity $dt\, \mu(dx)$. Write P_{ni} and V_{ni} for the marginal distributions and values of P_n. Let $\mathfrak{B}'$ be obtained from $\mathfrak{B}$ by finitely many of the following operations, all with values in a fixed $[0, H']$:*

(a) replace finitely many distributions of the individual variables Q_i by distributions Q_i' for which there are approximating distributions $Q_{ni}' \to Q_i'$ weakly;

(b) replace the Poisson marks above a continuity level $b_0 > a$ by their images under a possibly randomized map Φ, using an independent random seed ξ for each mark, such that the distribution of $\Phi(x, \xi)$ varies

weakly continuously with x at μ-almost every point, discarding image marks at or below a; or multiply the Poisson intensity on a μ-continuity band bounded away from a by $1+\theta$, where $|\theta| \le 1/4$;

(c) if $a > 0$, apply $x \mapsto (x-h)_+$, $0 < h < a$, to every value and lower the terminal reward to $a - h$.

These operations have finite product approximations: apply (a) to the corresponding variables, the value transformations in (b) to a residual set of variables $i > N_n$ for integer cutoffs N_n chosen along a diagonal, and (c) to all variables. For an intensity tilt, multiply each residual variable's band mass by $1+\theta$ and its complementary mass by $1 - \theta p/(1-p)$, where p is its band probability. The resulting finite instances P'_n converge to $\mathfrak{B}'$. For all weights $w, c \ge 0$, one has $V(P'_n; w, c) \to V(\mathfrak{B}'; w, c)$ and $\mathbb{E}_{P'_n}[X^*] \to \mathbb{E}_{\mathfrak{B}'}[X^*]$, and Lemma 3.1 gives

$$V(\mathfrak{B}'; w, c) \ge C_{\mathrm{PS}}[w\mathbb{E}_{\mathfrak{B}'}[X^*] + c] \qquad (w, c \ge 0).$$

Proof. Fix weights $w, c \ge 0$, and write $L_n(x) = \sum_i \Pr(V_{ni} > x)$ for the original tail counts. For (a), the replacement distributions converge weakly by assumption. In particular, the continuous mapping theorem applies to kernels weakly continuous Q_i-almost everywhere, and bounded convergence applies to bounded continuous likelihood tilts. Replacing a variable by its mean also satisfies these conditions: bounded weak convergence gives $\mathbb{E}_{P_{ni}}[V_{ni}] \to \mathbb{E}_{Q_i}[V_i]$, so the corresponding point masses converge.

For (b), Definition 2.2 gives residual-measure convergence for each fixed initial set of individual variables and continuity level $d > a$; the individual variables have summable exceedance probabilities there. Diagonalizing over countably many levels and continuous test functions gives $N_n \uparrow \infty$ so that residual measures converge to μ and $\max_{i>N_n} \Pr(V_{ni} > d) \to 0$ for every $d > a$.

Almost-everywhere continuity of the mark kernel preserves finite-measure convergence. An image above $d > a$ comes from a mark above $\min(d, b_0) > a$, so the maximum individual exceedance probability still tends to zero. Tail counts below a still diverge because the total mass changed is bounded.

For an intensity tilt on a band B bounded away from a, put $p_{ni} = P_{ni}(B)$ for $i > N_n$. These probabilities tend uniformly to zero, so the stated complementary scaling defines a probability distribution for all large n. Above any fixed $d > a$, its compensating change has total variation at most

$$|\theta| \max_{i>N_n} \frac{p_{ni}}{1-p_{ni}} \sum_{i>N_n} P_{ni}(V_{ni} > d) \longrightarrow 0.$$

Thus the limiting band intensity is multiplied by $1+\theta$, while other positive intensities stay unchanged. The probability multipliers are bounded, so the rare-variable condition above a and divergent tail counts below a still hold.

In (c), the changed tail counts satisfy $L'_n(y) = L_n(y+h)$ for $y \ge 0$, so they diverge for $0 \le y < a - h$ and stay bounded for $y > a - h$, while individual-variable values and Poisson marks are shifted continuously.

With cap H', the three comparisons from Lemma 3.11 give convergence of both scalar values and expected maxima: causal Poissonization of residual variables, weak continuity for finite initial sets of individual variables, and terminal-reward approximation. Passing Lemma 3.1 to the limit gives the final inequality. □

Proof of Lemma 3.12. Use the critical instance $\mathfrak{B}$, weights $w > 0, c \ge 0$ and terminal reward a from Lemma 3.12. If $a > 0$, restrict to policies that accept values above a or wait for the terminal reward. All individual identities are still observed. Apply Lemma E.1(c) with $h \in (0, a)$. From the shifted observations, recover original cloud marks and simulate original individual values from their conditional distributions. The forward transformation applies the deterministic shift. Corresponding policies have the same duration, and their reward in the shifted instance is lower by h. Thus the scalar value falls by wh, the expected maximum by h, and the zero critical gap by $w(1 - C_{\mathrm{PS}})h > 0$, contradicting Lemma E.1(c). □

Proof of Lemma 3.13. Convergence of values alone does not supply an optimizing policy. We construct one by showing that the limiting thresholds give unambiguous decisions and attain the limiting value.

Fix the limiting instance of Lemma 3.13, with cap H, individual variables V_i with distributions Q_i, Poisson mark measure μ, zero terminal reward and weights $w, c > 0$. For an integer $N \geq 1$ and cutoff $b > 0$, omitting individual variables beyond N costs at most $w\mathbb{E}[\max_{i>N} V_i]$; omitting Poisson marks at most b costs at most wb, uniformly over states and horizons. Indeed, simulate omitted observations independently as policy randomness and replace their selection by waiting; duration increases and reward decreases by at most the omitted maximum. Since

$$\mathbb{E}\left[\max_{i>N} V_i\right] \leq d + H \sum_{i>N} \Pr(V_i > d) \longrightarrow d \qquad (d > 0),$$

that maximum tends to zero. Continuation values of truncated instances converge uniformly over remaining states to a measurable limit, monotone in the horizon and unseen subset of individual variables.

To turn the limiting continuations into an optimizer, we need to rule out ties at positive thresholds with positive probability. We establish a positive lower bound on the derivative with respect to the remaining horizon first, using the optimizers of the truncated instances. At a state with horizon $u > 0$, write T for the optimal threshold policy's expected additional duration in a truncated instance divided by u, and q_i for its reach probability at individual variable i. For a positive level b, its expected reward is at most

$$b/4 + 2TH\left(\sum_i \Pr(V_i > b/4) + \mu((b/4, H])\right).$$

For individual variables use $q_i \leq 2T$. The Poisson process contributes an expected number $\mu((b/4, H])uT$ of reached band points by the compensator identity, which follows from the Mecke identity [22, Theorem 4.1]. When continuation values are bounded away from zero these bounds give a uniform positive lower bound on T. Increasing the horizon adds duration derivative cT and more Poisson opportunities. The resulting derivative bound passes to the uniform limit. Conditional on other arrival times, positive acceptance equalities have Lebesgue-null candidate times. Mecke's identity gives the same conclusion for Poisson candidates by conditioning on an inserted point.

Use these limiting thresholds. For $0 < \eta < 1$, before $1 - \eta$ they are at least $c\eta$, so only finitely many potentially acceptable occurrences appear almost surely, defining the rule unambiguously on $[0, 1)$. Uniform threshold convergence and the absence of equalities give convergence of decisions of policies for the truncated instances; omitted observations from individual variables in the relevant positive band have vanishing probability. For any $d > 0$, the terminal window changes duration by at most η and has expected maximum at most $d + H\eta L_d$, where $L_d = \sum_i \Pr(V_i > d) + \mu((d, H]) < \infty$ is the limiting expected tail count. Refine the truncated instance, then let $\eta \downarrow 0$ and $d \downarrow 0$: the rule attains the limiting value.

At a competing policy's first disagreement, dynamic programming charges its Bellman gap to regret. The derivative bound and terminal bound then give (3), including uniqueness of the selected outcome at zero regret. □

Perturbations of values require knowing which occurrence is selected, since nearby rewards can carry different perturbation tags. The following consequence strengthens outcome stability in this direction.

Corollary E.2. *Fix $b > 0$ and let π_* be the optimal threshold policy for an instance in a family of finite instances satisfying Theorem 3.8, or in a limiting instance with zero terminal reward of Lemma 3.13. Assign each occurrence a uniformly bounded tag depending only on its value and independent auxiliary randomness revealed at that occurrence, and set the tag to zero for values below b. Let Y_π be the tag of the occurrence selected by a competing policy π, or zero if none is selected. Then $\mathbb{E}[|Y_\pi - Y_{\pi_*}|] \to 0$ uniformly as the scalar regret tends to zero.*

Proof. Write $\overline{L}_d$ for a uniform bound on the expected count above $d > 0$, including the Poisson count in a

limiting instance. Different tags require different selected occurrences, at least one with value at least b. If the rewards differ by less than $b/2$, both values exceed $b/2$; for any $\eta > 0$, if the durations differ by less than η, their arrival times are within η. Such a pair has probability at most $K\eta(\overline{L}_{b/2} + \overline{L}^2_{b/2})$, including Poisson pairs. Outcome stability controls the two exceptions. Let regret tend to zero, then $\eta \downarrow 0$. Bounded tags give the conclusion, including independent auxiliary tags. □

F Nondegeneracy at a critical limit

Both sample bounds use a distribution perturbation that changes the fixed optimal policy's expected duration. If every such response vanished, each individual variable could be replaced by its mean and every finite positive band of Poisson marks deleted while preserving criticality. The resulting deterministic instance would contradict $C_{\mathrm{PS}} < \sqrt{3} - 1$. The proof has two distinct tasks: compute derivatives of the optimal value with the base policy fixed, then justify the contractions and deletions without changing either value or expected maximum.

Proof of Lemma 3.14. Use the base limiting instance P with cap H, positive weights w, c and optimal threshold policy π_0 from Lemma 3.14. For real θ near zero, realize the perturbed distributions P^θ, with $P^0 = P$, on a common observation space, using the tags below for perturbations of values. Write $V(\theta) = V(P^\theta; w, c)$. For an admissible policy π, let $r_0(\pi) = V(0) - \mathbb{E}_P[wX_{\tau_\pi} + c\tau_\pi]$ be its base regret. Choose a policy π_θ within θ^2 of optimal. The base optimizer supplies a lower comparison for the change in value, and the perturbed optimizer supplies an upper comparison:

$$\begin{aligned} \mathbb{E}_{P^\theta}[wX_{\tau_{\pi_0}} + c\tau_{\pi_0}] - \mathbb{E}_P[wX_{\tau_{\pi_0}} + c\tau_{\pi_0}] &\le V(\theta) - V(0), \\ V(\theta) - V(0) &\le \mathbb{E}_{P^\theta}[wX_{\tau_{\pi_\theta}} + c\tau_{\pi_\theta}] - \mathbb{E}_P[wX_{\tau_{\pi_\theta}} + c\tau_{\pi_\theta}] + \theta^2. \end{aligned} \tag{71}$$

If $\Delta_\theta(\pi) := \mathbb{E}_{P^\theta}[wX_{\tau_\pi} + c\tau_\pi] - \mathbb{E}_P[wX_{\tau_\pi} + c\tau_\pi] = \theta D(\pi) + o(\theta)$ uniformly in π, with bounded D, then

$$r_0(\pi_\theta) \le \Delta_\theta(\pi_\theta) - \Delta_\theta(\pi_0) + \theta^2 = O(|\theta|).$$

The perturbed optimizer is therefore nearly optimal at the base instance. If $D(\pi) \to D(\pi_0)$ as $r_0(\pi) \to 0$, divide (71) by positive and negative θ to obtain $V'(0) = D(\pi_0)$. This reduces the derivative of an optimized value to the response of one fixed policy.

Outcome stability proves the continuity of D required by this argument. For a bounded individual-variable likelihood tilt with centered score Z, bounded payoffs give the uniform expansion

$$\Delta_\theta(\pi) = \theta\mathbb{E}_P[Z(wX_{\tau_\pi} + c\tau_\pi)] + O(\theta^2), \qquad D(\pi) = \mathbb{E}_P[Z(wX_{\tau_\pi} + c\tau_\pi)].$$

Let ω be the outcome-stability modulus supplied by Lemma 3.13. Cauchy–Schwarz and bounded payoffs give

$$|D(\pi) - D(\pi_0)| \le K\|Z\|_2\sqrt{\omega(r_0(\pi))} \longrightarrow 0,$$

where K depends only on w, c, H. An intensity tilt on a finite positive Poisson band B has the same expansion, with score $Z = N_B - \mu(B)$, where μ is the mark measure and N_B is the band's point count, which has finite moments of every order.

For bounded continuous ϕ supported in $[b, H]$, $b > 0$, and small real h, apply $x \mapsto x + h\phi(x)$ to one individual variable or a finite positive band of Poisson marks. Reveal original values and, for symmetric perturbations, independent fair signs as tags at affected occurrences. Conditional sampling of a tag given its displayed value simulates any policy using tags without revealing future observations. Revealing tags therefore leaves the optimal value unchanged and puts all policies on a common observation space, including at $h = 0$.

For a fixed policy using tags, the stopping time stays the same, and selecting an affected occurrence changes the reward by $h\phi$. Thus $\Delta_h(\pi) = hD(\pi)$, with $D(\pi) = w\mathbb{E}_P[\phi(X_{\tau_\pi})\mathbf{1}_{\{\text{an affected occurrence is selected}\}}]$ and $|D| \le w\|\phi\|_\infty$. Corollary E.2, applied to this bounded tag supported on values at least b, gives $D(\pi) \to D(\pi_0)$ as base regret vanishes, proving the derivative under the perturbation of values.

For symmetric perturbations $x \mapsto x + h\varsigma\psi(x)$, with an independent fair sign ς at each affected occurrence and continuous $0 \le \psi \le 1$ supported in $[b, H]$, write ς_π for the sign at the affected occurrence selected by π, and set it to zero otherwise. The same argument gives $D(\pi) = w\mathbb{E}_P[\varsigma_\pi\psi(X_{\tau_\pi})]$. The base policy ignores the independent signs, so $D(\pi_0) = 0$ and $0 \le V(h) - V(0) \le hD(\pi_h) + h^2 = o(h)$. □

The contraction argument must preserve the prophet's expected maximum as well as the online value. At criticality, stationarity under perturbations of values relates a variable's selection probability to its probability of supplying the maximum. When vanishing duration response makes selection insensitive to a variable's value, this relation shows that mean contraction preserves the expected maximum.

Lemma F.1. *Let $\mathfrak{B}$ be a limit of finite instances in the sense of Definition 2.2, with cap H, individual variables V_i with distributions Q_i, Poisson mark measure μ, zero terminal reward and $\mathbb{E}_{\mathfrak{B}}[X^*] > 0$. Suppose it is critical at weights $w, c > 0$. Its positive maximum is attained by more than one occurrence with probability zero. For each individual variable i, let X^*_{-i} be the realized maximum of all competing values, including the Poisson marks. For the optimal threshold policy,*

$$a_i(x) := \Pr(\textit{select } i \mid V_i = x) = C_{\mathrm{PS}}\ \Pr(X^*_{-i} < x) =: C_{\mathrm{PS}} b_i(x)$$

for almost every supported positive x, with the Poisson process included in the competing maximum.

Proof. The operations of Lemma E.1 preserve nonnegativity of the scalar gap, which is zero at the critical instance. We first exclude tied positive maxima so that the derivative of the expected maximum can be assigned to its unique maximizing occurrence.

Ties. A positive-probability tie between independent values of individual variables, or involving Poisson points, requires a common atom. Thus some $x > 0$ is the maximum of at least two occurrences on an event E of positive probability. After restricting E if needed, symmetrically perturb a fixed participating individual variable, or the Poisson process if all tied occurrences are Poisson points. Choose $0 < \eta < x$ and continuous $0 \le \psi \le 1$ with $\psi(x) = 1$ and support in $[x - \eta, x + \eta] \cap [0, H]$. For small $h > 0$, apply $y \mapsto y + h\varsigma\psi(y)$ with independent fair signs ς. This is Lemma E.1(a) or (b), allowing an enlarged cap.

Conditional Jensen shows that mean-zero perturbations never decrease the expected maximum. On E, one affected tied value gives $\mathbb{E}[\max(x + h\varsigma, x)] = x + h/2$; if two are affected, their independent signs have expected maximum at least $1/2$. Hence $\mathbb{E}[X^*]$ increases by at least $h\Pr(E)/2$. By Lemma 3.14, the scalar value changes by only $o(h)$, making the gap at most $o(h) - wC_{\mathrm{PS}}h\Pr(E)/2 < 0$, contrary to Lemma E.1.

Location identity. Write $C_c((0, H])$ for the continuous functions with compact support in $(0, H]$. Perturb individual variable i by $y \mapsto y + h\phi(y)$ for $\phi \in C_c((0, H])$ and either sign of small h, an operation of type (a). By Lemma 3.14, the scalar derivative is $w\int a_i\phi\, dQ_i$. The maximum changes by at most $|h|\|\phi\|_\infty$; absence of positive ties and dominated convergence give prophet derivative $\int b_i\phi\, dQ_i$. The nonnegative gap vanishes at zero, so

$$\int_{(0,H]} (a_i - C_{\mathrm{PS}}b_i)\phi\, dQ_i = 0 \qquad (\phi \in C_c((0, H])).$$

A finite signed measure on $(0, H]$ that annihilates $C_c((0, H])$ is zero, giving $a_i = C_{\mathrm{PS}}b_i$ for Q_i-almost every positive x. □

Proof of Lemma 3.15. Use the critical limiting instance $\mathfrak{B}$, positive weights w, c and optimal threshold policy π_0 from Lemma 3.15. Write H for its cap, μ for its Poisson mark measure and $\tau = \tau_{\pi_0}$ for its stopping time. Suppose every duration derivative vanishes. With other values and arrival times fixed, τ is nonincreasing in

an individual value V_i: a higher value can only trigger selection at its arrival. Zero covariance with every bounded centered score makes $\mathbb{E}[\tau \mid V_i]$ constant. Write $\tau(x, \cdot)$ for the stopping time when V_i is set to x and the remaining randomness is unchanged. For supported $x < x'$, the nonnegative difference $\tau(x, \cdot) - \tau(x', \cdot)$ has zero mean. Apply its almost-sure equality on a countable dense subset and the extreme points, then use monotonicity between them. Distinct arrival times imply that the selected variable also does not depend on this value.

Consequently the conditional selection probability $a_i(x)$ of Lemma F.1 is constant on the support. Mean contraction also preserves the prophet's expected maximum. Indeed, that lemma makes the competing maximum X^*_{-i}'s cumulative distribution function constant between supported values, so $x \mapsto \mathbb{E}[\max(x, X^*_{-i})]$ is affine on their convex hull. The CDF is the slope of this expected-maximum function at its continuity points. Replacing V_i by its mean preserves the prophet expectation and the old policy's reward and duration: its monotone stopping rule is constant throughout that hull. Conversely, run any policy for the contracted distribution on the old distribution by substituting the mean for V_i. Thus the scalar optimum is unchanged.

We can repeat the contraction because the modified instance remains in the same class and the duration derivatives of its optimal threshold policy are still zero. Specifically, Lemma E.1(a), using point masses at the means of the corresponding finite approximating variables, shows that the contracted instance is still a limit of finite instances. It is still critical, so Lemma F.1 still excludes equal positive contracted means that can jointly maximize. A zero atom forces the constant selection probability to be zero; stationarity under perturbations of values then gives zero contribution to the maximum, with the same conclusion. The stopping time remains the same function of other variables and the Poisson process, preserving their zero score covariances. Uniqueness identifies it with the new optimal stopping time, allowing iteration.

For a finite positive Poisson band B, the Mecke identity and the Poisson perturbation formula [22, Theorems 4.1 and 19.1] write the duration derivative as the expected effect of inserting a point. This fixed Markov policy can only stop earlier, so zero derivative means that the stopping time is unchanged almost surely for almost every insertion location under the band's intensity. Such an insertion also leaves the selected reward unchanged. The fixed policy's weighted payoff therefore has zero derivative; criticality and the envelope identity force the expected maximum to have zero derivative as well. Intensity stationarity for the current maximum X^* gives

$$\int_B \mathbb{E}[(x - X^*)_+]\, \mu(dx) = 0.$$

Condition on the positive-probability event that the band is empty. Independence then forces the outside maximum to dominate almost every band mark, so deletion preserves value and expected maximum. For a μ-continuity band B, the map $x \mapsto x\mathbf{1}_{B^c}(x)$ is almost-everywhere continuous and Lemma E.1(b) preserves the convergence class. Delete successively larger finite continuity bands of the Poisson process. The policy ignores deleted points, preserving its stopping time and remaining score covariances; each instance is therefore still critical with zero duration responses.

For integers $N \geq 1$, let $\mathfrak{B}_N$ contract the first N individual variables and delete the Poisson process above a positive continuity level $\delta_N \downarrow 0$. It has the original value and expected maximum. Compare it with the instance $\mathfrak{B}_\infty$ in which every individual variable is contracted and the Poisson process is absent. Both have terminal reward zero. As in Lemma 3.13, the errors after omitting variables beyond N and the remaining Poisson process are bounded by their expected maxima (multiplied by w for scalar value). Before contraction, the remaining variables are controlled by their exceedance probabilities; after contraction, they are controlled by the largest remaining mean. For every $\delta > 0$, local finiteness bounds these maxima by

$$\delta + H \sum_{i>N} \Pr(V_i > \delta) + \delta_N \quad \text{and} \quad \sup_{i>N} \mathbb{E}[V_i] \leq \delta + H \sup_{i>N} \Pr(V_i > \delta),$$

respectively. Let $N \to \infty$, then $\delta \downarrow 0$. Values and expected maxima agree in the limit, so $\mathfrak{B}_\infty$ remains critical. Its positive deterministic values tend to zero and therefore have a largest member.

Ignore all but the largest deterministic value H_*. Normalize $H_* = 1$ and $w + c = 1$. The competitive benchmark is then one, so a lower bound on scalar value is also a lower bound on the scalar ratio. This deterministic variable arrives uniformly and has scalar value $c + \int_0^1 (w - cu)_+ \, du$, whose minimum is $\sqrt{3} - 1$ at $w/c = \sqrt{3} - 1$. Additional variables can only increase the optimal value, contradicting $C_{\mathrm{PS}} < \sqrt{3} - 1$. □

Every point-insertion contribution to the duration derivative is nonpositive. A finite band with nonzero response can therefore be enlarged to $[\delta, H]$, with $\delta > 0$ a continuity point, without cancelling its response. The enlargement gives the upper-band choice stated in Lemma 3.15.

Remark F.2. *The location identity in Lemma F.1 is essential to mean contraction: a convex function may agree with a line at two support points and lie strictly below it between them. Stationarity under likelihood perturbations alone does not justify the step.*

G Finite-bit implementation

We implement the planner and calibrated learner with bounded random-bit budgets. The main steps are to round the planner's transition uniforms, preserve arrival ranks when rounding clocks, and treat the outer tie-breaking marks separately. The last step lets the implementation error scale with the expected maximum even when the original values are unbounded.

Proposition G.1. *For rationally represented observations, the bounded learner and its outer reduction have implementations with polynomial bit complexity in n, m and the input representation length. They use a polynomial number of independent fair random bits on every outcome. The bounded implementation has $O(1/m)$ reward and reach deficits. The outer implementation has an $O(1/m)$ deficit in competitive ratio. The bounded learner uses two blocks of m samples per distribution. The outer reduction uses $3m + O(\log m)$.*

For the bounded bit implementation, fix a rational lower bound in $[\mu_{\min}/2, \mu_{\min}]$ and use it as the class parameter throughout Construction 11.1, including the choice of the rational endpoint b. Every originally eligible instance remains eligible, so Theorem 4.2 gives the same sample budget and $O(1/m)$ deficits, with constants depending only on the original $\mu_{\min}$. These rational choices are fixed constants of the implementation.

G.1 Rounding the planner

Rounding changes the transition operator. Its contraction lets us control the resulting value error through the error of one transition.

Let T be the right side of (57). In the sampler of Lemma 12.2, for an integer $b \geq 1$, replace each uniform by its downward b-bit rounding. Evaluate the categorical and Bernoulli probabilities as exact rationals. The resulting operator T_b has the same term zu/β_A, coefficient κ_A and functions $f_{i,B}$, with a rounded transition distribution. It still contracts by $1/4$ and maps functions taking values in $[0, 5]$ into that interval. Denote its fixed point by $\bar{v}_b$.

Put $h_b = 2^{-b}$ and let N bound the number of active identities in a query. Couple the ideal and rounded samplers using the same uniforms. The first-success categorical choice has at most N probability boundaries, so it changes with probability at most Nh_b. The later Bernoulli choices for D contribute at most Nh_b, as does choosing a uniform member of D. When these choices agree, the two maxima of uniforms differ by at most h_b. For a remaining identity, both its independent membership uniform and its threshold s move by at most h_b; a bound of $4h_b$ on a different membership decision is sufficient. A union bound thus bounds the probability of any different discrete set or index by $10Nh_b$.

On matching discrete transitions, $|s - s_b| \leq h_b$. For every fixed normalized-time policy, its payoff as a function of the horizon is affine with slope in $[0, z]$. Taking the supremum shows that $v_B^Q(u; 1, z)$ is

z-Lipschitz in u, and hence $\bar{v}_B$ is z/β_B-Lipschitz. The two integrands at the true $\bar{v}$ differ by at most $4h_b$ on matching transitions and by at most one otherwise. Consequently,

$$\|T\bar{v} - T_b\bar{v}\|_\infty \le \frac{1}{4}(10N+4)2^{-b}, \qquad \|\bar{v} - \bar{v}_b\|_\infty \le \frac{10N+4}{3}\,2^{-b}. \tag{72}$$

The second inequality follows by applying the contraction bound to the difference of the two fixed points.

For a top-level planner call, choose one precision b such that the last bound, multiplied by an upper bound on β_A, is at most $h/4$. By (58), it suffices to take

$$b = O\big(L(Q) + \log(N+1) + \log(1/h)\big).$$

Use this same rounded kernel at every recursive level and in both copies of every paired difference. The telescoping identity, independence, contraction, clipping, and (59) then hold with fixed point $\bar{v}_b$. Run the estimator with accuracy $h/2$ and the prescribed confidence. The deterministic error in (72) transfers that guarantee to the original value.

For completeness, its exact arithmetic also has polynomial bit length. Empirical probabilities have denominators at most m; the products β_A and first-success probabilities have bit length polynomial in n and $\log m$. A child horizon is the input horizon multiplied by at most r dyadic factors, adding $O(rb)$ bits. The recursive estimator has $O(16^r)$ nodes, each performing polynomially many rational operations in n, m. Its divisions use known rational quantities from the empirical distributions, sets, and averaging counts, never recursively estimated denominators. Multiplying the finitely many coefficient and leaf denominators gives a common denominator of polynomial bit length in the total number of operations and input length. The same bound holds for numerators. Rational arithmetic, positive parts, clipping, comparisons, and medians therefore give the claimed bit complexity.

The learner's other numerical tasks have the same property. The empirical expected maximum is computed from the sorted support values and the product of marginal cumulative distribution functions. Its value is a rational sum over those support intervals, with polynomially many terms and polynomial bit length. Coefficient grids can be chosen rational. Solving one linear piece of the empirical duration equation and computing its mixing probability are rational operations. Round the final mixture probability to a dyadic grid with error at most m^{-2}. Coupling the two coins changes the selected policy with probability at most m^{-2}, so each bounded reward, duration, and reach expectation changes by at most that amount.

G.2 Clocks and online execution

Arrival ranks determine which variables have been seen, while numerical timestamps enter the thresholds. We preserve the ranks and bound the effect of rounding the timestamps.

Use a power-of-two clock grid of size Q_{clk}. In the ideal rounded-clock protocol, generate a uniform permutation independently of n grid clocks, sort the clocks, and assign them successively in permutation order. Process equal timestamps in that same order. During the online phase the permutation is the exogenous arrival order; the simulated callback sees only current and past assigned times. During validation use an independent permutation with the same distribution. Thus validation and online execution have the same trajectory distribution.

Couple the sorted grid clocks to sorted continuous uniforms by downward rounding, using the same permutation in both executions. Let τ be the continuous stopping time under the implemented policy and $\widehat{\tau}$ the assigned grid timestamp. Both equal 1 if nothing is selected. Thus $|\tau - \widehat{\tau}| \le Q_{\mathrm{clk}}^{-1}$. The selected value is X_τ. Every timestamp moves by at most Q_{clk}^{-1}, and the arrival ranks and remaining identity sets are unchanged even when rounded times coincide. Compare the rounded-clock numerical policy with the exact empirical comparison policy on the coupled continuous clocks. On accurate queries, the threshold error at the rounded horizon is at most the policy tolerance h from Construction 11.1. The horizon Lipschitz bound adds at most

$4/Q_{\mathrm{clk}}$ when comparing with the exact threshold at the continuous horizon. Thus the first action disagreement on accurate queries lies in the boundary band of Lemma 11.5 with tolerance $h + 4/Q_{\mathrm{clk}}$.

To generate a validation permutation with bounded fair random bits, sort a separate set of n independent finite-bit keys and break key ties deterministically. These keys are independent of timestamps and planner randomness. Couple them to continuous keys, whose order is exactly uniform. The permutations can differ only when two grid keys coincide, an event of probability at most $n(n-1)/(2Q_{\mathrm{clk}})$. Across the m validation trajectories the total probability is at most $mn(n-1)/(2Q_{\mathrm{clk}})$. On the complementary coupling event the learned rule is identical to the ideal rounded-clock rule under the same remaining randomness. Normalized reward and each reach indicator are bounded by one, so this probability also bounds the extra expected loss. We analyze the ideal rounded protocol directly and transfer its guarantees through this coupling.

For example, choose the least power of two at least

$$m^3(n+1)^2 + \frac{m^2(1+\bar{L})^2}{z_-},$$

where $\bar{L}$ is the empirical count cutoff in Section 11. Enlarging the boundary band adds $O(m^{-2})$ to the bound on its probability. The validation-permutation error is also $O(m^{-2})$.

Use rounded stopping timestamps both in the validation curves and on the fresh trajectory defining the selected curve's mean. Conditional on the training information, the validation trajectories are independent copies of this trajectory, and the curves remain nondecreasing. The calibration moment bounds therefore apply. On the middle segments and the right extension, the conditional mean of the continuous stopping time is at most the selected curve's mean plus Q_{clk}^{-1}. The reward comparison in (54) thus loses at most an additional z_+/Q_{clk} from the duration term. For reach, the continuous stopping time dominates the rounded timestamp, and the last grid policy satisfies $X_\tau \geq z_+(1-\widehat{\tau})$ by the threshold floor. The same right-extension probability bound follows. With the exact-policy comparison, the proof of Theorem 4.2 gives $O(1/m)$ reward and reach deficits for the ideal rounded-clock protocol. The key and coin couplings preserve this order. The clock precision, permutation keys, and mixture coin all use polynomially many bits.

G.3 Outer tie-breaking marks and unbounded values

For bounded values, a small probability of a different action directly bounds the expected loss. For unbounded values, we use an exceptional event independent of the values so that its expected loss scales with the expected maximum. The independence of the tie-breaking marks supplies this property.

The cap-scale budget $q_m = 6000\lceil \log_2(2m) \rceil$ in Construction 5.1 is an integer multiple of 500 and is computed from integer bit length. Its sample count and concentration bounds therefore apply directly in the finite-bit model.

Reserve $N_{\mathrm{out}} = n(3m + q_m + 1)$ independent outer marks, one for each allowed training observation and online candidate, including observations on unused branches. Round the marks downward to a power-of-two grid of size $Q_{\mathrm{out}} \geq m^2 N_{\mathrm{out}}^2$. Let E be the event that two reserved grid marks coincide. Then

$$\Pr(E) \leq \frac{N_{\mathrm{out}}(N_{\mathrm{out}}-1)}{2Q_{\mathrm{out}}} \leq \frac{1}{2m^2}.$$

On E^c, the order of all marks is preserved. Every lexicographic cutoff comparison, and hence the entire algorithm under the same remaining randomness, is identical. The event E depends only on the marks and is independent of all sample and online values. Let π^{round} and π^{cont} be the policies using rounded and continuous outer marks, respectively, with the same remaining randomness. For original expected maximum $\mathbb{E}[X^*] < \infty$, their selected values satisfy

$$\mathbb{E}\left[\left|X_{\tau_{\pi^{\mathrm{round}}}} - X_{\tau_{\pi^{\mathrm{cont}}}}\right|\right] \leq \mathbb{E}\left[\max_i V_i \, \mathbf{1}_E\right] = \mathbb{E}[X^*]\Pr(E) \leq \frac{\mathbb{E}[X^*]}{2m^2}.$$

Thus this error remains proportional to the benchmark before any bounded normalization. Auxiliary marks introduced only to analyze the numerical cap order statistic need not be generated, since they leave its numerical value unchanged.

The bounded learner uses two independent sample blocks, while the outer budget is $3m + q_m$. All subsequent observations after outer rounding lie on a rational bounded grid. The preceding implementation adds $O(1/m)$ to the bounded deficits and $O(\mathbb{E}[X^*]/m^2)$ for the outer marks. The lifting argument in Lemma 5.4 therefore gives the same competitive-ratio order. The number of arithmetic operations and random bits is polynomial; when the original observations are rational, their representation lengths also give the stated polynomial bound for bit operations. This proves Proposition G.1 and the bit-complexity claim of Theorem 12.1.

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